

NONHYPERLINEAR GROUPS EXIST

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ABSTRACT. We construct a countable discrete group that is not hyperlinear. It is a wreath product built from a polynomial elementary subgroup of an explicit Laurent semidirect product, and a specified nonidentity product of two lamps vanishes under every homomorphism into a tracial matrix ultraproduct. We prove that commutants of finitely generated Kazhdan groups are internal in the original matrix dimensions, and that an internal subgroup commutant is normalized by a Kazhdan group generated by subgroup compressions. The argument constructs expanding matrix blocks from a quotient spectral gap, matches their full multiplicity algebras, and controls both matrix sizes and multiplicities under compression. The main result of this paper was obtained using the Danus system.

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1. INTRODUCTION

Hyperlinearity asks whether finite pieces of a group admit arbitrarily accurate unitary matrix models in normalized Hilbert–Schmidt norm. For $d \geq 1$ we write

$$\mathrm{tr}_d = \mathrm{Tr}/d, \quad \|x\|_{2,d} = (\mathrm{Tr}(x^*x)/d)^{1/2} \quad (x \in M_d(\mathbb{C})).$$

A discrete group H is hyperlinear if, for every finite symmetric $F \subset H$ containing the identity e and every $\varepsilon > 0$, there are $d \geq 1$ and a map $\phi : F \rightarrow U(d)$ with $\phi(e) = I_d$ such that

$$\|\phi(gh) - \phi(g)\phi(h)\|_{2,d} < \varepsilon \quad (g, h, gh \in F),$$

and

$$\|\phi(g) - \phi(h)\|_{2,d} - \sqrt{2} < \varepsilon \quad (g, h \in F, g \neq h).$$

The dimension is unrestricted. The problem is therefore to find an obstruction that survives every choice of matrix size. Spectral rigidity provides a possible source of such an obstruction, but the normalization allows errors to concentrate on subspaces of small relative dimension.

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Several approaches connect this problem with rigidity of approximate representations. Dogon [Dog23, Theorems 1.3 and 1.6] obtains nonhyperlinear central extensions under weak completely positive stability, in particular under flexible Hilbert–Schmidt stability, together with the stated Kazhdan and extension hypotheses. Dogon and Vigdorovich [DV25, Theorem 1.3] develop this approach for groups with a uniform spectral gap in finite-dimensional representations. If such a group is Hilbert–Schmidt stable and admits a central extension by $\mathbb{Z}$ with finite abelianization, they obtain a nonhyperlinear finite central extension of the base group. Their question on approximation of representations of $\mathrm{SL}_2(\mathbb{Z})$ by congruence representations through odd congruence quotients would supply a concrete application [DV25, Question 1.4 and Corollary 1.5]. These results make the stability premise a precise remaining input. Strict stability in the original dimension is a different demand: Becker and Lubotzky [BL20, Theorem 1.3(ii)] show that infinite hyperlinear Kazhdan groups fail strict Hilbert–Schmidt stability.

Other approximation problems also lead close to the group question. Taller and Vidick [TV25, Theorem 5.2 and Remark 5.3] prove that approximating the quantum value of a linear constraint system game is RE-hard, with a positive completeness loss. Their discussion of the group consequence requires perfect completeness [TV25, Section 1, p. 3], which [TV25, Theorem 5.2] does not establish. Manzoor [Man25, Theorems 1.1 and 1.2] constructs a non-co-hyperlinear invariant random subgroup and a probability-measure-preserving equivalence relation whose von Neumann algebra is not Connes embeddable. A nonembeddable trace obtained in this way need not be the canonical trace of a discrete group. Thus these results leave distinct interfaces to the finite-local group obstruction considered here.

Our construction uses rigidity under subgroup compression. Kun’s expander decomposition for sofic approximations of Kazhdan groups [Kun16, Theorem 1] supplies a model for turning a spectral gap into finite components. Alekseev and Thom [AT26] describe centralizers of such permutation approximations and ask for a unitary analogue in [AT26, Open Problem 6.2(a)]. Kun and Thom [KT26] prove normalization of permutation centralizers under subgroup compression and use it to construct explicit nonsofic wreath products. The polynomial and Laurent groups used below are supplied by [KT26, Theorem E and Section 5.2]. Eckhardt [Eck26, Theorem 4.3] obtains a related obstruction to operator-norm matrix approximation. To obtain a normalized Hilbert–Schmidt obstruction, one needs finite matrix algebras realizing the entire unitary commutant and a normalization theorem for these algebras. Permutation components and operator-norm estimates do not themselves supply those two statements.

We show that a specific wreath product from the Kun–Thom family is not hyperlinear. Let $\mathbb{F}_2$ be the field of two elements and put

$$R_+ = \mathbb{F}_2[x_1, x_2, x_3], \quad R = \mathbb{F}_2[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}].$$

For a unital ring B , let $\mathrm{EL}_3(B)$ be the subgroup of $\mathrm{GL}_3(B)$ generated by the matrices $I_3 + bE_{ij}$, where $b \in B$, $i \neq j$, and E_{ij} is the usual matrix unit. For a column vector $v \in \mathbb{Z}^3$ write $x^v = x_1^{v_1} x_2^{v_2} x_3^{v_3}$. A matrix $A \in \mathrm{SL}_3(\mathbb{Z})$ acts by $\alpha_A(x^v) = x^{Av}$, extended to the Laurent ring and matrix entries. Define

$$\Gamma = \mathrm{EL}_3(R_+), \quad G = \mathrm{EL}_3(R) \rtimes_{\alpha} \mathrm{SL}_3(\mathbb{Z}), \quad W = \left(\bigoplus_{G/\Gamma} C_2 \right) \rtimes G.$$

Here C_2 is the group of order two, Γ lies in the first factor of G , and G acts on the left cosets by left translation. For a coset x write a_x for its nonidentity lamp. Set

$$t = (I_3, I_3 + E_{12}), \quad \gamma_0 = (I_3 + x_2 E_{12}, I_3), \quad b = a_{\gamma_0 t \Gamma} a_{t \Gamma}.$$

For a nonprincipal ultrafilter ω on $\mathbb{N}$ and positive integers d_n , the tracial matrix ultraproduct

$$\mathcal{M} = \prod_{n \rightarrow \omega} (M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$$

is the algebra of uniformly operator-norm-bounded sequences modulo sequences whose normalized 2-norm tends to zero along ω .

Theorem 1.1. *The group W defined above is countable and $b \neq e$. For every nonprincipal ultrafilter ω on $\mathbb{N}$, every sequence $d_n \geq 1$, and every homomorphism $\rho : W \rightarrow U(\mathcal{M})$, one has $\rho(b) = I$.*

There exist a finite symmetric $F \subset W$ containing e and $\eta > 0$ such that for every $d \geq 1$ and every $\phi : F \rightarrow U(d)$ with $\phi(e) = I_d$, at least one of the following holds:

- (i) *Some $g, h, gh \in F$ satisfy $\|\phi(gh) - \phi(g)\phi(h)\|_{2,d} \geq \eta$.*
- (ii) *Some distinct $g, h \in F$ satisfy $|\|\phi(g) - \phi(h)\|_{2,d} - \sqrt{2}| \geq \eta$.*

In particular, W is not hyperlinear.

This is Theorem 8.3. The construction fixes the field, the matrix rank and the number of variables, as well as the element that every matrix-ultraproduct homomorphism kills. The finite set F and tolerance η are obtained by a compactness argument; their existence is uniform over all dimensions. Nontriviality of b is checked directly in the lamp group.

The first analytical ingredient realizes Kazhdan commutants in the original coordinate dimensions. A group is Kazhdan if it has property (T), the uniform displacement property for unitary representations off their invariant vectors. Discrete Kazhdan groups are finitely generated; see [BHV08, Theorem 1.3.1] for this fact and general background. For unital $*$ -subalgebras $A_n \subseteq M_{d_n}(\mathbb{C})$, write $[A_n]_\omega$ for the classes of uniformly operator-norm-bounded sequences with entries in A_n . A subalgebra of $\mathcal{M}$ of this form is called *internal*.

Theorem 1.2. *Let H be a finitely generated discrete group with property (T), let ω be a nonprincipal ultrafilter on $\mathbb{N}$, let $d_n \geq 1$, and let $\pi : H \rightarrow U(\mathcal{M})$ be a homomorphism. Then there are unital $*$ -subalgebras $A_n \subseteq M_{d_n}(\mathbb{C})$ such that*

$$\pi(H)' \cap \mathcal{M} = [A_n]_\omega.$$

This is Theorem 6.7. It applies to an arbitrary homomorphism and to the specified ultrafilter and dimensions. It answers [AT26, Open Problem 6.2(a)] without the dimension change allowed there. The conclusion concerns the commutant of the ultraproduct representation; it does not assert that its coordinate lifts are exact group representations.

The second ingredient converts the one-sided inclusions furnished by subgroup compression into normalization. We write $Z(\mathcal{D})$ for the center of a von Neumann algebra $\mathcal{D}$.

Theorem 1.3. *Let H be a countable group with property (T), let $\Lambda \leq H$, and suppose that $H = \langle \Lambda, t_1, \dots, t_s \rangle$ for finitely many t_i satisfying $t_i \Lambda t_i^{-1} \leq \Lambda$. Let $\pi : H \rightarrow U(\mathcal{M})$ be a homomorphism into a tracial matrix ultraproduct with specified ω and dimensions d_n . Suppose $\mathcal{D} = \pi(\Lambda)' \cap \mathcal{M} = [A_n]_\omega$ is internal. Then $\pi(H)$ normalizes $\mathcal{D}$, and there are unital abelian subalgebras $B_n \subseteq Z(A_n)$ satisfying*

$$Z(\mathcal{D}) \cap \pi(H)' = [B_n]_\omega.$$

This is Theorem 7.4. Only the subgroup commutant is assumed internal. In the wreath-product application, the preceding theorem supplies this assumption for Γ . Normalization then forces the two lamps defining b to have the same image.

The main contribution is the passage from quotient spectral gap to finite matrix structure with errors independent of the number and ranks of the blocks. We adapt Kun's exceptional-set and smoothing method to projections, select minimum-rank pieces with a positive leakage penalty, and use a monotone resolvent construction to control both their summed covariance and leakage on their actual ranges. Equivariant rounding uses finite-algebra projection comparison from Houdayer–Vaes [HV13, Lemma 2.1] and the near-identity conjugacy estimate recorded by Ioana [Ioa25, Lemma 2.2(1)]. The resulting expanding blocks must then be matched across unequal ranks. Bounded polar intertwiners allow property (T) to perform this matching, and uniform conditional-expectation estimates assemble the entire commutant. Finally, normalization requires a separate argument: an internal invariant center, a relative finite dilation, and two bounded median transforms controlling both matrix sizes and representation multiplicities. The

two transforms turn integer branching inequalities into equality on a large corner. The group pair and the lamp construction are those of Kun–Thom; the new analytical statements make their obstruction apply to arbitrary normalized-Hilbert–Schmidt matrix models.

Section 2 fixes the trace and ultraproduct conventions. Sections 3 and 4 prove uniform projection repair and equivariant rounding. Section 5 constructs the expanding blocks, and Section 6 matches them and proves internality. Section 7 proves normalization under subgroup compression. Section 8 applies these results to the explicit group and obtains the finite-local obstruction.

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2. NOTATION AND CONVENTIONS

Set-up 2.1. Let ω be a specified nonprincipal ultrafilter on $\mathbb{N}$ and let $d_n \geq 1$. Write $\text{tr}_d = \text{Tr}/d$ and $\|x\|_{2,d} = \text{tr}_d(x^*x)^{1/2}$. The tracial matrix ultraproduct $\mathcal{M} = \prod_{n \rightarrow \omega} (M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ is the quotient of uniformly operator-norm-bounded sequences by those whose normalized 2-norm tends to zero along ω . Its trace is τ . An unmarked coordinate 2-norm uses the denominator d_n , including for matrices supported on a corner. A norm with subscript $2, r$ uses that corner’s rank r instead. Rectangular Hilbert–Schmidt norms have the explicitly indicated ambient denominator. Limits along ω are denoted by $\lim_\omega$; auxiliary tolerances are fixed before such limits unless explicitly stated otherwise.

Definition 2.2. A unital von Neumann subalgebra $\mathcal{A} \subseteq \mathcal{M}$ is *internal* if there are unital $*$ -subalgebras $A_n \subseteq M_{d_n}(\mathbb{C})$ for which $\mathcal{A} = [A_n]_\omega$, the classes of uniformly operator-norm-bounded sequences with their n th entries in A_n . No change of the dimensions d_n is part of this definition.

Set-up 2.3. For a fixed positive integer h and unitaries $u_{j,n} \in U(d_n)$, write $u_j = [u_{j,n}]$ and $\mathcal{C} = \{u_1, \dots, u_h\}' \cap \mathcal{M}$. Put

$$T_n(x) = \frac{x}{2} + \frac{1}{4h} \sum_{j=1}^h (u_{j,n} x u_{j,n}^* + u_{j,n}^* x u_{j,n}), \quad \mathcal{E}_n(x) = \frac{1}{4h} \sum_{j=1}^h \|[u_{j,n}, x]\|_{2,d_n}^2.$$

The corresponding quotient map is T . We say the tuple has *quotient spectral gap* $a \in (0, 1)$ if

$$\langle (I - T)\xi, \xi \rangle \geq a \|\xi\|_2^2 \quad (\xi \in L^2(\mathcal{M}) \ominus L^2(\mathcal{C})).$$

Here $L^2(\mathcal{M})$ is the trace Hilbert space, not the full Hilbert ultraproduct of the coordinate Hilbert–Schmidt spaces. For a coordinate linear map L , write $\|L\|_{\infty \rightarrow 2, n} = \sup_{\|x\|_{\text{op}} \leq 1} \|L(x)\|_{2, d_n}$.

Definition 2.4. A *positive operator-valued partition* is a finite list of positive matrices summing to I ; a *projection-valued partition* is such a list consisting of projections. Zero outcomes are permitted. All energies and squared distances of lists are summed over outcomes. A unitary list $v_1, \dots, v_k$ on an r -dimensional space has *scalar gap* $c > 0$ if

$$\frac{1}{4k} \sum_{j=1}^k \|[v_j, x]\|_{2,r}^2 \geq c \|x\|_{2,r}^2 \quad (x \in M_r(\mathbb{C}), \text{Tr } x = 0).$$

A group has *property (T)* if its unitary representations have a uniform displacement bound off invariant vectors for one finite Kazhdan set. For a finitely generated such group we use a nonempty finite symmetric generating list, with repetitions allowed.

Definition 2.5. Projections p, q in a von Neumann algebra are *equivalent*, written $p \sim q$, if there is a partial isometry v with $v^*v = p$ and $vv^* = q$. A *MASA* is a maximal abelian self-adjoint von Neumann subalgebra. In a finite tracial von Neumann algebra, $E_{\mathcal{A}}$ denotes the trace-preserving conditional expectation onto a unital von Neumann subalgebra $\mathcal{A}$.

Definition 2.6. A discrete group H is *hyperlinear* if for every finite symmetric $F \subset H$ containing the identity e and every $\varepsilon > 0$ there exist $d \geq 1$ and $\phi : F \rightarrow U(d)$ with $\phi(e) = I_d$ such that

$$\|\phi(gh) - \phi(g)\phi(h)\|_{2,d} < \varepsilon \quad (g, h, gh \in F),$$

and

$$\|\phi(g) - \phi(h)\|_{2,d} - \sqrt{2} < \varepsilon \quad (g, h \in F, g \neq h).$$

3. UNIFORM REPAIR FROM QUOTIENT SPECTRAL GAP

We retain the notation of Section 2. We first lift the conditional expectation onto $\mathcal{C}$ by a single sequence of averaging maps. We then obtain one large corner on which every low-energy projection admits a repair. Throughout, the ultrafilter ω is the specified one, and coordinate norms use the ambient denominator d_n .

Proposition 3.1. *Let $\omega, d_n, h, u_{j,n}$ be as in the standing setting and suppose that their quotient spectral gap is $a \in (0, 1)$. Then there are integers $\ell_n \rightarrow_{\omega} \infty$ such that $\Phi_n = T_n^{\ell_n}$ induces $E_{\mathcal{C}}$ on every operator-norm-bounded sequence. For every sequence of unital $*$ -subalgebras $A_n \subseteq M_{d_n}(\mathbb{C})$,*

$$[A_n]_{\omega} = \mathcal{C} \iff \|\Phi_n - E_{A_n}\|_{\infty \rightarrow 2, n} \rightarrow_{\omega} 0.$$

If the u_j are the images of a homomorphism from a group with property (T), and $\kappa \in (0, 1]$ is a Kazhdan displacement constant for the chosen generating list of length h , then the quotient gap holds with $a = \kappa^2/(4h)$.

The averaging lengths are chosen uniformly over the whole operator unit ball. Thus the resulting lift applies to every bounded sequence, including sequences chosen after the lengths have been fixed.

Proof. We identify the Hilbert-space fixed vectors, transfer estimates on the operator unit ball, and choose the averaging lengths. We then prove the algebra criterion and the property (T) assertion.

Step 1. In this step, we identify the fixed space of T on $L^2(\mathcal{M})$ with $L^2(\mathcal{C})$ and the orthogonal projection onto it with $E_{\mathcal{C}}$, and we obtain the spectral bounds (3.2) from the quotient spectral gap. Each T_n is a convex combination of inner automorphisms. It is unital, completely positive, trace preserving, and contractive in operator norm and in 2-norm. As a Hilbert-space operator it is self-adjoint and satisfies $0 \leq T_n \leq I$: the operator $2T_n - I$ is a self-adjoint average of unitary operators. The same assertions hold for T on $L^2(\mathcal{M})$. Expansion of the commutators gives

$$(3.1) \quad \langle (I - T_n)x, x \rangle = \frac{1}{4h} \sum_{j=1}^h \|[u_{j,n}, x]\|_{2, d_n}^2 = \mathcal{E}_n(x).$$

The analogous identity holds for every vector in $L^2(\mathcal{M})$.

The spectral theorem gives strong convergence of T^m to the orthogonal projection P onto $\ker(I - T)$. If $x \in \mathcal{M}$, the sequence $T^m(x)$ stays in the operator-norm ball of radius $\|x\|_{\text{op}}$. This ball is closed in 2-norm: an ultraweakly convergent subnet of a bounded net has the same limit as its 2-norm limit, as is seen by testing against $y \mapsto \tau(y^* \cdot)$ for $y \in \mathcal{M}$. Consequently $P(x) \in \mathcal{M}$. The quotient version of (3.1) implies $P(x) \in \mathcal{C}$. Conversely, T fixes every element of $\mathcal{C}$. Density of $\mathcal{M}$ in its trace Hilbert space now gives

$$\ker(I - T) = L^2(\mathcal{C}), \quad P|_{\mathcal{M}} = E_{\mathcal{C}}.$$

For the second equality, the trace-preserving expectation is the orthogonal projection in 2-norm: for $c \in \mathcal{C}$, $\tau(c^*(x - E_{\mathcal{C}}(x))) = 0$ by bimodularity and trace preservation.

Put $r = 1 - a$. On $L^2(\mathcal{C})^\perp$ the spectrum of T lies in $[0, r]$. In the following bounds the expectation also denotes its extension to $L^2(\mathcal{M})$. For integers $1 \leq j \leq k$,

$$(3.2) \quad \|T^j - E_{\mathcal{C}}\|_{2 \rightarrow 2} \leq r^j, \quad \|T^k - T^j\|_{2 \rightarrow 2} \leq r^j.$$

Indeed, the second difference vanishes on $L^2(\mathcal{C})$ and its absolute value on $[0, r]$ is $t^j(1 - t^{k-j}) \leq r^j$.

Step 2. In this step, we prove the uniform transfer identity (3.3) on the operator unit ball, choose the averaging lengths $\ell_n \rightarrow_\omega \infty$, and show that $\Phi_n = T_n^{\ell_n}$ induces $E_{\mathcal{C}}$ on every operator-norm-bounded sequence. Suppose linear maps $L_n : M_{d_n}(\mathbb{C}) \rightarrow M_{d_n}(\mathbb{C})$ have uniform bounds in both operator-to-operator norm and 2-to-2 norm, and let L_ω be their induced map on $\mathcal{M}$. Then

$$(3.3) \quad \sup_{\substack{x \in \mathcal{M} \\ \|x\|_{\text{op}} \leq 1}} \|L_\omega(x)\|_2 = \lim_{\omega} \|L_n\|_{\infty \rightarrow 2, n}.$$

Every contraction in $\mathcal{M}$ has contraction representatives: clip the singular values of any bounded representative at 1. Functional calculus in the quotient shows that this leaves its class unchanged. These representatives prove the upper bound in (3.3). For the reverse bound, choose in each coordinate a contraction attaining the displayed coordinate norm, using compactness of the matrix unit ball, and take its class in $\mathcal{M}$.

Apply (3.3) to $L_n = T_n^k - T_n^j$, for fixed $1 \leq j \leq k$. Both required bounds are at most 2, and (3.2) gives

$$\lim_{\omega} \|T_n^k - T_n^j\|_{\infty \rightarrow 2, n} \leq r^j.$$

We may therefore choose decreasing sets $\Omega_m \in \omega$, with $\Omega_m \subseteq \{n : n \geq m\}$, such that

$$(3.4) \quad \|T_n^k - T_n^j\|_{\infty \rightarrow 2, n} \leq r^j + \frac{1}{m} \quad (n \in \Omega_m, 1 \leq j \leq k \leq m).$$

Only finitely many estimates are imposed for each m , so finite intersections in the given ultrafilter suffice. Set

$$\ell_n = \max(\{m : 1 \leq m \leq n, n \in \Omega_m\} \cup \{1\}).$$

For every fixed m , membership in Ω_m implies $\ell_n \geq m$; hence $\ell_n \rightarrow_\omega \infty$. For fixed j , on the ω -large set where $n \in \Omega_1$ and $\ell_n \geq j$, (3.4) yields

$$\|T_n^{\ell_n} - T_n^j\|_{\infty \rightarrow 2, n} \leq r^j + \frac{1}{\ell_n}.$$

The maps $\Phi_n = T_n^{\ell_n}$ are contractions in both norms and induce a map Φ on $\mathcal{M}$. By (3.3) and (3.2), every contraction $x \in \mathcal{M}$ satisfies

$$\|\Phi(x) - E_{\mathcal{C}}(x)\|_2 \leq \|\Phi(x) - T^j(x)\|_2 + \|T^j(x) - E_{\mathcal{C}}(x)\|_2 \leq 2r^j.$$

Letting j tend to infinity gives $\Phi = E_{\mathcal{C}}$. The supremum in (3.4) makes this identity valid on the entire operator unit ball; scaling gives the assertion for every operator-norm-bounded sequence.

Step 3. In this step, we prove the finite-algebra criterion, namely that $[A_n]_\omega = \mathcal{C}$ if and only if $\|\Phi_n - E_{A_n}\|_{\infty \rightarrow 2, n} \rightarrow_\omega 0$, by comparing Φ with the map on $\mathcal{M}$ induced by the coordinate expectations E_{A_n} . Let $\mathcal{D} = [A_n]_\omega$. The coordinate expectations are contractions in both norms, so they induce a map Q on $\mathcal{M}$. Its range is exactly $\mathcal{D}$: its outputs have bounded representatives in A_n , and it fixes every such sequence. Coordinate orthogonality also gives

$$\tau(c^*(x - Q(x))) = 0 \quad (c \in \mathcal{D}, x \in \mathcal{M}).$$

If $\mathcal{D} = \mathcal{C}$, these identities characterize the orthogonal projection onto $\mathcal{C}$, and hence $Q = E_{\mathcal{C}} = \Phi$. Equation (3.3), applied to $\Phi_n - E_{A_n}$, gives the required convergence of norms. Conversely, that convergence implies $Q = \Phi$, so their ranges coincide and $\mathcal{D} = \mathcal{C}$.

Step 4. In this step, we prove the property (T) case, in which the Kazhdan displacement inequality for the conjugation representation on $L^2(\mathcal{M})$ gives the quotient gap $a = \kappa^2/(4h)$,

and conclude the proof. Write $u_j = \pi(s_j)$ for the given homomorphism and generating list. Conjugation defines a unitary representation

$$\rho(g)\xi = \pi(g)\xi\pi(g)^* \quad \text{on } L^2(\mathcal{M}).$$

Its invariant space is $L^2(\mathcal{C})$ by the fixed-space identification above and generation by the s_j . The displacement inequality and the energy identity imply

$$\langle (I - T)\xi, \xi \rangle = \frac{1}{4h} \sum_{j=1}^h \|\rho(s_j)\xi - \xi\|_2^2 \geq \frac{\kappa^2}{4h} \|\xi\|_2^2 \quad (\xi \perp L^2(\mathcal{C})).$$

This proves the stated quotient gap. $\square$

The next result makes the repair estimates relative to the trace of the input projection. The corner and the error bound are chosen before that input, so they also cover projections of very small rank.

Theorem 3.2. *Let $\omega, d_n, h, u_{j,n}$ have quotient spectral gap $a \in (0, 1)$, and put $L = 36/a^2$. Then there are projections $e_n \in M_{d_n}(\mathbb{C})$ and numbers $\eta_n > 0$ with $\text{tr}_{d_n}(1 - e_n) \rightarrow_\omega 0$ and $\eta_n \rightarrow_\omega 0$ such that, for every n and every projection $p \leq e_n$ with $\mathcal{E}_n(p) \leq \text{tr}_{d_n}(p)/(4L)$, there is a projection $f \in M_{d_n}(\mathbb{C})$ satisfying*

$$\|f - p\|_{2,d_n}^2 \leq L\mathcal{E}_n(p), \quad \frac{\text{tr}_{d_n}(p)}{4} \leq \text{tr}_{d_n}(f) \leq \frac{9\text{tr}_{d_n}(p)}{4}, \quad \mathcal{E}_n(f) \leq \eta_n \text{tr}_{d_n}(f).$$

The same e_n, η_n work for all eligible projections, with no lower bound on their relative ranks.

The uniformity comes from removing the word neighborhoods of projections on which a relative averaging estimate fails. Their complement supports all the inputs to the theorem.

Proof. We first transfer the quotient gap with a uniform additive error. For each fixed target energy, we remove a small exceptional subspace to turn this into a relative estimate. Averaging and a spectral cut then give the repair. A final diagonal choice makes the target energy tend to zero along ω .

Step 1. In this step, we transfer the quotient gap to the coordinate maps T_n and obtain the uniform additive estimate (3.5) for every coordinate projection, with errors $\delta_{k,n}$ tending to zero along ω for each fixed k . Put $r = 1 - a$ and $D_n = I - T_n$. The spectral bounds (3.2) in the proof of Proposition 3.1 imply

$$\|T^k(I - T)\xi\|_2 \leq r^k \|(I - T)\xi\|_2 \quad (\xi \in L^2(\mathcal{M}), \ k \geq 1),$$

because $(I - T)\xi$ is perpendicular to $L^2(\mathcal{C})$. For each fixed integer $k \geq 1$, define

$$\delta_{k,n} = \max \left\{ 0, \sup_{\substack{p \in M_{d_n}(\mathbb{C}) \\ p = p^* = p^2}} (\|T_n^k D_n(p)\|_{2,d_n} - r^k \|D_n(p)\|_{2,d_n}) \right\}.$$

Then $\delta_{k,n} \rightarrow_\omega 0$. Otherwise, on an ω -large set one could choose projections p_n with a fixed positive violation. Their class $[p_n] \in \mathcal{M}$ would violate the preceding Hilbert-space inequality, since fixed powers of the coordinate maps and their 2-norms pass to the quotient. Thus

$$(3.5) \quad \|T_n^k D_n(p)\|_{2,d_n} \leq r^k \|D_n(p)\|_{2,d_n} + \delta_{k,n} \quad \text{for every coordinate projection } p.$$

Step 2. In this step, we construct, for a fixed target energy η , one corner $e_{n,\eta}$ with $\text{tr}_{d_n}(1 - e_{n,\eta}) \rightarrow_\omega 0$ such that every nonzero projection $p \leq e_{n,\eta}$ satisfies the relative estimates (3.10), by removing word neighborhoods of bad projections of maximum rank. Fix $\eta \in (0, 1]$. Choose an integer $K \geq 1$ and $\zeta > 0$ such that

$$(3.6) \quad r^K \leq \frac{\eta^2}{1152}, \quad 0 < \zeta \leq \min \left\{ \frac{1}{K}, \frac{\eta^2}{1152} \right\}.$$

These choices are made before taking any ultrafilter limit. Fix a coordinate n and $1 \leq k \leq K$. Call a projection q bad for these choices if

$$(3.7) \quad \|T_n^k D_n(q)\|_{2,d_n} \geq (r^k + \zeta) \|D_n(q)\|_{2,d_n}, \quad \|D_n(q)\|_{2,d_n} \geq \eta \|q\|_{2,d_n}.$$

The zero projection is bad. Choose a bad projection $q_{k,n}$ of maximum rank; such a rank exists in the finite set $\{0, \dots, d_n\}$. Combining (3.5) and (3.7) gives

$$\eta \zeta \|q_{k,n}\|_{2,d_n} \leq \delta_{k,n}, \quad \text{tr}_{d_n}(q_{k,n}) \leq \frac{\delta_{k,n}^2}{\eta^2 \zeta^2}.$$

Let $B_{k,n}$ be the projection onto the span of $w(\text{ran } q_{k,n})$, where w ranges over words of length at most $2k + 2$ in the symmetric list

$$I, u_{1,n}, u_{1,n}^*, \dots, u_{h,n}, u_{h,n}^*.$$

Padding words with I gives at most $(2h + 1)^{2k+2}$ words, and hence

$$(3.8) \quad \text{tr}_{d_n}(B_{k,n}) \leq (2h + 1)^{2k+2} \frac{\delta_{k,n}^2}{\eta^2 \zeta^2}.$$

No nonzero projection $p \leq 1 - B_{k,n}$ can be bad. To prove this, put $q = q_{k,n}$ and, for a projection v , let $H(v)$ be the span of the subspaces $w(\text{ran } v)$ for words of length at most $k + 1$. The spaces $H(q)$ and $H(p)$ are orthogonal: for such words w_1, w_2 , $\xi \in \text{ran } q$, and $v \in \text{ran } p$,

$$\langle w_1 \xi, w_2 v \rangle = \langle w_2^* w_1 \xi, v \rangle = 0,$$

since $w_2^* w_1$ has length at most $2k + 2$. The matrices $D_n(v)$ and $T_n^k D_n(v)$ have both their ranges and adjoint ranges in $H(v)$: each is a linear combination of conjugates of v by words of length at most $k + 1$. It follows that

$$\begin{aligned} \|D_n(q + p)\|_{2,d_n}^2 &= \|D_n(q)\|_{2,d_n}^2 + \|D_n(p)\|_{2,d_n}^2, \\ \|T_n^k D_n(q + p)\|_{2,d_n}^2 &= \|T_n^k D_n(q)\|_{2,d_n}^2 + \|T_n^k D_n(p)\|_{2,d_n}^2. \end{aligned}$$

Also $p \perp q$. If p were bad, adding the squares of the inequalities in (3.7) would make $q + p$ bad, contradicting the maximal rank of q .

Let $e_{n,\eta}$ be the projection onto the orthogonal complement of the span of the ranges of $B_{1,n}, \dots, B_{K,n}$. The rank bound (3.8) gives

$$(3.9) \quad \text{tr}_{d_n}(1 - e_{n,\eta}) \leq \sum_{k=1}^K (2h + 1)^{2k+2} \frac{\delta_{k,n}^2}{\eta^2 \zeta^2} \xrightarrow{\omega} 0.$$

Every nonzero projection $p \leq e_{n,\eta}$ satisfies one of the following alternatives:

$$(3.10) \quad \begin{aligned} &\|D_n(p)\|_{2,d_n} < \eta \|p\|_{2,d_n}, \quad \text{or} \\ &\|T_n^k D_n(p)\|_{2,d_n} \leq (r^k + \zeta) \|D_n(p)\|_{2,d_n} \quad (1 \leq k \leq K). \end{aligned}$$

Indeed, if the first alternative fails, the second inequality in (3.7) holds for every k , so the first inequality there must fail for each k . This establishes (3.10) simultaneously for every such p .

Step 3. In this step, we reduce to a nonzero projection $p \leq e_{n,\eta}$ in the second alternative of the relative estimates, show that the average $z = T_n^K(p)$ stays close to p as in (3.11), and bound its energy as in (3.12). Let $p \leq e_{n,\eta}$ satisfy the low-energy hypothesis of the theorem. For $p = 0$ choose $f = 0$. In the first alternative of (3.10), choose $f = p$; then

$$\mathcal{E}_n(p) = \langle D_n(p), p \rangle \leq \|D_n(p)\|_{2,d_n} \|p\|_{2,d_n} < \eta \text{tr}_{d_n}(p),$$

and all the required bounds hold. It remains to treat a nonzero p in the second alternative. Set $z = T_n^K(p)$. It is a positive contraction and $\text{tr}_{d_n}(z) = \text{tr}_{d_n}(p)$. Since D_n is a positive Hilbert-space contraction,

$$\|D_n(p)\|_{2,d_n}^2 \leq \langle D_n(p), p \rangle = \mathcal{E}_n(p), \quad \|D_n(p)\|_{2,d_n} \leq \|p\|_{2,d_n}.$$

Telescoping and (3.10) give

$$\begin{aligned}
 \|z - p\|_{2,d_n} &\leq \sum_{j=0}^{K-1} \|T_n^j D_n(p)\|_{2,d_n} \\
 (3.11) \quad &\leq \left(\sum_{j=0}^{K-1} r^j + (K-1)\zeta \right) \|D_n(p)\|_{2,d_n} \\
 &\leq \frac{2}{a} \|D_n(p)\|_{2,d_n}.
 \end{aligned}$$

Here $K\zeta \leq 1$ and $a < 1$. Put $\theta = r^K + \zeta$, so that $\theta \leq \eta^2/576$ by (3.6). The estimate at $k = K$ gives

$$\|D_n(z)\|_{2,d_n} = \|T_n^K D_n(p)\|_{2,d_n} \leq \theta \sqrt{\text{tr}_{d_n}(p)}.$$

Since $\|z\|_{2,d_n} \leq \|p\|_{2,d_n}$, we obtain

$$(3.12) \quad \mathcal{E}_n(z) = \langle D_n(z), z \rangle \leq \theta \text{tr}_{d_n}(p).$$

Step 4. In this step, we choose a spectral cut $f = 1_{[t,1]}(z)$ of small relative energy by integrating over $t \in [1/3, 2/3]$, bound its distance to p , and deduce the three estimates of Theorem 3.2 for the fixed target η in the corner $e_{n,\eta}$. Choose an orthonormal eigenbasis $\xi_1, \dots, \xi_{d_n}$ for z , with eigenvalues $\lambda_i \in [0, 1]$, and put

$$w_{i\ell} = \frac{1}{4hd_n} \sum_{j=1}^h |\langle u_{j,n} \xi_\ell, \xi_i \rangle|^2.$$

Every row and column of these weights sums to $1/(4d_n)$. The commutator formula reads

$$\mathcal{E}_n(z) = \sum_{i,\ell} w_{i\ell} (\lambda_i - \lambda_\ell)^2.$$

For $t \in [1/3, 2/3]$, let $f_t = 1_{[t,1]}(z)$. The corresponding formula for $\mathcal{E}_n(f_t)$ replaces each eigenvalue by its indicator of $[t, 1]$. The integral of the squared difference of two such indicators is at most $|\lambda_i - \lambda_\ell|$, and it is zero unless at least one of the two eigenvalues is at least $1/3$. Writing $\mathcal{I} = \{i : \lambda_i \geq 1/3\}$, we have

$$\sum_{\substack{i,\ell \\ i \in \mathcal{I} \text{ or } \ell \in \mathcal{I}}} w_{i\ell} \leq \frac{|\mathcal{I}|}{2d_n} \leq \frac{3}{2} \text{tr}_{d_n}(z).$$

Weighted Cauchy–Schwarz therefore gives

$$\int_{1/3}^{2/3} \mathcal{E}_n(f_t) dt \leq \sqrt{\frac{3}{2} \text{tr}_{d_n}(z) \mathcal{E}_n(z)}.$$

The interval has length $1/3$, so some $t \in [1/3, 2/3]$ satisfies

$$(3.13) \quad \mathcal{E}_n(f_t) \leq 3 \sqrt{\frac{3}{2} \text{tr}_{d_n}(z) \mathcal{E}_n(z)} \leq 6 \sqrt{\text{tr}_{d_n}(z) \mathcal{E}_n(z)}.$$

We also need a distance bound for this cut, without assuming that p commutes with z . Let $\beta_i = \langle p \xi_i, \xi_i \rangle$. Then $0 \leq \beta_i \leq 1$ and

$$\|z - p\|_{2,d_n}^2 = \frac{1}{d_n} \sum_i (\lambda_i^2(1 - \beta_i) + (1 - \lambda_i)^2 \beta_i).$$

The contribution of i to $d_n \|f_t - p\|_{2,d_n}^2$ is β_i when $\lambda_i < t$, and $1 - \beta_i$ when $\lambda_i \geq t$. In the first case $(1 - \lambda_i)^2 \geq 1/9$; in the second, $\lambda_i^2 \geq 1/9$. Thus every $t \in [1/3, 2/3]$ satisfies

$$\|f_t - p\|_{2,d_n}^2 \leq 9 \|z - p\|_{2,d_n}^2.$$

Choose t as in (3.13) and put $f = f_t$. Equation (3.11) yields

$$\|f - p\|_{2,d_n}^2 \leq \frac{36}{d^2} \mathcal{E}_n(p) = L \mathcal{E}_n(p).$$

The hypothesis on p gives $\|f - p\|_{2,d_n} \leq \|p\|_{2,d_n}/2$. The triangle inequality and its reverse imply

$$\frac{1}{2} \|p\|_{2,d_n} \leq \|f\|_{2,d_n} \leq \frac{3}{2} \|p\|_{2,d_n}.$$

Squaring proves the two trace bounds. Finally, (3.13) and (3.12) give

$$\mathcal{E}_n(f) \leq 6\sqrt{\theta} \operatorname{tr}_{d_n}(p) \leq \frac{\eta}{4} \operatorname{tr}_{d_n}(p) \leq \eta \operatorname{tr}_{d_n}(f).$$

We have proved all three repair estimates for the fixed target η , in the single corner $e_{n,\eta}$. The distance bound controls the trace of f , while the spectral cut controls its energy. Every estimate retains the factor $\operatorname{tr}_{d_n}(p)$; no positive lower bound on this trace has been used.

Step 5. In this step, we define e_n and η_n by one diagonal choice among the corners $e_{n,1/m}$ constructed for the targets $\eta = 1/m$, verify that $\operatorname{tr}_{d_n}(1 - e_n) \rightarrow_\omega 0$ and that the target η_n tends to zero along ω , and conclude the proof. For each integer $m \geq 1$, apply the fixed-target construction with $\eta = 1/m$, choosing its parameters once, and denote the resulting projections by $e_{n,1/m}$. By (3.9), $\operatorname{tr}_{d_n}(1 - e_{n,1/m}) \rightarrow_\omega 0$ for every fixed m . Define decreasing sets in the specified ultrafilter by

$$\Lambda_m = \bigcap_{i=1}^m \left\{ n : \operatorname{tr}_{d_n}(1 - e_{n,1/i}) \leq \frac{1}{m} \right\}.$$

For $n \in \Lambda_1$, let

$$m(n) = \max\{m : 1 \leq m \leq n, n \in \Lambda_m\}, \quad e_n = e_{n,1/m(n)}, \quad \eta_n = \frac{1}{m(n)}.$$

For $n \notin \Lambda_1$, put $e_n = 0$ and $\eta_n = 1$. For every fixed J , the set $\Lambda_J \cap \{n : n \geq J\}$ belongs to ω and has $m(n) \geq J$. Hence $\eta_n \rightarrow_\omega 0$. On Λ_1 , the definition of $m(n)$ also gives

$$\operatorname{tr}_{d_n}(1 - e_n) \leq \frac{1}{m(n)},$$

so $\operatorname{tr}_{d_n}(1 - e_n) \rightarrow_\omega 0$. The fixed-target result supplies the three bounds for every eligible $p \leq e_n$ at each $n \in \Lambda_1$. At other indices the only input is $p = 0$, for which $f = 0$ works. Thus the same choices e_n, η_n work for all the projections asserted in the theorem, at every coordinate. $\square$

4. EQUIVARIANT ROUNDING OF OPERATOR-VALUED PARTITIONS

We use the normalized traces and summed partition distances of Section 2. We round a positive operator-valued partition while making its total commutator energy small. The number of outcomes may grow with the dimension. The proof keeps all outcomes in a single finite auxiliary algebra and returns the resulting projections to the original matrix algebra.

We use the following projection selection result [HV13, Lemma 2.1].

Lemma 4.1 (Houdayer–Vaes). *Let N be a von Neumann algebra with a faithful normal semifinite trace σ , and let $A \subseteq N$ be a MASA such that $\sigma|_A$ is semifinite. Then every projection $p \in N$ with $\sigma(p) < \infty$ is equivalent in N to a projection in A .*

In a finite algebra the restriction of the trace to a MASA is finite, so the semifiniteness hypothesis on the restriction is automatic. The equivalence in this lemma takes place inside the specified algebra. We also use the following finite-trace case of [Ioa25, Lemma 2.2(1)].

Lemma 4.2 (Ioana). *Let (N, σ) be a tracial von Neumann algebra and let $p, q \in N$ be equivalent projections. Then there is a unitary $v \in N$ with $vpv^* = q$ and $\|v - 1\|_{2,\sigma} \leq 3\|p - q\|_{2,\sigma}$.*

The estimate is unchanged if the trace is multiplied by a positive constant. We will apply it in a finite corner, with that corner's projection as its identity.

Theorem 4.3. *For every integer $h \geq 1$ and $\varepsilon > 0$ there is $\delta > 0$ such that the following holds for all $d, m \geq 1$. Let $u_1, \dots, u_h \in U(d)$, put $\mathcal{E}_u(x) = (4h)^{-1} \sum_j \| [u_j, x] \|_{2,d}^2$, and let $a_1, \dots, a_m \in M_d(\mathbb{C})$ be a positive operator-valued partition. If $\sum_i \mathcal{E}_u(a_i) \leq \delta$, then there is a projection-valued partition $q_1, \dots, q_m$ in the same $M_d(\mathbb{C})$ such that*

$$\sum_i \mathcal{E}_u(q_i) \leq \varepsilon, \quad \sum_i \|q_i - a_i\|_{2,d}^2 \leq 5000 \sum_i \text{tr}_d(a_i - a_i^2) + \varepsilon.$$

The error in distance is controlled by the failure of the input effects to be projections. This error need not tend to zero for the output energy to tend to zero. The theorem requires neither spectral gap nor relations among the unitaries.

Proof. We first cut each effect to a projection with small total energy. We then use a finite auxiliary space to move the range of their column operator into an algebra that respects every outcome. Lifting this range and its partial isometry gives a partition in the original dimension. Separate estimates control its total energy and distance.

Step 1. In this step, we choose spectral cutoff projections of the a_i and prove the summed bounds (4.1), (4.2), and (4.3) on their total energy, their distance to the a_i , and the deviation of their sum from I_d . For a positive contraction $b \in M_d(\mathbb{C})$, let $b_\theta = \mathbf{1}_{[\theta,1]}(b)$ for $\theta \in J = [1/3, 2/3]$. In an eigenbasis of b , with eigenvalues λ_r , we have, for $1 \leq j \leq h$,

$$\int_J \| [u_j, b_\theta] \|_{2,d}^2 d\theta \leq \frac{1}{d} \sum_{r,s} |\lambda_r - \lambda_s| |(u_j)_{rs}|^2 \mathbf{1}_{\{\max(\lambda_r, \lambda_s) \geq 1/3\}}.$$

Cauchy–Schwarz bounds the right-hand side by $\| [u_j, b] \|_{2,d} (6 \text{tr}_d(b))^{1/2}$. Indeed, the square of the second factor is at most

$$\frac{3}{d} \sum_{r,s} (\lambda_r + \lambda_s) |(u_j)_{rs}|^2 = 6 \text{tr}_d(b).$$

Summing over j , applying Cauchy–Schwarz again, and dividing by $|J| = 1/3$, we obtain a threshold $\theta \in J$ such that

$$\mathcal{E}_u(b_\theta) \leq \frac{3\sqrt{6}}{2} \sqrt{\text{tr}_d(b) \mathcal{E}_u(b)} \leq 4 \sqrt{\text{tr}_d(b) \mathcal{E}_u(b)}.$$

Choose such a threshold for each a_i , and write f_i for its spectral projection. Put

$$v_d = \sum_i \text{tr}_d(a_i - a_i^2), \quad \beta_d = \sum_i \mathcal{E}_u(a_i), \quad S_d = \sum_i f_i.$$

Since $\sum_i \text{tr}_d(a_i) = 1$, the preceding estimate gives

$$(4.1) \quad \sum_i \mathcal{E}_u(f_i) \leq 4\sqrt{\beta_d}.$$

For $\lambda \in [0, 1]$ and $\theta \in J$, the two cases $\lambda < \theta$ and $\lambda \geq \theta$ give

$$\begin{aligned} \mathbf{1}_{[\theta,1]}(\lambda) &\leq 3\lambda, & |\mathbf{1}_{[\theta,1]}(\lambda) - \lambda| &\leq 3\lambda(1 - \lambda), \\ |\mathbf{1}_{[\theta,1]}(\lambda) - \lambda|^2 &\leq 2\lambda(1 - \lambda). \end{aligned}$$

Consequently,

$$(4.2) \quad 0 \leq S_d \leq 3I_d, \quad |\text{tr}_d(S_d) - 1| \leq 3v_d, \quad \sum_i \|f_i - a_i\|_{2,d}^2 \leq 2v_d.$$

We also need the total overlap of these projections. The order inequality $\sum_{k \neq i} f_k \leq 3(I_d - a_i)$ implies

$$\sum_{i \neq k} \text{tr}_d(f_i f_k) \leq 3 \sum_i \text{tr}_d(f_i (I_d - a_i)) \leq 9v_d.$$

Here $\text{tr}_d(f_i f_k) = \text{tr}_d(f_i f_k f_i) \geq 0$, and the last inequality follows from $f_i \leq 3a_i$ and the commutation of f_i with a_i . Expanding the square now yields

$$(4.3) \quad \|S_d - I_d\|_{2,d}^2 = \sum_{i \neq k} \text{tr}_d(f_i f_k) + 1 - \text{tr}_d(S_d) \leq 12v_d.$$

Step 2. In this step, we construct, for a sequence of data with total energy tending to zero along ω , auxiliary spaces K_n of dimension at most $4d_n$ carrying unitaries $\hat{u}_{j,n}$ that commute with every summand projection and satisfy (4.5). Fix h and a sequence of data $(d_n, m_n, u_{j,n}, a_{i,n})$ as in the theorem. Write $u_n = (u_{1,n}, \dots, u_{h,n})$, and suppose that $\sum_i \mathcal{E}_{u_n}(a_{i,n}) \rightarrow_\omega 0$. Set

$$v_n = \sum_{i=1}^{m_n} \text{tr}_{d_n}(a_{i,n} - a_{i,n}^2), \quad v = \lim_\omega v_n \in [0, 1].$$

Apply the cutoffs above at every coordinate, obtaining $f_{i,n}$, and put $S_n = \sum_i f_{i,n}$. Equation (4.1) gives $\sum_i \mathcal{E}_{u_n}(f_{i,n}) \rightarrow_\omega 0$.

Let $H_n = \mathbb{C}^{d_n}$ and form the finite-dimensional space

$$K_n = H_n \oplus \bigoplus_{i=1}^{m_n} f_{i,n} H_n.$$

Write E_n for its first-summand projection, $F_{i,n}$ for the projection onto the i th auxiliary summand, and $F_n = \sum_i F_{i,n}$. Thus $E_n + F_n = I_{K_n}$. By (4.2),

$$(4.4) \quad d_n \leq \dim K_n = d_n + \sum_i \text{rank}(f_{i,n}) \leq 4d_n.$$

On $B(K_n)$ we use the finite trace $\sigma_n = \text{Tr}_{K_n}/d_n$ and the corresponding norm $\|\cdot\|_{2,\sigma_n}$. For a rectangular map $X : H_n \rightarrow K_n$ this notation means $\|X\|_{2,\sigma_n}^2 = \text{Tr}_{H_n}(X^* X)/d_n$. In particular, every norm on the physical corner still uses d_n .

Define $W_n : H_n \rightarrow F_n K_n$ by $W_n \xi = (f_{i,n} \xi)_i$, and also regard it as an element of $F_n B(K_n) E_n$. Then

$$W_n^* W_n = S_n, \quad \|W_n\|_{\text{op}} \leq \sqrt{3}, \quad \sigma_n(F_n) = \text{tr}_{d_n}(S_n).$$

For each i, j, n , extend the polar part of $f_{i,n} u_{j,n} f_{i,n}$ to a unitary $w_{j,i,n}$ on $f_{i,n} H_n$. The extension exists because this compression is a square matrix. Zero summands are simply omitted. Put $c_{j,i,n} = f_{i,n} u_{j,n} f_{i,n}$. Each singular value s of this contraction satisfies $(1-s)^2 \leq 1-s^2$, so

$$\begin{aligned} \|w_{j,i,n} - c_{j,i,n}\|_{2,\sigma_n}^2 &\leq \frac{1}{d_n} \text{Tr}_{f_{i,n} H_n}(f_{i,n} - c_{j,i,n}^* c_{j,i,n}) \\ &= \|(I - f_{i,n}) u_{j,n} f_{i,n}\|_{2,d_n}^2. \end{aligned}$$

The other off-diagonal corner has the same squared norm, since $\text{Tr}(c_{j,i,n}^* c_{j,i,n}) = \text{Tr}(c_{j,i,n} c_{j,i,n}^*)$. The right supports $f_{i,n}$ and $I - f_{i,n}$ are orthogonal. Decomposing along these supports therefore gives

$$\begin{aligned} \|w_{j,i,n} f_{i,n} - f_{i,n} u_{j,n}\|_{2,\sigma_n}^2 &= \|w_{j,i,n} - c_{j,i,n}\|_{2,\sigma_n}^2 + \|f_{i,n} u_{j,n} (I - f_{i,n})\|_{2,d_n}^2 \\ &\leq \|[u_{j,n}, f_{i,n}]\|_{2,d_n}^2. \end{aligned}$$

Set

$$\hat{u}_{j,n} = u_{j,n} \oplus \bigoplus_i w_{j,i,n} \in U(K_n).$$

These unitaries commute exactly with E_n and every $F_{i,n}$, and

$$(4.5) \quad \|\hat{u}_{j,n} W_n - W_n u_{j,n}\|_{2,\sigma_n}^2 \leq \sum_i \|[u_{j,n}, f_{i,n}]\|_{2,d_n}^2 \rightarrow_\omega 0.$$

Their restriction to the physical summand is the original $u_{j,n}$.

Step 3. In this step, we construct, by an equivalence of projections in the finite invariant corner FDF of the commutant D of the $\hat{u}_j$, the partial isometry Z of (4.7), whose range projection

commutes with the abelian algebra $\mathcal{L}$. Let N be the tracial ultraproduct of the $B(K_n)$ with their normalized matrix traces. By (4.4), the formula

$$\sigma([x_n]) = \lim_{\omega} \frac{\text{Tr}_{K_n}(x_n)}{d_n}$$

defines a faithful normal finite trace on N , a positive scalar multiple of its normalized trace. The two traces define the same null ideal. Let

$$D = \{\widehat{u}_1, \dots, \widehat{u}_h\}' \cap N, \quad \widehat{u}_j = [\widehat{u}_{j,n}].$$

Write $E = [E_n]$, $F = [F_n]$, and $W = [W_n]$. These elements belong to D , by (4.5); moreover, $\sigma(E) = 1$ and $W = FWE$.

We retain all labels by using the entire internal abelian algebra

$$L_n = \left\{ \sum_{i=1}^{m_n} \alpha_i F_{i,n} : \alpha_i \in \mathbb{C} \right\}, \quad \mathcal{L} = [L_n]_{\omega} \subseteq FNF.$$

Here $[L_n]_{\omega}$ includes every uniformly bounded sequence of such operators. The coordinate conditional expectations onto L_n induce a 2-norm continuous projection onto $\mathcal{L}$, so its unit ball is 2-norm closed. Thus $\mathcal{L}$ is a von Neumann algebra with unit F . Every element of $\mathcal{L}$ commutes with every $\widehat{u}_j$, and hence $\mathcal{L} \subseteq FDF$.

Take the polar decomposition $W = V|W|$ inside D , and put $e = V^*V \leq E$ and $f = VV^* \leq F$. On E , the inequality $(\sqrt{s} - 1)^2 \leq (s - 1)^2$ for $s \geq 0$, together with the contribution of the kernel at $s = 0$, gives

$$(4.6) \quad \begin{aligned} \sigma(E - e) &\leq 12v, & \|V - W\|_{2,\sigma}^2 &\leq 12v, \\ \sigma(F - f) &= \sigma(F) - \sigma(e) \leq 15v. \end{aligned}$$

The first two estimates use (4.3). For the last one use $|\sigma(F) - 1| \leq 3v$ and $\sigma(e) = 1 - \sigma(E - e)$.

Suppose first that $F \neq 0$. Choose a MASA $\mathcal{A}$ of the finite algebra FDF containing $\mathcal{L}$. Its trace $\sigma|_{FDF}$ is faithful, normal, and finite, as is its restriction to $\mathcal{A}$. Lemma 4.1 therefore gives a projection $p_0 \in \mathcal{A}$ equivalent to $F - f$ inside FDF . Put $\tilde{f} = F - p_0$. This equivalence preserves the dimension on every central component. More precisely, for the normalized center-valued trace Θ of FDF we have

$$\Theta(p_0) = \Theta(F - f), \quad \Theta(\tilde{f}) = \Theta(f).$$

These identities follow from $\Theta(x^*x) = \Theta(xx^*)$ applied to a partial isometry implementing the equivalence. In particular, $\sigma(F - \tilde{f}) = \sigma(F - f) \leq 15v$, and

$$\|f - \tilde{f}\|_{2,\sigma}^2 = \|(F - f) - p_0\|_{2,\sigma}^2 \leq \sigma(F - f) + \sigma(p_0) \leq 30v.$$

Apply Lemma 4.2 to the equivalent projections p_0 and $F - f$ in FDF , normalizing its trace and then rescaling the estimate. We obtain a unitary U with identity F such that

$$Up_0U^* = F - f, \quad U\tilde{f}U^* = f, \quad \|U - F\|_{2,\sigma} \leq 3\sqrt{30v}.$$

This also explicitly implements the equivalence of the complements. Define $Z = U^*V$. Then

$$(4.7) \quad \begin{aligned} Z &\in D, \quad Z^*Z = e, \quad ZZ^* = \tilde{f}, \quad [\tilde{f}, \mathcal{L}] = 0, \\ \|Z - W\|_{2,\sigma} &\leq (\sqrt{270} + \sqrt{12})\sqrt{v} \leq 20\sqrt{v}. \end{aligned}$$

The distance bound follows by inserting V and using (4.6). If $F = 0$, set $\tilde{f} = U = Z = 0$; then $V = W = e = 0$, and all the bounds and identities in (4.7) still hold.

Step 4. In this step, we construct projections g_n lifting $\tilde{f}$, of rank at most d_n and commuting with every label projection $F_{i,n}$, and partial isometries t_n lifting Z with range g_nK_n , and define the projection-valued partition (4.9) in $M_{d_n}(\mathbb{C})$. Choose positive contraction representatives

$y_n \in F_n B(K_n) F_n$ of $\tilde{f}$. They can be obtained by taking self-adjoint representatives, compressing by F_n , and clipping the spectrum to $[0, 1]$. Define the pinching map

$$P_n(y) = \sum_i F_{i,n} y F_{i,n}.$$

For $y \in F_n B(K_n) F_n$, averaging over independent scalar phases $\zeta_i \in \mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ gives

$$(4.8) \quad 2\|y - P_n(y)\|_{2,\sigma_n}^2 = \int_{\mathbb{T}^{m_n}} \left\| \left[y, \sum_i \zeta_i F_{i,n} \right] \right\|_{2,\sigma_n}^2 d\zeta.$$

The identity follows by expanding the off-diagonal blocks: the average of $|\zeta_i - \zeta_k|^2$ is 2 for $i \neq k$. If $\|y_n - P_n(y_n)\|_{2,\sigma_n}$ did not tend to zero, at each coordinate of an ω -large set we could select phases witnessing a fixed positive lower bound in (4.8). Their sequence would represent an element of $\mathcal{L}$ with nonzero commutator with $\tilde{f}$. This contradicts (4.7). Hence $P_n(y_n)$ still represents $\tilde{f}$.

The spectral projections

$$\tilde{f}_n = \mathbf{1}_{[1/2,1]}(P_n(y_n))$$

also represent $\tilde{f}$ and commute with every $F_{i,n}$. Indeed, $P_n(y_n)$ is a positive contraction whose idempotence defect tends to zero in 2-norm, and $|\mathbf{1}_{[1/2,1]}(s) - s| \leq 2|s - s^2|$ on $[0, 1]$. This construction uses all coordinate label unitaries at once.

Since $\sigma(\tilde{f}) = \sigma(e) \leq 1$, we have

$$\frac{(\text{rank}(\tilde{f}_n) - d_n)_+}{d_n} \rightarrow_\omega 0.$$

Remove exactly this excess rank, when present, from $\tilde{f}_n$ within its individual label blocks. This gives projections g_n satisfying

$$\begin{aligned} g_n &\leq \tilde{f}_n, & [g_n, F_{i,n}] &= 0 \quad (1 \leq i \leq m_n), \\ \text{rank}(g_n) &\leq d_n, & \|g_n - \tilde{f}_n\|_{2,\sigma_n} &\rightarrow_\omega 0. \end{aligned}$$

Choose contraction representatives $z_n \in F_n B(K_n) E_n$ of Z , and put $z'_n = g_n z_n$. These still represent Z , so $\|g_n - z'_n z_n'^*\|_{2,\sigma_n} \rightarrow_\omega 0$. Extend the polar part of z'_n to a partial isometry t_n from H_n onto all of $g_n K_n$. There is enough room for the extension because $\text{rank}(g_n) \leq d_n$. For every left singular value $s \in [0, 1]$, including the zero singular values, the extension contributes $(1-s)^2 \leq 1-s^2$ to the squared error. Thus

$$\|t_n - z'_n\|_{2,\sigma_n}^2 \leq \sigma_n(g_n - z'_n z_n'^*) \rightarrow_\omega 0.$$

The trace tends to zero by Cauchy–Schwarz and $\sigma_n(I_{K_n}) \leq 4$. Consequently $[t_n] = Z$ and

$$t_n t_n^* = g_n, \quad e'_n = t_n^* t_n \leq E_n, \quad [e'_n] = e.$$

Set $g_{i,n} = F_{i,n} g_n$ and $Q_{i,n} = t_n^* g_{i,n} t_n \in E_n B(K_n) E_n = M_{d_n}(\mathbb{C})$. The $g_{i,n}$ are orthogonal projections summing to g_n ; hence the $Q_{i,n}$ are orthogonal projections summing to e'_n . Since $m_n \geq 1$, we can complete this family by setting

$$(4.9) \quad q_{1,n} = Q_{1,n} + E_n - e'_n, \quad q_{i,n} = Q_{i,n} \quad (2 \leq i \leq m_n).$$

This is a projection-valued partition in exactly $M_{d_n}(\mathbb{C})$.

Step 5. In this step, we deduce from $Z \in D$, $\tilde{f} \in D$, and the exact commutation of the $\hat{u}_{j,n}$ with every label projection that the total energy of the $q_{i,n}$ tends to zero along ω . The membership $Z \in D$ implies, for every j ,

$$\rho_{j,n} = \hat{u}_{j,n} t_n - t_n u_{j,n}, \quad \|\rho_{j,n}\|_{2,\sigma_n} \rightarrow_\omega 0.$$

Also $[g_n] = \tilde{f} \in D$, so $\|[\hat{u}_{j,n}, g_n]\|_{2,\sigma_n} \rightarrow_\omega 0$. The exact commutation of $\hat{u}_{j,n}$ with every label projection gives the summed identity

$$(4.10) \quad \sum_i \|[\hat{u}_{j,n}, g_{i,n}]\|_{2,\sigma_n}^2 = \|[\hat{u}_{j,n}, g_n]\|_{2,\sigma_n}^2.$$

Indeed, each commutator on the left is supported on its own $F_{i,n}$ block.

Put $\rho_{j,n}^- = \hat{u}_{j,n}^* t_n - t_n u_{j,n}^*$. Then $\rho_{j,n}^- = -\hat{u}_{j,n}^* \rho_{j,n} u_{j,n}^*$, so it has the same 2-norm as $\rho_{j,n}$. Expanding the commutator gives

$$[u_{j,n}, Q_{i,n}] = -(\rho_{j,n}^-)^* g_{i,n} t_n + t_n^* [\hat{u}_{j,n}, g_{i,n}] t_n + t_n^* g_{i,n} \rho_{j,n}.$$

For the first term, orthogonality and $\|t_n\|_{\text{op}} \leq 1$ give

$$\sum_i \|(\rho_{j,n}^-)^* g_{i,n} t_n\|_{2,\sigma_n}^2 \leq \sum_i \|(\rho_{j,n}^-)^* g_{i,n}\|_{2,\sigma_n}^2 \leq \|\rho_{j,n}\|_{2,\sigma_n}^2.$$

The third term satisfies the same bound. For the middle term, (4.10) gives a bound by $\|[\hat{u}_{j,n}, g_n]\|_{2,\sigma_n}^2$. Therefore

$$\sum_i \| [u_{j,n}, Q_{i,n}] \|_{2,d_n}^2 \leq 3(2\|\rho_{j,n}\|_{2,\sigma_n}^2 + \|[\hat{u}_{j,n}, g_n]\|_{2,\sigma_n}^2).$$

The remainder has small commutator as well, since

$$[u_{j,n}, e'_n] = -(\rho_{j,n}^-)^* t_n + t_n^* \rho_{j,n}, \quad \|[u_{j,n}, E_n - e'_n]\|_{2,d_n} \leq 2\|\rho_{j,n}\|_{2,\sigma_n}.$$

Using (4.9) we conclude that

$$\sum_i \| [u_{j,n}, q_{i,n}] \|_{2,d_n}^2 \leq 20\|\rho_{j,n}\|_{2,\sigma_n}^2 + 6\|[\hat{u}_{j,n}, g_n]\|_{2,\sigma_n}^2 \rightarrow_\omega 0.$$

Summing over the fixed list of h unitaries proves $\sum_i \mathcal{E}_{u_n}(q_{i,n}) \rightarrow_\omega 0$. All these bounds sum over the entire partition. The trace of the remainder may tend to a positive number; its commutator tends to zero because its class is $E - e \in D$.

Step 6. In this step, we compare t_n with W_n on every label block and deduce the bound (4.11), in which the variance v controls the total distance from the $q_{i,n}$ to the $a_{i,n}$. Let $t_{i,n} = F_{i,n} t_n$ and $W_{i,n} = F_{i,n} W_n$, as maps from H_n to the i th auxiliary summand. Both are contractions, and

$$t_{i,n}^* t_{i,n} = Q_{i,n}, \quad W_{i,n}^* W_{i,n} = f_{i,n}.$$

Expanding the difference of these two products gives

$$\sum_i \|Q_{i,n} - f_{i,n}\|_{2,d_n}^2 \leq 4 \sum_i \|t_{i,n} - W_{i,n}\|_{2,\sigma_n}^2 = 4\|t_n - W_n\|_{2,\sigma_n}^2.$$

By (4.7), the ultralimit is at most $1600v$. The remainder in (4.9) has squared norm tending to $\sigma(E - e) \leq 12v$, and (4.2) bounds the squared distance from $(f_{i,n})_i$ to $(a_{i,n})_i$ by $2v_n$. The triangle inequality in the Hilbert direct sum over all m_n outcomes now yields

$$(4.11) \quad \lim_\omega \sum_i \|q_{i,n} - a_{i,n}\|_{2,d_n}^2 \leq (40 + \sqrt{12} + \sqrt{2})^2 v \leq 5000v.$$

This includes $v = 0$. Thus the variance pays for the distance and for the possibly positive trace of the invariant remainder, while the total energy vanishes independently.

Step 7. In this step, we show by contradiction, applying the preceding construction to a sequence of hypothetical counterexamples, that δ can be chosen uniformly in the dimension d and the number of outcomes m , and conclude the proof. If the theorem failed for some fixed h and $\varepsilon > 0$, then for each n we could choose data with arbitrary d_n, m_n and $\sum_i \mathcal{E}_{u_n}(a_{i,n}) \leq 1/n$ for which no partition satisfies both conclusions. The construction above gives partitions with total energy tending to zero and, by (4.11),

$$\lim_\omega \left(\sum_i \|q_{i,n} - a_{i,n}\|_{2,d_n}^2 - 5000v_n \right) \leq 0.$$

Both required inequalities therefore hold on an ω -large set, contradicting the choice of the data. This proves the existence of δ depending only on h and ε . $\square$

5. PHYSICAL SCALAR-EXPANDING DECOMPOSITIONS

We retain the matrix norms and energies of the standing setting. We show that, after a perturbation tending to zero, the generators preserve scalar-expanding blocks covering arbitrarily large trace. The number of blocks and their ranks may vary without restriction. Here physical blocks are subspaces of the original coordinate space; the auxiliary spaces used in their construction are removed before the conclusion.

Theorem 5.1. *Let $\omega, d_n, h, u_{j,n}$ have quotient spectral gap $a \in (0, 1)$. For every $\varepsilon > 0$ there is a number $c = c(a, h, \varepsilon) > 0$ and, for every n , a projection-valued partition $R_{0,n}, R_{1,n}, \dots, R_{m_n,n}$ of I_{d_n} and unitaries $z_{j,n}^+, z_{j,n}^- \in U(d_n)$ such that:*

- (i) *Every $z_{j,n}^\pm$ commutes with every $R_{i,n}$, and $\sum_{j,\pm} \|z_{j,n}^\pm - u_{j,n}\|_{2,d_n}^2 \rightarrow_\omega 0$.*
- (ii) *$\lim_\omega \text{tr}_{d_n}(R_{0,n}) \leq \varepsilon$.*
- (iii) *On every nonzero $R_{i,n}$ with $i \geq 1$, the doubled list of restrictions of the $z_{j,n}^\pm$ has scalar gap c in the normalized trace of that block.*

There is no restriction on m_n or on the nonzero block ranks.

Each original generator occurs twice in the perturbed list. Both copies have its original ultraproduct image. We first prove the finite estimates that keep all errors independent of the number of blocks.

Lemma 5.2. *Let $d, h \geq 1$ and $m \geq 0$ be integers, let $u_1, \dots, u_h \in U(d)$, and let $f_1, \dots, f_m \in M_d(\mathbb{C})$ be projections. Put*

$$\mathcal{E}_u(x) = \frac{1}{4h} \sum_j \|[u_j, x]\|_{2,d}^2, \quad S_i = \sum_{k \leq i} f_k, \quad S_0 = 0.$$

For $\mu > 0$ define

$$D_i^{(\mu)} = (I + \mu S_i)^{-1}, \quad B_i^{(\mu)} = D_{i-1}^{(\mu)} - D_i^{(\mu)}.$$

Then $B_i^{(\mu)} \geq 0$, $\text{rank } B_i^{(\mu)} \leq \text{rank } f_i$, and the following estimates hold:

$$(5.1) \quad \mathcal{E}_u(D_l^{(\mu)}) \leq \mu \sum_{i \leq l} \mathcal{E}_u(f_i) \quad (0 \leq l \leq m),$$

$$(5.2) \quad \mathcal{E}_u(D_m^{(\mu)}) + \sum_i \mathcal{E}_u(B_i^{(\mu)}) \leq 72\mu \sum_i \mathcal{E}_u(f_i).$$

For every fixed $\mu > 0$ there is $C_\mu < \infty$, independent of all dimensions and ranks, such that

$$(5.3) \quad \sum_i \|[u_j, D_{i-1}^{(\mu)}]f_i\|_{2,d}^2 \leq C_\mu \sum_i \|[u_j, f_i]\|_{2,d}^2 \quad (1 \leq j \leq h).$$

Consequently, for any sequence of these data, with $u_n = (u_{1,n}, \dots, u_{h,n})$, h fixed, $\sup_n \sum_i \text{tr}_{d_n}(f_{i,n}) < \infty$, and $\sum_i \mathcal{E}_{u_n}(f_{i,n}) \rightarrow_\omega 0$, every continuous $\psi : [0, \infty) \rightarrow \mathbb{C}$ having a finite limit at infinity satisfies

$$(5.4) \quad \sum_i \|[u_{j,n}, \psi(S_{i-1,n})]f_{i,n}\|_{2,d_n}^2 \rightarrow_\omega 0 \quad (1 \leq j \leq h).$$

The last estimate controls a different density at each label. Its bound uses the total energy of the projections, with no factor counting them.

Proof. We obtain the static estimate from a rectangular resolvent, factor the increments by a block Cholesky decomposition, and then recover the localized estimate from the diagonal factors. Every rectangular Hilbert–Schmidt norm in the proof has denominator d . Zero projections may be omitted, and the assertions for an empty family follow from $D_0^{(\mu)} = I$.

Step 1. In this step, we bound the inclusion defect δ_u in (5.5), show that the static resolvent map Q_μ is $\sqrt{\mu}$ -Lipschitz, and deduce the static estimate (5.1) for every prefix. Write $H = \mathbb{C}^d$, $K_i = f_i H$, and $K = \bigoplus_i K_i$. Let $F_i : H \rightarrow K_i$ be the row f_i , and let F be the column with rows F_i . Thus $F^* F = S_m$. For a unitary u choose a unitary v_i on K_i completing the polar partial isometry of $f_i u f_i$, and put $V = \bigoplus_i v_i$. The two source subspaces $f_i H$ and $(I - f_i)H$, together with $(1 - s)^2 \leq 1 - s^2$ for $0 \leq s \leq 1$, give

$$(5.5) \quad \delta_u^2 := \|VF - Fu\|_2^2 \leq \sum_i \|[u, f_i]\|_{2,d}^2.$$

Indeed the error on the first subspace is at most $\text{tr}_d(f_i) - \|f_i u f_i\|_{2,d}^2$, and the error on the second is exactly that quantity. Their sum is $\|[u, f_i]\|_{2,d}^2$. The inclusion defects for v_i^*, u^* have the same norms, since

$$v_i^* F_i - F_i u^* = -v_i^* (v_i F_i - F_i u) u^*.$$

The map $Z \mapsto Q_\mu(Z) = (I + \mu Z^* Z)^{-1}$ is $\sqrt{\mu}$ -Lipschitz in this norm. Along a differentiable path,

$$\dot{Q}_\mu = -\mu Q_\mu (\dot{Z}^* Z + Z^* \dot{Z}) Q_\mu.$$

Singular values give $\|Q_\mu\|_{\text{op}} \leq 1$ and $\|Z Q_\mu\|_{\text{op}} \leq (2\sqrt{\mu})^{-1}$. Each of the two terms is therefore bounded by $\sqrt{\mu} \|\dot{Z}\|_2/2$. Integrating along a line proves the assertion. Since $Q_\mu(VF u^*) = u Q_\mu(F) u^*$, (5.5) proves (5.1), also for each prefix.

Step 2. In this step, we construct the block lower triangular factorization C , show that $\mathcal{J}$ is 3-Lipschitz and equivariant, prove (5.7) for $Z = F$, and deduce positivity and the rank bound for $B_i^{(\mu)}$ together with (5.2). For an arbitrary $Z : H \rightarrow K$, factor

$$I_K + \mu Z Z^* = C(Z) C(Z)^*$$

with $C(Z)$ block lower triangular and all diagonal blocks positive definite. Successive positive Schur complements give existence and uniqueness. They also give differentiability along differentiable paths, since inverses and positive square roots of positive definite matrices are differentiable. Set $\mathcal{J}(Z) = C(Z)^{-1} Z$. We claim that $\mathcal{J}$ is 3-Lipschitz, uniformly in μ and the block sizes.

Put

$$\Xi = C^{-1} \dot{C}, \quad \Upsilon = C^{-1} \mu (\dot{Z} Z^* + Z \dot{Z}^*) C^{-*}.$$

Differentiating the factorization gives $\Xi + \Xi^* = \Upsilon$. The strictly lower blocks of Ξ equal those of Υ . On a diagonal block write $C_{ii} = G > 0$ and $\dot{C}_{ii} = J = J^*$. Then $\Xi_{ii} = G^{-1} J$ and $\Upsilon_{ii} = G^{-1} J + J G^{-1}$. In an eigenbasis of G , with eigenvalues $g_r > 0$,

$$(\Xi_{ii})_{rs} = \frac{g_s}{g_r + g_s} (\Upsilon_{ii})_{rs}.$$

Summing squared entries proves $\|\Xi\|_2 \leq \|\Upsilon\|_2$. Also $\|C^{-1}\|_{\text{op}} \leq 1$, and

$$\mu \mathcal{J} \mathcal{J}^* = I - C^{-1} C^{-*} \leq I.$$

Hence $\|\mathcal{J}\|_{\text{op}} \leq \mu^{-1/2}$ and

$$\|\Upsilon\|_2 = \|\mu(C^{-1} \dot{Z} \mathcal{J}^* + \mathcal{J} \dot{Z}^* C^{-*})\|_2 \leq 2\sqrt{\mu} \|\dot{Z}\|_2.$$

The identity $\dot{\mathcal{J}} = C^{-1} \dot{Z} - \Xi \mathcal{J}$ now gives $\|\dot{\mathcal{J}}\|_2 \leq 3\|\dot{Z}\|_2$. Integration proves the claim. Uniqueness of the positive diagonal factors also gives the equivariance identities

$$C(VZu^*) = VC(Z)V^*, \quad \mathcal{J}(VZu^*) = V\mathcal{J}(Z)u^*.$$

Apply this factorization to $Z = F$ and put $W = \sqrt{\mu} \mathcal{J}(F)$. Write $H_i = F_i D_{i-1}^{(\mu)} F_i^*$ as an operator on K_i . Block elimination gives

$$(5.6) \quad W_i = \sqrt{\mu} (I_{K_i} + \mu H_i)^{-1/2} F_i D_{i-1}^{(\mu)}, \quad C_{ii} = (I_{K_i} + \mu H_i)^{1/2}.$$

Here is the elimination explicitly. If $F_{<i}$ is the column of preceding rows, its contribution to the diagonal Schur complement is removed using

$$I - \mu F_{<i}^* (I + \mu F_{<i} F_{<i}^*)^{-1} F_{<i} = D_{i-1}^{(\mu)}.$$

The resulting complement is $I_{K_i} + \mu H_i$. In the i th row of $CW = \sqrt{\mu}F$, subtracting the preceding rows similarly leaves $\sqrt{\mu}F_i D_{i-1}^{(\mu)}$. This proves (5.6). The inverse identity, verified by multiplication, is

$$(5.7) \quad B_i^{(\mu)} = \mu D_{i-1}^{(\mu)} F_i^* (I_{K_i} + \mu H_i)^{-1} F_i D_{i-1}^{(\mu)} = W_i^* W_i.$$

It proves positivity and the rank bound. Telescoping gives $W^* W = I - D_m^{(\mu)} \leq I$.

Lipschitz continuity and equivariance yield

$$(5.8) \quad \sum_i \|v_i W_i - W_i u\|_2^2 \leq 9\mu \delta_u^2.$$

The same bound holds for v_i^*, u^* . Expanding

$$[u, W_i^* W_i] = (u W_i^* - W_i^* v_i) W_i + W_i^* (v_i W_i - W_i u)$$

and using $\|W_i\|_{\text{op}} \leq 1$ gives $\sum_i \|[u, B_i^{(\mu)}]\|_2^2 \leq 36\mu \delta_u^2$. The identical expansion for the entire contraction W gives $\|[u, D_m^{(\mu)}]\|_2^2 \leq 36\mu \delta_u^2$. Summing over the original generators proves (5.2).

Step 3. In this step, we bound the commutators of the unitaries v_i with the diagonal factors C_{ii} in terms of δ_u^2 and deduce the localized covariance estimate (5.3) with an explicit constant C_μ . Equation (5.6) implies

$$W_i F_i^* = g_\mu(H_i), \quad g_\mu(x) = \frac{\sqrt{\mu}x}{\sqrt{1+\mu x}}.$$

The expansion

$$[v_i, W_i F_i^*] = (v_i W_i - W_i u) F_i^* + W_i (u F_i^* - F_i^* v_i)$$

and the inclusion estimates give

$$\sum_i \|[v_i, g_\mu(H_i)]\|_2^2 \leq 2(9\mu + 1)\delta_u^2.$$

On $[0, 1]$ we have

$$g'_\mu(x) = \frac{\sqrt{\mu}(1+\mu x/2)}{(1+\mu x)^{3/2}} \geq \ell_\mu := \frac{\sqrt{\mu}}{(1+\mu)^{3/2}} > 0.$$

In an eigenbasis of H_i , comparison of divided differences therefore bounds $\sum_i \|[v_i, H_i]\|_2^2$ by $2(9\mu + 1)\ell_\mu^{-2}\delta_u^2$. Since $x \mapsto \sqrt{1+\mu x}$ has Lipschitz constant at most $\mu/2$, it follows that

$$\sum_i \|[v_i, C_{ii}]\|_2^2 \leq M_\mu \delta_u^2, \quad M_\mu = \frac{\mu^2(9\mu + 1)}{2\ell_\mu^2}.$$

Taking adjoints in (5.6) gives $D_{i-1}^{(\mu)} F_i^* = \mu^{-1/2} W_i^* C_{ii}$. Consequently

$$u D_{i-1}^{(\mu)} F_i^* - D_{i-1}^{(\mu)} F_i^* v_i = \mu^{-1/2} ((u W_i^* - W_i^* v_i) C_{ii} + W_i^* [v_i, C_{ii}]).$$

The sum of its squared norms is at most $2\mu^{-1}(9\mu(1+\mu) + M_\mu)\delta_u^2$. Subtract $D_{i-1}^{(\mu)}(u F_i^* - F_i^* v_i)$ to obtain (5.3), for example with

$$C_\mu = 36(1+\mu) + 4M_\mu/\mu + 2.$$

Restriction to K_i has the same Hilbert–Schmidt norm as right multiplication by f_i , so this is the asserted estimate.

Step 4. In this step, we show that finite linear combinations of 1 and the resolvents $(x+s)^{-1}$ are uniformly dense in the continuous functions on $[0, \infty]$ and on $[0, 1]$, deduce (5.4) from (5.3), and conclude the proof. Finite linear combinations of 1 and $(x+s)^{-1}$, $s > 0$, are uniformly dense in the continuous functions on $[0, \infty]$. To see this, set $y = (1+x)^{-1}$. A continuous function of

y on $[0, 1]$ is uniformly approximated by its Bernstein polynomials: if $\varpi(\rho)$ is its modulus of continuity, the error is at most

$$\varpi(\rho) + \frac{\|\psi\|_\infty}{2N\rho^2}.$$

This follows by splitting the binomial average into $|k/N - y| \leq \rho$ and its complement, whose probability is at most $1/(4N\rho^2)$. First choose ρ , then N . For each integer $r \geq 1$, the identity

$$\left. \frac{\partial^{r-1}}{\partial s^{r-1}} (x+s)^{-1} \right|_{s=1} = (-1)^{r-1} (r-1)! (1+x)^{-r}$$

approximates y^r uniformly by finite differences of resolvents. Uniformity follows from the bounded next derivative for $x \geq 0$ and s near 1. This proves the density assertion. The same argument applies on a compact interval such as $[0, 1]$, by extending the corresponding function of y continuously to $[0, 1]$.

Each resolvent is a scalar multiple of a density covered by (5.3). For an approximant with uniform error at most ρ , the error in the direct sum of commutators, after right multiplication by f_i , has squared norm at most $4\rho^2 \sum_i \text{tr}_{d_n}(f_i)$. Fix the approximant first, take the ultrafilter limit using the localized estimates, and then let ρ decrease to zero. This proves (5.4). $\square$

Lemma 5.3. *Let $d \geq 1$ and $m \geq 0$ be integers, and let $I = D_0 \geq D_1 \geq \dots \geq D_m \geq 0$ be matrices in $M_d(\mathbb{C})$. Put $B_i = D_{i-1} - D_i$. For every continuous $g : [0, 1] \rightarrow \mathbb{C}$ and every $\rho > 0$ there is a finite $K_{g,\rho}$, independent of d, m and the chain, such that every projection $e \in M_d(\mathbb{C})$ satisfies*

$$(5.9) \quad \sum_i \text{tr}_d((I-e)g(D_{i-1})B_i g(D_{i-1})^*) \leq 2K_{g,\rho} \text{tr}_d(I-e) + 2\rho^2.$$

In particular, for any sequence of such chains and projections with $\text{tr}_{d_n}(I - e_n) \rightarrow_\omega 0$, the sum on the left tends to zero for every fixed g .

This estimate concerns the ranges of amplitudes with squared modulus B_i . It remains valid for positive matrices dominated by a fixed multiple of the matrices inside the sum.

Proof. Fix $s > 0$. Write $A = D_{i-1} + sI$, so that $A - B_i = D_i + sI > 0$. The inverse identity gives

$$(A - B_i)^{-1} - A^{-1} - A^{-1}B_iA^{-1} = A^{-1}B_i(A - B_i)^{-1}B_iA^{-1} \geq 0.$$

After summing and telescoping, we obtain

$$(5.10) \quad \sum_i (D_{i-1} + sI)^{-1} B_i (D_{i-1} + sI)^{-1} \leq s^{-1}I.$$

For the constant function the corresponding sum is $\sum_i B_i = I - D_m \leq I$. If $r(x) = c_0 + \sum_{k=1}^N c_k(x + s_k)^{-1}$, the triangle inequality in the direct sum of the vectors $B_i^{1/2}r(D_{i-1})^*\xi$ gives

$$\sum_i r(D_{i-1})B_i r(D_{i-1})^* \leq K_r I, \quad K_r = \left(|c_0| + \sum_{k=1}^N |c_k| s_k^{-1/2} \right)^2.$$

The approximation argument in **Step 4** of the proof of Lemma 5.2 supplies such an r with $\|g - r\|_\infty \leq \rho$. Moreover,

$$\sum_i \|(g - r)(D_{i-1})B_i^{1/2}\|_{2,d}^2 \leq \rho^2 \sum_i \text{tr}_d(B_i) \leq \rho^2.$$

Multiply the amplitudes on the left by $I - e$ and apply $\|X + Y\|_2^2 \leq 2\|X\|_2^2 + 2\|Y\|_2^2$. This proves (5.9) with $K_{g,\rho} = K_r$. For the limiting assertion, keep r fixed while taking the ultrafilter limit, and then let ρ decrease to zero. $\square$

Lemma 5.4. *Let $d, h \geq 1$ and $m \geq 0$ be integers, and let $P_0, \dots, P_m$ be a projection-valued partition in $M_d(\mathbb{C})$. Let $w_1, \dots, w_h \in U(d)$ commute with every P_i , and put*

$$\mathcal{E}_w(x) = \frac{1}{4h} \sum_j \|[w_j, x]\|_{2,d}^2.$$

Suppose $0 < \vartheta \leq 1$, $\xi_i \geq 0$ for $1 \leq i \leq m$, and every projection $q \leq P_i$ of rank at most $\text{rank}(P_i)/2$ satisfies

$$(5.11) \quad \mathcal{E}_w(q) \geq \vartheta \text{tr}_d(q) - \xi_i.$$

Put $\xi = \sum_{i=1}^m \xi_i$. Then there are projections $R_i \leq P_i$ for $i \geq 1$ and unitaries $z_j^+, z_j^- \in U(d)$ such that, with $R_0 = I - \sum_{i=1}^m R_i$, the following hold:

(i) Every $z_j^\pm$ commutes with every R_i , and

$$\text{tr}_d(R_0) - \text{tr}_d(P_0) \leq \frac{8\xi}{\vartheta}, \quad \sum_{i=1}^m \mathcal{E}_w(R_i) \leq \frac{\xi}{7}.$$

(ii) The total change satisfies

$$\sum_{j,\pm} \|z_j^\pm - w_j\|_{2,d}^2 \leq \frac{16h\xi}{7}.$$

(iii) On every nonzero R_i with $i \geq 1$, the doubled list of restrictions of the $z_j^\pm$ has scalar gap $\vartheta^2/64$ in the normalized trace of that block.

The additive defects are charged once to each original block. No lower bound on the trace of an individual block is required.

Proof. We remove low-energy subprojections, prove a Cheeger estimate for the remaining compressed contractions, and realize those contractions as averages of two nearby unitaries.

Step 1. In this step, we construct each R_i by pruning inside one block P_i , either discarding the block or successively removing subprojections that satisfy (5.12), and prove (5.14) together with the trace and energy estimates for the R_i . Fix $P = P_i \neq 0$, and write $\rho_P = \text{tr}_d(P)$. If $\xi_i > \vartheta\rho_P/8$, take $R_i = 0$. This loses less than $8\xi_i/\vartheta$ trace and contributes zero energy. Otherwise put $\kappa = \vartheta/8$ and start with $Q = P$. For $x = QxQ$, define the compressed energy

$$\mathcal{E}_{w,Q}(x) = \frac{1}{4h} \sum_j \|[Qw_jQ, x]\|_{2,d}^2.$$

Whenever a nonzero projection $p \leq Q$ satisfies

$$(5.12) \quad \text{tr}_d(p) \leq \text{tr}_d(Q)/2, \quad \mathcal{E}_{w,Q}(p) < \kappa \text{tr}_d(p),$$

remove it and replace Q by $Q - p$. The rank strictly decreases, and no removal exhausts Q , so the process terminates with a nonzero projection R .

Let $p_1, \dots, p_l$ be any initial segment of the removed projections, and let Q_k be the remainder just before removing p_k . Set $A = \sum_{k=1}^l p_k$ and $\mu_A = \text{tr}_d(A)$. The entries joining A to $P - A$ split orthogonally over the p_k . For each k , those entries are among the entries joining p_k to $Q_k - p_k$. Since P reduces every w_j , this gives

$$(5.13) \quad \mathcal{E}_w(A) = \mathcal{E}_w(P - A) \leq \sum_{k=1}^l \mathcal{E}_{w,Q_k}(p_k) \leq \kappa \mu_A.$$

No initial segment has $\mu_A \geq \rho_P/4$. Indeed, at the first such segment the previous removed trace μ_{old} is less than $\rho_P/4$, and (5.12) implies

$$\rho_P/4 \leq \mu_A \leq \mu_{\text{old}} + (\rho_P - \mu_{\text{old}})/2 < 5\rho_P/8.$$

The smaller-rank member q of $A, P - A$ then has trace between $\rho_P/4$ and $\rho_P/2$. By the hypothesis on this block,

$$\mathcal{E}_w(A) = \mathcal{E}_w(q) \geq \vartheta \rho_P/4 - \xi_i \geq \vartheta \rho_P/8.$$

But (5.13) bounds the same quantity by a number less than $5\vartheta \rho_P/64$, a contradiction.

Apply (5.11) to the final removed projection $A = P - R$, whose trace is less than $\rho_P/4$. Together with (5.13), it gives

$$(\vartheta - \kappa) \operatorname{tr}_d(A) \leq \xi_i, \quad \mathcal{E}_w(R) \leq \frac{\xi_i}{7}, \quad \operatorname{tr}_d(P - R) \leq \frac{8\xi_i}{7\vartheta}.$$

At termination we also have

$$(5.14) \quad \mathcal{E}_{w,R}(p) \geq \kappa \operatorname{tr}_d(p) \quad \left(p \leq R, \quad \operatorname{rank} p \leq \frac{\operatorname{rank} R}{2} \right).$$

Apply the procedure in every nonzero block, and put $R_i = 0$ when $P_i = 0$. The discarded blocks and the pruned blocks together give the two estimates in (i).

Step 2. In this step, we prove the expansion estimate (5.15) for the quadratic form $\mathcal{Q}$ of the compressed contractions $a_j = R w_j R$ on a retained block, first for positive matrices of small support and then for every trace-zero x . Fix a retained nonzero $R = R_i$, set $r = \operatorname{rank} R$, and use $\operatorname{tr}_r = \operatorname{Tr}/r$ on its range. The matrices $a_j = R w_j R$ on that range are contractions. Define

$$\mathcal{Q}(x) = \frac{1}{8h} \sum_j (\| [a_j, x] \|_{2,r}^2 + \| [a_j^*, x] \|_{2,r}^2).$$

For self-adjoint x the two commutators have equal norms. Dividing (5.14) by $\operatorname{tr}_d(R)$ therefore gives $\mathcal{Q}(p) \geq \kappa \operatorname{tr}_r(p)$ whenever $\operatorname{tr}_r(p) \leq 1/2$.

Let $y \geq 0$ have support of normalized trace at most $1/2$. In an eigenbasis of y , write its eigenvalues as y_b . The projections $p_s = 1_{(s,\infty)}(y^2)$ satisfy the same support bound. The layer-cake formula and Cauchy–Schwarz give

$$\begin{aligned} \kappa \operatorname{tr}_r(y^2) &\leq \int_0^\infty \mathcal{Q}(p_s) ds \\ &= \frac{1}{4hr} \sum_{j,b,k} |y_b^2 - y_k^2| |(a_j)_{bk}|^2 \leq \mathcal{Q}(y)^{1/2} \mathcal{H}(y)^{1/2}, \end{aligned}$$

where

$$\mathcal{H}(y) = \frac{1}{4hr} \sum_{j,b,k} (y_b + y_k)^2 |(a_j)_{bk}|^2.$$

Each row and column of a contraction has squared Euclidean norm at most one. Thus $(s+t)^2 \leq 2(s^2 + t^2)$ gives $\mathcal{H}(y) \leq \operatorname{tr}_r(y^2)$, and hence $\mathcal{Q}(y) \geq \kappa^2 \operatorname{tr}_r(y^2)$.

For a self-adjoint x choose a median b of its eigenvalues. Both $y_+ = (x - bR)_+$ and $y_- = (bR - x)_+$ have support trace at most $1/2$. The scalar inequality

$$|s - t|^2 \geq |(s - b)_+ - (t - b)_+|^2 + |(b - s)_+ - (b - t)_+|^2$$

applied entry by entry yields $\mathcal{Q}(x) \geq \mathcal{Q}(y_+) + \mathcal{Q}(y_-)$. If $\operatorname{tr}_r(x) = 0$, we obtain

$$(5.15) \quad \mathcal{Q}(x) \geq \kappa^2 \operatorname{tr}_r((x - bR)^2) \geq \kappa^2 \|x\|_{2,r}^2.$$

For completeness, the operator of this quadratic form is

$$\mathcal{L}(x) = \frac{1}{8h} \sum_j ((a_j^* a_j + a_j a_j^*)x + x(a_j^* a_j + a_j a_j^*) - 2a_j^* x a_j - 2a_j x a_j^*).$$

It is self-adjoint for the Hilbert–Schmidt inner product and preserves adjoints. For self-adjoint A, B the number $\operatorname{tr}_r(A\mathcal{L}(B))$ is real. Therefore $\mathcal{Q}(A + iB) = \mathcal{Q}(A) + \mathcal{Q}(B)$. The real and imaginary parts of a trace-zero matrix are trace zero, so (5.15) holds for every trace-zero x .

Step 3. In this step, we construct the doubled unitaries $z_j^\pm$ whose averages are the compressed contractions, deduce the scalar gap from (5.15), bound their cost $\sum_{j,\pm} \|z_j^\pm - w_j\|_{2,d}^2$ by $16h\xi/7$, and conclude the proof. On each nonzero member R of the final partition, complete the polar partial isometry of $a = Rw_jR$ to a unitary v and set

$$z_{j,R}^\pm = v(|a| \pm i(R - |a|^2)^{1/2}).$$

These are unitaries and $(z_{j,R}^+ + z_{j,R}^-)/2 = a$. Their direct sums give $z_j^\pm$ commuting with every R_i . Convexity applied both to a and to a^* gives, for any x on a retained block,

$$\frac{1}{8h} \sum_{j,\pm} \|[z_{j,R}^\pm, x]\|_{2,r}^2 \geq \mathcal{Q}(x).$$

Indeed $\|[z^*, x]\|_{2,r} = \|[z, x]\|_{2,r}$ for a unitary z , by $[z^*, x] = -z^*[z, x]z^*$. Combining the two convexity inequalities gives the displayed bound, including for complex x . Equation (5.15) proves (iii).

Put $C_j = \sum_R Rw_jR$ and $L_j = \|w_j - C_j\|_{2,d}^2$, where the sum is over the final partition. Orthogonality of its matrix blocks gives

$$\begin{aligned} \|z_j^\pm - C_j\|_{2,d}^2 &= \sum_R \text{tr}_d(R - (Rw_jR)^*(Rw_jR)) = L_j, \\ \|z_j^\pm - w_j\|_{2,d}^2 &= 2L_j. \end{aligned}$$

Writing $\mathcal{B} = \sum_R \mathcal{E}_w(R)$, the same block expansion gives $\mathcal{B} = (2h)^{-1} \sum_j L_j$. The original P_i reduce w_j , so the only final boundary inside P_i joins R_i to $P_i - R_i$. Consequently

$$\mathcal{E}_w(R_0) = \sum_{i \geq 1} \mathcal{E}_w(R_i), \quad \mathcal{B} = 2 \sum_{i \geq 1} \mathcal{E}_w(R_i) \leq 2\xi/7.$$

Summing the edit costs over both signs proves $\sum_{j,\pm} \|z_j^\pm - w_j\|_{2,d}^2 = 8h\mathcal{B} \leq 16h\xi/7$, as required. $\square$

Proof of Theorem 5.1. We select projections of minimum rank using a penalty for leakage out of the repair corner. Their resolvent increments form an almost projection-valued partition. Rounding and conditioning its amplitudes produce physical blocks with the original rank certificates. The range estimate transfers those certificates with a total positive defect, which Lemma 5.4 removes.

Step 1. In this step, we define, for a given t with $0 < t \leq 1/256$, the fixed parameters of the construction, all independent of the coordinate ranks, together with the target trace bound $F(t)$, the target scalar gap $\beta^2/256$, and the norm conventions. Fix $0 < t \leq 1/256$. Every limit below is taken with this t fixed. Set

$$\begin{aligned} L &= 36/a^2, & \beta &= \frac{t}{4L}, & \gamma &= \sqrt{t}, & \lambda &= t^{-1}, \\ k_t &= \frac{\lambda}{1+\lambda}(1-2\sqrt{t})^2, & \sigma &= \frac{\gamma}{\lambda(1-\gamma)}, & A_* &= \frac{1+t}{k_t}, \\ h_0 &= \log 2 - \frac{1}{2}, & h_1 &= \log(4/3) - \frac{1}{4}, & b_t &= \frac{2(t+\sigma)}{k_t}, \\ v_t &= \frac{b_t}{h_0}, & s_t &= 2t + \frac{b_t}{h_1}, & q_t &= v_t + s_t, & F(t) &= 2((5002 + 4/\gamma)q_t + s_t). \end{aligned}$$

These numbers are positive and independent of all coordinate ranks. We construct blocks with discarded trace at most $F(t)$ in the ultrafilter limit and scalar gap $\beta^2/256$. Unless a block norm is explicitly indicated, every norm in the proof, including a rectangular norm, uses the ambient denominator d_n . In calculations at a single coordinate, write $d = d_n$ and $H = \mathbb{C}^d$, and omit n from the projections and densities.

Step 2. In this step, we define the leakage $\ell(p)$ of a projection p out of the repair corner e_n and show that every nonzero p satisfying the test (5.17) with leakage penalty has a repair f

satisfying (5.18) and (5.19). Theorem 3.2 supplies e_n, η_n with $\text{tr}_{d_n}(I - e_n) \rightarrow_\omega 0$ and $\eta_n \rightarrow_\omega 0$. Fix a coordinate and write $e = e_n$. For a projection p , put $\ell(p) = \text{tr}_d(p(I - e))$. There is a projection $q \leq e$ such that

$$(5.16) \quad \|p - q\|_{2,d}^2 \leq 2\ell(p), \quad \text{tr}_d(p) - 2\ell(p) \leq \text{tr}_d(q) \leq \text{tr}_d(p).$$

Take $q = 1_{[1/2,1]}(epe)$. To verify the estimates, diagonalize pep on $p\mathbb{C}^d$. If a unit eigenvector has eigenvalue $\nu > 0$, its image under e , divided by $\sqrt{\nu}$, is a unit vector in $e\mathbb{C}^d$. These images are orthonormal. For $\nu \geq 1/2$ retain that image in q ; its contribution to $d\|p - q\|_{2,d}^2$ is $2(1 - \nu)$. For $\nu < 1/2$, including $\nu = 0$, discard it; its contribution is $1 \leq 2(1 - \nu)$. The number discarded is at most $2\sum(1 - \nu) = 2d\ell(p)$. This proves (5.16).

The energy is a squared norm and satisfies $\mathcal{E}_n(x) \leq \|x\|_{2,d_n}^2$, by $\|u_{j,n}, x\|_{2,d_n} \leq 2\|x\|_{2,d_n}$. Consequently

$$\mathcal{E}_n(q) \leq 2\mathcal{E}_n(p) + 4\ell(p).$$

Suppose $p \neq 0$ and

$$(5.17) \quad \mathcal{E}_n(p) + 3\ell(p) < \beta \text{tr}_{d_n}(p).$$

Then $\ell(p) < \beta \text{tr}_{d_n}(p)/3$, so $\text{tr}_{d_n}(q) \geq \text{tr}_{d_n}(p)/2 > 0$. Since $\beta \leq 1/(16L)$,

$$\mathcal{E}_n(q) < 2\beta \text{tr}_{d_n}(p) \leq \frac{\text{tr}_{d_n}(p)}{8L} \leq \frac{\text{tr}_{d_n}(q)}{4L}.$$

The individual repair theorem therefore gives a projection f with

$$(5.18) \quad \begin{aligned} \|f - p\|_{2,d_n}^2 &\leq 2L\mathcal{E}_n(q) + 2\|q - p\|_{2,d_n}^2 \\ &\leq 4L\mathcal{E}_n(p) + (8L + 4)\ell(p) \leq 4L(\mathcal{E}_n(p) + 3\ell(p)) < t \text{tr}_{d_n}(p), \\ \mathcal{E}_n(f) &\leq \eta_n \text{tr}_{d_n}(f). \end{aligned}$$

We used $L \geq 1$ in the last comparison. For projections, $|\text{tr}_d(f) - \text{tr}_d(p)| \leq \|f - p\|_{2,d}^2$, because $0 \leq \text{tr}_d(fp) \leq \min\{\text{tr}_d(f), \text{tr}_d(p)\}$. Thus this repair also satisfies

$$(5.19) \quad (1 - t) \text{rank } p < \text{rank } f < (1 + t) \text{rank } p.$$

Step 3. In this step, we prove the coarea estimate (5.20), choose recursively in high-density windows b_i projections p_i of minimum rank satisfying (5.17) with repairs f_i , and obtain the total bounds (5.21) and (5.22). We use the following coarea estimate for a positive contraction D :

$$(5.20) \quad \int_0^1 \mathcal{E}_n(1_{[\theta,1]}(D)) d\theta \leq \frac{1}{2} \sqrt{\mathcal{E}_n(D)}.$$

In an eigenbasis of D , with eigenvalues d_b , let $w_{bk} = (4hd_n)^{-1} \sum_j |(u_{j,n})_{bk}|^2$. The weights sum to $1/4$. The energy of D is $\sum_{b,k} w_{bk}(d_b - d_k)^2$, while the integral is $\sum_{b,k} w_{bk}|d_b - d_k|$. Cauchy-Schwarz proves (5.20).

At coordinate n , start with $S_0 = 0$ and $D_0 = I$. Given the preceding choices, put $D_{i-1} = (I + \lambda S_{i-1})^{-1}$ and choose

$$\theta_i \in [1 - \gamma, 1 - \gamma/2], \quad b_i = 1_{[\theta_i,1]}(D_{i-1}), \quad \mathcal{E}_n(b_i) \leq \frac{\sqrt{\mathcal{E}_n(D_{i-1})}}{\gamma}.$$

Such a threshold exists by (5.20), since the interval has length $\gamma/2$. If a nonzero $p \leq b_i$ satisfies (5.17), choose one of minimum positive rank, denote it by p_i , and repair it to f_i as above. Set $S_i = S_{i-1} + f_i$ and $D_i = (I + \lambda S_i)^{-1}$. If no such p exists, stop. The minimum is over a nonempty set of integers in $\{1, \dots, d_n\}$, so strictness of the test causes no existence problem.

Put $B_i = D_{i-1} - D_i$. The inverse identity (5.7) and $f_i D_{i-1} f_i \leq I_{f_i \mathbb{C}^{d_n}}$ imply

$$\begin{aligned} \text{tr}_{d_n}(B_i) &\geq \frac{\lambda}{1 + \lambda} \|D_{i-1} f_i\|_{2,d_n}^2 \\ &\geq \frac{\lambda}{1 + \lambda} (1 - \gamma - \sqrt{t})^2 \text{tr}_{d_n}(p_i) = k_t \text{tr}_{d_n}(p_i). \end{aligned}$$

For the second inequality, D_{i-1} is bounded below by $1 - \gamma$ on $b_i \mathbb{C}^{d_n}$ and $\|D_{i-1}(f_i - p_i)\|_2 \leq \sqrt{t \operatorname{tr}_{d_n}(p_i)}$. This argument does not require p_i to commute with D_{i-1} . The positive increments telescope to $I - D_l \leq I$ at every prefix. Since each p_i has rank at least one, the recursion stops after at most d_n/k_t selections. Writing $m = m_n$ for their number, we have

$$(5.21) \quad \sum_i \operatorname{tr}_{d_n}(p_i) \leq k_t^{-1}, \quad A_n := \sum_i \operatorname{tr}_{d_n}(f_i) \leq A_*, \quad J_n := \sum_i \mathcal{E}_n(f_i) \leq \eta_n A_* \rightarrow_\omega 0.$$

The static estimate (5.1) shows that every chosen window, including the final one $b = b_{m+1}$, has energy at most $\sqrt{\lambda \eta_n A_*}/\gamma$. Failure of the selection test for $p = b$, with the same inequality valid if $b = 0$, gives

$$\beta \operatorname{tr}_{d_n}(b) \leq \mathcal{E}_n(b) + 3 \operatorname{tr}_{d_n}(b(I - e_n)) \leq \frac{\sqrt{\lambda \eta_n A_*}}{\gamma} + 3 \operatorname{tr}_{d_n}(I - e_n).$$

Define the right side divided by β to be ζ_n . Then $\zeta_n \rightarrow_\omega 0$. The kernel of S_m is contained in b , and $\operatorname{rank} S_m \leq \sum_i \operatorname{rank} f_i$. Hence

$$(5.22) \quad \operatorname{tr}_{d_n}(b) \leq \zeta_n, \quad A_n \geq 1 - \zeta_n.$$

Step 4. In this step, we define the second resolvent scale $D_i^{(1)}$, bound A_n and s_n by means of Φ_1 , deduce the variance bound (5.28), and obtain the energy bound (5.29) from Lemma 5.2. From the actual threshold used in the selection,

$$(5.23) \quad p_i \leq 1_{[0, \sigma]}(S_{i-1}), \quad \sigma = \frac{\gamma}{\lambda(1 - \gamma)}.$$

Introduce $D_i^{(1)} = (I + S_i)^{-1}$ without changing the recursion, and let $H_i^{(1)} = f_i D_{i-1}^{(1)} f_i$ on $f_i H$. On the spectral window in (5.23), $I - D_{i-1}^{(1)} \leq \sigma I$. By (5.18),

$$\begin{aligned} \operatorname{tr}_d(f_i(I - D_{i-1}^{(1)})) &= \|(I - D_{i-1}^{(1)})^{1/2} f_i\|_{2,d}^2 \\ &\leq 2\sigma \operatorname{tr}_d(p_i) + 2\|f_i - p_i\|_{2,d}^2 \leq 2(t + \sigma) \operatorname{tr}_d(p_i). \end{aligned}$$

The determinant identity $\det(I + XY) = \det(I + YX)$ gives

$$\frac{\det(I + S_i)}{\det(I + S_{i-1})} = \det_{f_i H}(I_{f_i H} + H_i^{(1)}).$$

After taking logarithms and telescoping, use $\log(1 + x) \geq \log 2 - (1 - x)$ for $0 \leq x \leq 1$ to obtain

$$(5.24) \quad \Phi_1 := \operatorname{tr}_d \log(I + S_m) = \sum_i \frac{1}{d} \operatorname{Tr}_{f_i H} \log(I_{f_i H} + H_i^{(1)}) \geq A_n \log 2 - b_t.$$

Scalar Jensen on the eigenvalues of S_m gives $\Phi_1 \leq \log(1 + A_n)$. The function $v \mapsto v \log 2 - \log(1 + v)$ vanishes at 1 and has derivative at least h_0 for $v \geq 1$. It follows that

$$(5.25) \quad A_n \leq 1 + v_t.$$

For $A_n < 1$ this inequality holds as well.

Let $X = (I + S_m)/2$ and $\phi(x) = x - 1 - \log x$. Equations (5.22) and (5.24) imply

$$\operatorname{tr}_d(\phi(X)) = \frac{A_n - 1}{2} + \log 2 - \Phi_1 \leq -h_0(A_n - 1) + b_t \leq b_t + h_0 \zeta_n.$$

On the spectral subspace $S_m < 1/2$ we have $\phi(X) \geq \phi(3/4) = h_1$. On its complement $D_m = (I + \lambda S_m)^{-1} \leq 2/\lambda$. Thus

$$(5.26) \quad s_n := \operatorname{tr}_{d_n}(D_m) \leq 2/\lambda + (b_t + h_0 \zeta_n)/h_1, \quad \lim_\omega s_n \leq s_t.$$

The list $(D_m, B_1, \dots, B_m)$ is a positive operator-valued partition. Since $0 \leq B_i \leq I$ and $\operatorname{rank} B_i \leq \operatorname{rank} f_i$,

$$(5.27) \quad \sum_i \operatorname{tr}_{d_n}(B_i - B_i^2) \leq \sum_i \left(\frac{\operatorname{rank} f_i}{d_n} - \operatorname{tr}_{d_n}(B_i) \right) = A_n - 1 + s_n.$$

For the scalar inequality used here, $x - x^2 \leq 1 - x$ on $[0, 1]$; apply it on the support of B_i and then enlarge the rank bound. Also

$$\sum_i \operatorname{tr}_{d_n}(B_i - B_i^2) = \operatorname{tr}_{d_n}(D_m - D_m^2) + \sum_{i \neq k} \operatorname{tr}_{d_n}(B_i B_k) \geq \operatorname{tr}_{d_n}(D_m - D_m^2).$$

Therefore the full variance and total energy satisfy

$$(5.28) \quad \mathcal{V}_n := \operatorname{tr}_{d_n}(D_m - D_m^2) + \sum_i \operatorname{tr}_{d_n}(B_i - B_i^2), \quad \lim_{\omega} \mathcal{V}_n \leq 2q_t,$$

$$(5.29) \quad \mathcal{E}_n(D_m) + \sum_i \mathcal{E}_n(B_i) \leq 72\lambda J_n \rightarrow_{\omega} 0.$$

The last line is (5.2) at $\mu = \lambda$.

Step 5. In this step, we obtain by rounding a projection-valued partition $Q_0, \dots, Q_m$ from $D_m, B_1, \dots, B_m$, construct polar parts V_i with physical supports $P_i \leq Q_i$ and ranges under the windows b_i , bound $\lim_{\omega} \operatorname{tr}_{d_n}(P_0)$ by $F(t)$, and prove (5.37) and (5.38). Apply Theorem 4.3 to $(D_m, B_1, \dots, B_m)$. Equation (5.29) allows a choice of projection-valued partitions $(Q_0, Q_1, \dots, Q_m)$ such that

$$(5.30) \quad \sum_{i=0}^m \mathcal{E}_n(Q_i) \rightarrow_{\omega} 0, \quad H_n := \|Q_0 - D_m\|_{2,d_n}^2 + \sum_{i \geq 1} \|Q_i - B_i\|_{2,d_n}^2 \leq 5000\mathcal{V}_n + o_{\omega}(1).$$

To make the choice precise, first take the fixed tolerances $1/r$ in that theorem and their positive input tolerances δ_r . At coordinate n use the largest $r \leq n$ for which the input energy is at most $\min_{k \leq r} \delta_k$; if none exists, use an arbitrary projection-valued partition with the same labels. The chosen r tends to infinity along ω , which proves (5.30). All applications of the rounding theorem are thus at fixed tolerances.

The total label mismatch is

$$(5.31) \quad \ell_n := 1 - \operatorname{tr}_{d_n}(Q_0 D_m) - \sum_{i \geq 1} \operatorname{tr}_{d_n}(Q_i B_i) = \frac{\mathcal{V}_n + H_n}{2}, \quad \lim_{\omega} \ell_n \leq 5001q_t.$$

This identity follows by expanding the squared distances and using both partition sums. Each mismatch term is nonnegative.

Let $g : [0, 1] \rightarrow [0, 1]$ be zero on $[0, 1 - \gamma/2]$, one on $[1 - \gamma/4, 1]$, and affine between these intervals. Then

$$b_i g(D_{i-1}) = g(D_{i-1}), \quad 1 - g(x)^2 \leq 4(1 - x)/\gamma.$$

The first identity uses the original threshold θ_i chosen in the recursion. Since $B_i \leq D_{i-1}$,

$$\operatorname{tr}_{d_n}(B_i(I - D_{i-1})) \leq \operatorname{tr}_{d_n}(B_i - B_i^2).$$

Consequently

$$(5.32) \quad K_n := \sum_i \operatorname{tr}_{d_n}((I - g(D_{i-1}))^2 B_i), \quad \lim_{\omega} K_n \leq 4q_t/\gamma.$$

Here we used (5.27) and the bounds for A_n and s_n .

Define contractions on the original physical space by

$$Y_i = g(D_{i-1})B_iQ_i \quad (i \geq 1).$$

Their right supports lie in the mutually orthogonal Q_i , their left supports lie in b_i , and $\operatorname{rank} Y_i \leq \operatorname{rank} f_i$. We claim that

$$(5.33) \quad \sum_i \mathcal{E}_n(Y_i) \rightarrow_{\omega} 0.$$

The commutator terms involving B_i and Q_i are controlled by (5.29) and (5.30). For the remaining term use the amplitudes in (5.6) with $\mu = \lambda$. Writing $X_i = W_i^*$, we have $B_i = X_i X_i^*$ and $\|X_i\|_{\text{op}} \leq 1$. For each original generator $u = u_{j,n}$,

$$\|[u, g(D_{i-1})]B_i\|_2 \leq \|[u, g(D_{i-1})]X_i\|_2 \leq \sqrt{\lambda} \|[u, g(D_{i-1})]D_{i-1}f_i\|_2.$$

Now

$$[u, g(D)]D = [u, g(D)D] - g(D)[u, D].$$

Both $g(D)D$ and D are fixed continuous functions of S_{i-1} with limits at infinity. Equation (5.4), using the total bounds (5.21), shows that the squared norms in the last display tend to zero after summing over i . Expanding $[u, Y_i]$ and summing over j proves (5.33).

The total defect from an isometry, including the zero label, is

$$M_n := \text{tr}_{d_n}(Q_0) + \sum_{i \geq 1} \text{tr}_{d_n}(Q_i - Y_i^* Y_i).$$

Insert B_i and B_i^2 between Q_i and $Y_i^* Y_i$. The exact identity and its bound are

$$\begin{aligned} M_n &= \ell_n + \text{tr}_{d_n}(Q_0 D_m) + \sum_i \text{tr}_{d_n}(Q_i (B_i - B_i^2)) \\ &\quad + \sum_i \text{tr}_{d_n}(Q_i B_i (I - g(D_{i-1})^2) B_i) \\ &\leq \ell_n + s_n + \sum_i \text{tr}_{d_n}(B_i - B_i^2) + K_n. \end{aligned}$$

For the last sum first use $Q_i \leq I$, then $B_i^2 \leq B_i$ under the positive trace. Equations (5.26), (5.27), (5.31), and (5.32) give

$$(5.34) \quad \lim_{\omega} M_n \leq (5002 + 4/\gamma)q_t + s_t = F(t)/2.$$

Set $G_i = Y_i^* Y_i$, so $0 \leq G_i \leq Q_i$ and $\sum_i \text{tr}_{d_n}(G_i) \leq 1$. The product rule gives $\sum_i \mathcal{E}_n(G_i) \leq 4 \sum_i \mathcal{E}_n(Y_i) \rightarrow_{\omega} 0$. There is a common $\alpha_n \in [1/3, 1/2]$ such that the projections $P_i = 1_{[\alpha_n, 1]}(G_i)$ satisfy

$$(5.35) \quad P_i \leq Q_i, \quad \sum_i \mathcal{E}_n(P_i) \rightarrow_{\omega} 0, \quad \sum_i \text{tr}_{d_n}(Q_i - P_i) \leq 2 \sum_i \text{tr}_{d_n}(Q_i - G_i).$$

Here is the coarea estimate with the labels included. In eigenbases of the G_i , only pairs with at least one eigenvalue at least $1/3$ contribute when integrating the projection energy over $[1/3, 1/2]$. The indicator of such a pair is at most three times the sum of its two eigenvalues. Unitarity of each generator shows that the total weight of these pairs, summed also over i , is at most $\frac{3}{2} \sum_i \text{tr}_{d_n}(G_i) \leq 3/2$. Cauchy-Schwarz therefore bounds the integral by $(\frac{3}{2} \sum_i \mathcal{E}_n(G_i))^{1/2}$. Choose a common threshold with energy at most six times that bound. For the trace estimate, each eigenvalue discarded inside Q_i is below $\alpha_n \leq 1/2$ and hence has $1 - x \geq 1/2$. This proves all of (5.35).

Take the polar part of Y_i on this spectral subspace:

$$V_i = Y_i(G_i|_{P_i \mathbb{C}^{d_n}})^{-1/2} P_i, \quad V_i^* V_i = P_i, \quad \Pi_i := V_i V_i^* \leq b_i.$$

For $P_i = 0$ set $V_i = 0$. The rank bound is

$$(5.36) \quad \text{rank } P_i \leq \text{rank } f_i < (1 + t) \text{rank } p_i < 2 \text{rank } p_i.$$

To estimate the energies, define one fixed Lipschitz function

$$\chi(x) = \begin{cases} 3\sqrt{3}x, & 0 \leq x \leq 1/3, \\ x^{-1/2}, & 1/3 \leq x \leq 1. \end{cases}$$

It has $\chi(0) = 0$, $\|\chi\|_\infty \leq \sqrt{3}$ and Lipschitz constant at most $3\sqrt{3}$, and $V_i = Y_i\chi(G_i)P_i$. Entrywise comparison in an eigenbasis gives $\|[u, \chi(G_i)]\|_2 \leq 3\sqrt{3}\|[u, G_i]\|_2$. The product rule, (5.33), and (5.35) now imply

$$(5.37) \quad \sum_i \mathcal{E}_n(V_i) \rightarrow_\omega 0.$$

Put $P_0 = I - \sum_{i \geq 1} P_i$. Orthogonality and (5.34) give

$$\mathrm{tr}_{d_n}(P_0) = \mathrm{tr}_{d_n}(Q_0) + \sum_i \mathrm{tr}_{d_n}(Q_i - P_i) \leq 2M_n, \quad \lim_\omega \mathrm{tr}_{d_n}(P_0) \leq F(t).$$

In the matrix blocks of this partition, every entry counted by $\mathcal{E}_n(P_0)$ is also counted by $\sum_{i \geq 1} \mathcal{E}_n(P_i)$. Thus $\mathcal{E}_n(P_0) \leq \sum_{i \geq 1} \mathcal{E}_n(P_i) \rightarrow_\omega 0$.

The range projections, which need not be orthogonal, satisfy

$$\Pi_i \leq 3Y_i Y_i^* \leq 3g(D_{i-1})B_i g(D_{i-1}).$$

The first inequality uses $G_i \geq 1/3$ on P_i ; the second uses $Q_i \leq I$ and $B_i^2 \leq B_i$. Lemma 5.3 applied to the actual repair corners e_n consequently yields

$$(5.38) \quad \sum_i \mathrm{tr}_{d_n}(\Pi_i(I - e_n)) \rightarrow_\omega 0.$$

This is the needed estimate on the range ends $V_i V_i^*$. It follows from the monotone densities even though these ranges are only known to lie in b_i .

Step 6. In this step, we construct unitaries $w_{j,n}$ commuting with every P_i and satisfying (5.39), and transfer the minimum-rank selection test through V_i to every small-rank projection $q \leq P_i$, with additive defects whose sum ξ_n tends to zero along ω . Pinch each $u_{j,n}$ to $(P_0, P_1, \dots, P_m)$ and complete the polar partial isometry in each block. The resulting unitaries $w_{j,n}$ commute with this partition and satisfy

$$(5.39) \quad \sum_j \|w_{j,n} - u_{j,n}\|_{2,d_n}^2 \rightarrow_\omega 0.$$

For the estimate, put $C_j = \sum_i P_i u_{j,n} P_i$ and $L_j = \|u_{j,n} - C_j\|_{2,d_n}^2$. The singular-value inequality $(1-s)^2 \leq 1-s^2$ gives $\|w_{j,n} - C_j\|_{2,d_n}^2 \leq L_j$. The diagonal and off-diagonal parts are orthogonal, so $\|w_{j,n} - u_{j,n}\|_{2,d_n}^2 \leq 2L_j$. Finally $\sum_i \mathcal{E}_n(P_i) = (2h)^{-1} \sum_j L_j \rightarrow_\omega 0$, proving (5.39).

Write

$$\mathcal{E}_{w,n}(x) = \frac{1}{4h} \sum_j \| [w_{j,n}, x] \|_{2,d_n}^2,$$

and define

$$\begin{aligned} \Delta_{j,i}^+ &= u_{j,n} V_i - V_i w_{j,n}, & \Delta_{j,i}^- &= u_{j,n}^* V_i - V_i w_{j,n}^*, \\ Z_i &= \frac{1}{4h} \sum_j ((\Delta_{j,i}^+)^* \Delta_{j,i}^+ + (\Delta_{j,i}^-)^* \Delta_{j,i}^-), & H_i &= V_i^* (I - e_n) V_i. \end{aligned}$$

Both Z_i and H_i are positive and supported on P_i . The total traces tend to zero:

$$(5.40) \quad \sum_i \mathrm{tr}_{d_n}(Z_i) \rightarrow_\omega 0, \quad \sum_i \mathrm{tr}_{d_n}(H_i) \rightarrow_\omega 0.$$

For the first assertion expand $\Delta_{j,i}^+ = [u_{j,n}, V_i] + V_i(u_{j,n} - w_{j,n})$. Because the initial projections $V_i^* V_i = P_i$ are orthogonal,

$$\sum_i \|V_i(u_{j,n} - w_{j,n})\|_2^2 = \sum_i \mathrm{tr}_{d_n}((u_{j,n} - w_{j,n})^* P_i (u_{j,n} - w_{j,n})) \leq \|u_{j,n} - w_{j,n}\|_2^2.$$

Equations (5.37) and (5.39) prove the claim. The adjoint generators obey the same estimate, since $\|[u_{j,n}^*, V_i]\|_2 = \|[u_{j,n}, V_i]\|_2$ and $\|u_{j,n}^* - w_{j,n}^*\|_2 = \|u_{j,n} - w_{j,n}\|_2$. The second assertion in (5.40) is exactly (5.38) by cyclicity of trace.

Let $q \leq P_i$ be any projection of rank at most $\text{rank}(P_i)/2$. Then $y = V_i q V_i^*$ is a projection under b_i with the same rank as q , and (5.36) gives $\text{rank } y < \text{rank } p_i$. The actual minimum-rank selection rule therefore implies

$$(5.41) \quad \beta \text{tr}_{d_n}(q) \leq \mathcal{E}_n(y) + 3 \text{tr}_{d_n}(y(I - e_n)).$$

For $q = 0$ this also holds. To compare energies, use the exact identity

$$[u_{j,n}, V_i q V_i^*] = V_i [w_{j,n}, q] V_i^* + \Delta_{j,i}^+ q V_i^* - V_i q (\Delta_{j,i}^-)^*.$$

The first term has norm $\|[w_{j,n}, q]\|_2$, since $w_{j,n}$ reduces P_i . The squared norms of the other two terms, summed with weight $(4h)^{-1}$, have sum $\text{tr}_{d_n}(q Z_i)$. The triangle inequality in the direct sum over j gives

$$\sqrt{\mathcal{E}_n(y)} \leq \sqrt{\mathcal{E}_{w,n}(q)} + \sqrt{2 \text{tr}_{d_n}(q Z_i)}.$$

Squaring yields $\mathcal{E}_n(y) \leq 2\mathcal{E}_{w,n}(q) + 4 \text{tr}_{d_n}(q Z_i)$. Also $\text{tr}_{d_n}(y(I - e_n)) = \text{tr}_{d_n}(q H_i)$. Thus (5.41) gives the simultaneous all-test inequalities

$$\mathcal{E}_{w,n}(q) \geq \frac{\beta}{2} \text{tr}_{d_n}(q) - \xi_{i,n}, \quad \xi_{i,n} = 2 \text{tr}_{d_n}(Z_i) + \frac{3}{2} \text{tr}_{d_n}(H_i), \quad \xi_n := \sum_i \xi_{i,n} \xrightarrow{\omega} 0.$$

Step 7. In this step, we obtain the blocks R_i and the doubled unitaries $z_j^\pm$ by pruning the defect ξ_n with Lemma 5.4, show that $F(t) \rightarrow 0$ as t decreases to zero, choose the tolerance t with $F(t) \leq \varepsilon$, and conclude the proof. Apply Lemma 5.4 at each coordinate with $\vartheta = \beta/2$. It gives $R_i \leq P_i$, $R_0 = I - \sum_{i \geq 1} R_i$, and doubled unitaries $z_j^\pm$ reducing this partition, with

$$\text{tr}_{d_n}(R_0) - \text{tr}_{d_n}(P_0) \leq 16\xi_n/\beta, \quad \sum_{j,\pm} \|z_{j,n}^\pm - w_{j,n}\|_{2,d_n}^2 \leq 16h\xi_n/7.$$

Every retained nonzero block has scalar gap $\vartheta^2/64 = \beta^2/256$ in its own normalized trace. The displayed loss and (5.34) give $\lim_{\omega} \text{tr}_{d_n}(R_0) \leq F(t)$. Moreover

$$\sum_{j,\pm} \|z_{j,n}^\pm - u_{j,n}\|_{2,d_n}^2 \leq 32h\xi_n/7 + 4 \sum_j \|w_{j,n} - u_{j,n}\|_{2,d_n}^2 \xrightarrow{\omega} 0.$$

All partitions and unitaries have been constructed in the original $M_{d_n}(\mathbb{C})$, including at coordinates outside any prescribed member of the ultrafilter.

Finally $\sigma = t^{3/2}/(1 - \sqrt{t}) = O(t^{3/2})$, and k_t stays bounded away from zero as t decreases to zero. Thus $b_t, v_t, s_t, q_t = O(t)$ and $F(t) = O(\sqrt{t}) \rightarrow 0$. Given the prescribed ε , choose one fixed $t \in (0, 1/256]$ with $F(t) \leq \varepsilon$, and take $c = \beta^2/256 > 0$. The range and covariance approximations above are made at this fixed t before taking their ultrafilter limits. The trace estimate proves (ii), the perturbation estimate and construction prove (i), and the pruning gap proves (iii). $\square$

6. INTERNALITY OF KAZHDAN COMMUTANTS

We retain the matrix-ultraproduct conventions of Section 2. We first match scalar-expanding blocks, including blocks of different dimensions. We then exhaust the relative commutant by internal corners and compare their conditional expectations uniformly on the operator-norm unit ball. The preceding decomposition has a small discarded trace at each fixed tolerance, and its scalar gap may depend on that tolerance. We therefore apply matching in each retained corner before passing to the exhaustion. The subscript 1 denotes the trace norm, with the same trace normalization as the corresponding 2-norm.

Lemma 6.1 (Powers–Størmer). *Let (N, σ) be a tracial von Neumann algebra. For positive $x, y \in N$ one has $\|x - y\|_{2,\sigma}^2 \leq \|x^2 - y^2\|_{1,\sigma}$.*

This is the Powers–Størmer inequality in [Ioa25, eq. (2.1)]. We use it to control the modulus of an approximate intertwiner, with the same trace on both sides.

Definition 6.2. Let $h, r, t \geq 1$ be integers, let $U = (U_1, \dots, U_h) \in U(r)^h$, and let $V = (V_1, \dots, V_h) \in U(t)^h$. Write $\|Z\|_{\text{HS}}^2 = \text{Tr}(Z^*Z)$ for the ordinary Hilbert–Schmidt norm of a rectangular matrix. Define the *mixed Rayleigh minimum* by

$$\lambda(U, V) = \min_{0 \neq Z: \mathbb{C}^t \rightarrow \mathbb{C}^r} \frac{\sum_{j=1}^h \|U_j Z - Z V_j\|_{\text{HS}}^2}{4h \|Z\|_{\text{HS}}^2}.$$

Put $U_0 = I_r$ and $V_0 = I_t$. With the infimum taken over isometries $\iota: \mathbb{C}^r \rightarrow \ell^2(\mathbb{N})$ and $j: \mathbb{C}^t \rightarrow \ell^2(\mathbb{N})$, define the *stabilized orbit distance* and its normalized version by

$$d_*(U, V) = \inf_{\iota, j} \left(\sum_{j=0}^h \|\iota U_j \iota^* - j V_j j^*\|_{\text{HS}}^2 \right)^{1/2}, \quad \Delta(U, V) = \frac{d_*(U, V)}{\sqrt{r+t}}.$$

The identity coordinate records the difference in ranks. Only finite-rank operators enter the distance, and the two ranges span a space of dimension at most $r+t$. Thus the infimum can be taken in $\mathbb{C}^{r+t}$ and is attained by compactness. The mixed minimum is attained on a Hilbert–Schmidt unit sphere. Both quantities are symmetric in the two tuples; for λ this follows by taking adjoints.

Lemma 6.3. Let $h, r, t, b \geq 1$ be integers and let $U \in U(r)^h$, $V \in U(t)^h$, and $Y \in U(b)^h$. Then

$$(6.1) \quad \frac{|r-t|}{r+t} \leq \Delta(U, V)^2,$$

$$(6.2) \quad \Delta(U, Y) \leq \sqrt{\frac{r+t}{r+b}} \Delta(U, V) + \sqrt{\frac{t+b}{r+b}} \Delta(V, Y).$$

If $\Delta(U, V) < 1$, then

$$(6.3) \quad \lambda(U, V) \leq \frac{\Delta(U, V)^2}{2h(1 - \Delta(U, V)^2)}.$$

These estimates compare rank, proximity of tuples, and the existence of approximate rectangular intertwiners.

Proof. For minimizing isometries ι, j , put $P = \iota \iota^*$ and $Q = j j^*$. Then

$$\|P - Q\|_{\text{HS}}^2 = r + t - 2 \text{Tr}(PQ) \geq |r - t|,$$

which proves (6.1). To prove the triangle inequality for d_* , choose embeddings for the pairs (U, V) and (V, Y) and use a unitary of $\ell^2(\mathbb{N})$ to make the two embeddings of the middle space agree. The triangle inequality in the direct sum of $h+1$ Hilbert–Schmidt spaces gives $d_*(U, Y) \leq d_*(U, V) + d_*(V, Y)$. Division by $\sqrt{r+b}$ gives (6.2).

Finally, put $Z = \iota^* j$. We have

$$\|Z\|_{\text{HS}}^2 = \text{Tr}(PQ) \geq \frac{(1 - \Delta(U, V)^2)(r+t)}{2}.$$

Moreover, $U_j Z - Z V_j$ is the compression by ι^* and j of $\iota U_j \iota^* - j V_j j^*$. Hence

$$\sum_{j=1}^h \|U_j Z - Z V_j\|_{\text{HS}}^2 \leq (r+t) \Delta(U, V)^2.$$

Their quotient gives (6.3). □

Lemma 6.4. Let H be a discrete group with property (T), let $s_1, \dots, s_h$ be a nonempty finite symmetric generating list, and let $c \in (0, 1)$. There is $\lambda_0 > 0$, depending only on these data, with the following property. Let ω be a nonprincipal ultrafilter on $\mathbb{N}$, let $r_n, t_n \geq 1$ be integers, and let $U_n \in U(r_n)^h$ and $V_n \in U(t_n)^h$ have scalar gap c . Let F_h be the free group on h generators.

Suppose that, for every word $\mathfrak{r}$ in the kernel of the map $F_h \rightarrow H$ sending the free generators to $s_1, \dots, s_h$,

$$\|\mathfrak{r}(U_n) - I_{r_n}\|_{2,r_n} \rightarrow_\omega 0, \quad \|\mathfrak{r}(V_n) - I_{t_n}\|_{2,t_n} \rightarrow_\omega 0.$$

If $\lambda(U_n, V_n) \leq \lambda_0$ for every n , then

$$\Delta(U_n, V_n) \rightarrow_\omega 0, \quad \frac{|r_n - t_n|}{r_n + t_n} \rightarrow_\omega 0.$$

A fixed small mixed Rayleigh minimum therefore forces asymptotic unitary matching after a negligible rank correction. The coordinate tuples need only satisfy the group relations asymptotically.

Proof. We first turn a Hilbert–Schmidt intertwiner into a bounded isometry. Property (T) then produces an exact intertwiner in a tracial ultraproduct, which we lift back to the two blocks.

Choose a Kazhdan displacement constant $\kappa \in (0, 1]$ for the given list. Such a constant exists because the elements of a finite Kazhdan set are words of bounded length in this list, and displacements are bounded by the sum along each word. Put $a_* = \kappa^2/(4h)$ and choose $\lambda_0 > 0$ so small that

$$(6.4) \quad \frac{\sqrt{\lambda_0}}{c} < \frac{1}{4}, \quad \left(\sqrt{\lambda_0} + \sqrt{2/c} \lambda_0^{1/4} \right)^2 < \frac{a_*}{16}.$$

Interchanging the tuples at each coordinate if necessary, we may assume $r_n \geq t_n$.

Step 1. In this step, we construct a bounded approximate intertwiner at one coordinate, namely an isometry J extending the polar part of a minimizing Z , and we prove the rank and modulus control (6.5) and the intertwining bound (6.6). At one coordinate suppress n , put $\lambda = \lambda(U, V)$, and choose a minimizing Z with $\text{Tr}(ZZ^*) = r$. Set $b = (ZZ^*)^{1/2}$ and $e_j = \|U_j Z - ZV_j\|_{\text{HS}}/\sqrt{r}$, so that $\sum_j e_j^2 = 4h\lambda$. Expanding the difference of the two products and using trace-norm Cauchy–Schwarz gives

$$\|U_j Z Z^* U_j^* - Z Z^*\|_{1,r} \leq 2e_j.$$

Indeed, take $A = U_j Z$, $B = ZV_j$ and write $AA^* - BB^* = (A - B)A^* + B(A - B)^*$; both A and B have Hilbert–Schmidt norm $\sqrt{r}$. Lemma 6.1 therefore gives

$$\frac{1}{4h} \sum_j \| [U_j, b] \|_{2,r}^2 \leq \frac{1}{2h} \sum_j e_j \leq \sqrt{\lambda}.$$

With $m = \text{tr}_r(b)$, the scalar gap applied to $b - mI_r$ yields

$$1 - m^2 \leq \frac{\sqrt{\lambda}}{c} =: \delta.$$

Since $\text{rank } b \leq t$, Cauchy–Schwarz gives $m^2 \leq t/r$. Also $0 \leq m \leq 1$, and hence

$$(6.5) \quad \frac{t}{r} \geq 1 - \delta, \quad \|b - I_r\|_{2,r}^2 = 2(1 - m) \leq 2\delta.$$

Write the polar decomposition as $Z = bJ_0$. Extend J_0 to an isometry $J : \mathbb{C}^t \rightarrow \mathbb{C}^r$, using $r \geq t$. Its range projection contains $\text{supp } b$, so

$$\frac{\|Z - J\|_{\text{HS}}^2}{r} = \frac{\text{Tr}((b - JJ^*)^2)}{r} \leq \|b - I_r\|_{2,r}^2.$$

Together with (6.5), the triangle inequality for the list of intertwining errors gives

$$(6.6) \quad \left(\frac{1}{4hr} \sum_j \|U_j J - JV_j\|_{\text{HS}}^2 \right)^{1/2} \leq \sqrt{\lambda} + \frac{\|Z - J\|_{\text{HS}}}{\sqrt{r}} \leq \sqrt{\lambda} + \sqrt{2/c} \lambda^{1/4}.$$

Step 2. In this step, we obtain an exact intertwiner in the quotient, namely a partial isometry z in the commutant $\mathcal{D}$ inside the tracial ultraproduct $\mathcal{N}$ with $z^*z = q$ and $zz^* = p$, and we deduce $t_n/r_n \rightarrow_\omega 1$. Let $\mathcal{N}$ be the tracial ultraproduct of $M_{r_n+t_n}(\mathbb{C})$, and denote its trace by σ .

The diagonal tuples $U_n \oplus V_n$ define a homomorphism $\rho : H \rightarrow U(\mathcal{N})$ by the relation hypotheses. Let p_n, q_n project onto the two summands and write $p = [p_n]$, $q = [q_n]$ and $\mathcal{D} = \rho(H)' \cap \mathcal{N}$. Regard J_n as an upper-right rectangular matrix and put $v = [J_n]$. This is an element of $\mathcal{N}$, since $\|J_n\|_{\text{op}} = 1$. Equations (6.4)–(6.6) imply

$$\|v\|_{2,\sigma}^2 = \lim_{\omega} \frac{t_n}{r_n + t_n} \geq \frac{3}{8}, \quad \frac{1}{4h} \sum_j \|\rho(s_j), v\|_{2,\sigma}^2 < \frac{a_*}{16}.$$

By Proposition 3.1, the quotient action has gap a_* off $L^2(\mathcal{D})$. Thus $\|v - E_{\mathcal{D}}(v)\|_{2,\sigma} \leq 1/4$, and $w = E_{\mathcal{D}}(v)$ is nonzero. Since $p, q \in \mathcal{D}$, bimodularity gives $w = pwq$.

The diagonal corners $p\mathcal{D}p$ and $q\mathcal{D}q$ are scalar. For the first corner, take bounded representatives x_n supported on p_n . Their commutators tend to zero in the r_n -normalized norm because $r_n/(r_n + t_n) \geq 1/2$. The scalar gap makes $x_n - \text{tr}_{r_n}(x_n)I_{r_n}$ tend to zero in that norm, and the bounded scalars have an ultralimit. The argument for q uses $t_n/(r_n + t_n) \geq 3/8$. Consequently $w^*w = \beta q$ and $ww^* = \alpha p$ for positive scalars α, β . The identity $(ww^*)w = w(w^*w)$ gives $\alpha = \beta$, and equality of traces gives $\sigma(p) = \sigma(q)$. In particular $t_n/r_n \rightarrow_{\omega} 1$, and $z = w/\sqrt{\alpha}$ satisfies

$$z^*z = q, \quad zz^* = p, \quad z \in \mathcal{D}.$$

Step 3. In this step, we transfer z back to the blocks by choosing rectangular contraction representatives X_n and isometries W_n close to them, deduce $\Delta(U_n, V_n) \rightarrow_{\omega} 0$ and the rank conclusion, and conclude the proof. Choose rectangular contraction representatives X_n of z . One obtains them by compressing a bounded representative and clipping its singular values at one. Functional calculus shows that this leaves its quotient class unchanged. Then

$$\|X_n^*X_n - I_{t_n}\|_{2,t_n} \rightarrow_{\omega} 0.$$

Extend the polar part of X_n to an isometry $W_n : \mathbb{C}^{t_n} \rightarrow \mathbb{C}^{r_n}$. The scalar inequality $(1 - s)^2 \leq (1 - s^2)^2$ for $0 \leq s \leq 1$ gives

$$\frac{\|X_n - W_n\|_{\text{HS}}^2}{r_n} \leq \frac{\text{Tr}((I_{t_n} - X_n^*X_n)^2)}{r_n} \rightarrow_{\omega} 0.$$

It follows that $\|U_{j,n}W_n - W_nV_{j,n}\|_{\text{HS}}/\sqrt{r_n} \rightarrow_{\omega} 0$ for each j . Embed the smaller block by W_n into the larger one. Splitting the domain at $W_nW_n^*$ gives

$$\|U_{j,n} - W_nV_{j,n}W_n^*\|_{\text{HS}}^2 = \|U_{j,n}W_n - W_nV_{j,n}\|_{\text{HS}}^2 + r_n - t_n.$$

For the identity coordinate the squared difference is $r_n - t_n$. Summing these equalities proves $\Delta(U_n, V_n) \rightarrow_{\omega} 0$. The rank conclusion follows as well. $\square$

Proposition 6.5. *Let H be a discrete group with property (T), let $s_1, \dots, s_h$ be a nonempty finite symmetric generating list, and let $\pi : H \rightarrow U(\mathcal{M})$ be a homomorphism for the specified ω and dimensions d_n . Let $u_{j,n}$ be unitary lifts of $\pi(s_j)$. Suppose that $c \in (0, 1)$ and unitaries $v_{j,n} \in U(d_n)$ satisfy*

$$\sum_{j=1}^h \|u_{j,n} - v_{j,n}\|_{2,d_n}^2 \rightarrow_{\omega} 0.$$

Suppose that each $v_{j,n}$ preserves an orthogonal decomposition

$$\mathbb{C}^{d_n} = K_{0,n} \oplus \bigoplus_{\alpha \in I_n} K_{\alpha,n}, \quad \frac{\dim K_{0,n}}{d_n} \rightarrow_{\omega} 0,$$

where I_n is finite and all $K_{\alpha,n}$ are nonzero. Assume that the restricted list on each $K_{\alpha,n}$, $\alpha \in I_n$, has scalar gap c . Then $\pi(H)' \cap \mathcal{M}$ is internal in the dimensions d_n .

Neither the number of blocks nor their ranks need be bounded. The proof produces coherent multiplicity spaces and a uniform gap on every rectangular sector outside their commutant.

Proof. We remove blocks on which the group relations are inaccurate, cluster the remaining blocks by their orbit distances, and align each cluster with one representative. We then sum the estimates over all rectangular sectors. Write $\mathcal{C} = \pi(H)' \cap \mathcal{M}$ as in the standing setting, put $u_n = (u_{j,n})_{j=1}^h$ and $v_n = (v_{j,n})_{j=1}^h$, and write $v_n^\alpha = (v_{1,n}|_{K_{\alpha,n}}, \dots, v_{h,n}|_{K_{\alpha,n}})$ and $r_{\alpha,n} = \dim K_{\alpha,n}$. Let λ_0 be given by Lemma 6.4. Choose $\rho \in (0, 1/4)$ such that

$$(6.7) \quad \frac{\rho^2}{2h(1-\rho^2)} < \lambda_0.$$

Step 1. In this step, we choose the retained blocks by discarding blocks with inaccurate group relations, whose total relative dimension tends to zero, and we prove the uniform estimate (6.8) for the relation errors on the retained blocks. The tuple v_n represents the same homomorphism as u_n . Indeed, telescoping a fixed word bounds the change in its normalized 2-norm by its length times the largest generator change, with the same bound for inverse letters. Let F_h be the free group on h generators. Enumerate the kernel of $F_h \rightarrow H$ as $\mathbf{r}_1, \mathbf{r}_2, \dots$, allowing repetitions. For every $k \geq 1$, the set

$$\mathcal{F}_k = \left\{ n : \sum_{\ell=1}^k \|\mathbf{r}_\ell(v_n) - I_{d_n}\|_{2,d_n}^2 \leq k^{-4} \right\}$$

belongs to ω . Let k_n be the largest $k \leq n$ such that $n \in \bigcap_{i=1}^k \mathcal{F}_i$, using $k_n = 1$ when this set is empty. Then $k_n \rightarrow_\omega \infty$ and $n \in \mathcal{F}_{k_n}$ on an ω -large set.

Discard every block α for which

$$\|\mathbf{r}_\ell(v_n^\alpha) - I_{r_{\alpha,n}}\|_{2,r_{\alpha,n}} > k_n^{-1} \quad \text{for some } \ell \leq k_n.$$

The sum of their relative ranks is at most

$$k_n^2 \sum_{\ell=1}^{k_n} \|\mathbf{r}_\ell(v_n) - I_{d_n}\|_{2,d_n}^2 \leq k_n^{-2}.$$

This follows by expanding the squared norms blockwise and bounding the indicator of a discarded block by k_n^2 times the sum of its squared relation errors. Together with $K_{0,n}$, the discarded space has relative dimension tending to zero. If J_n indexes the retained blocks, then for every fixed ℓ ,

$$(6.8) \quad \sup_{\alpha \in J_n} \|\mathbf{r}_\ell(v_n^\alpha) - I_{r_{\alpha,n}}\|_{2,r_{\alpha,n}} \rightarrow_\omega 0.$$

The set J_n is nonempty on an ω -large set, to which we restrict for the following construction.

Step 2. In this step, we show that the relation $\Delta(v_n^\alpha, v_n^\beta) < \rho$ on the retained blocks is transitive, so that these blocks split into clusters of Δ -diameter at most ε_n , where $\varepsilon_n \rightarrow_\omega 0$, and we prove the separation (6.9) between distinct clusters. Let ε_n be the maximum of $\Delta(v_n^\alpha, v_n^\beta)$ over retained pairs with $\lambda(v_n^\alpha, v_n^\beta) \leq \lambda_0$. Such pairs exist because a block can be paired with itself. Then $\varepsilon_n \rightarrow_\omega 0$. Otherwise, on an ω -large set one could choose pairs with distance bounded below by a positive constant. Equation (6.8) would allow Lemma 6.4 to be applied to those pairs, contradicting that lower bound. Off the chosen set, both tuples may be completed by the identical one-dimensional trivial tuple. This argument controls the maximum over all pairs.

Declare $\alpha \sim \beta$ when $\Delta(v_n^\alpha, v_n^\beta) < \rho$. By (6.3) and (6.7), related blocks have distance at most ε_n . Restrict further to coordinates where $\varepsilon_n < \min\{\rho/5, 1/2\}$. If $\alpha \sim \beta$ and $\beta \sim \zeta$, then (6.1) puts the rank ratios of both pairs between $1/2$ and 2 . The two square-root coefficients in (6.2) are then less than 2 , so

$$\Delta(v_n^\alpha, v_n^\zeta) \leq 4\varepsilon_n < \rho.$$

Thus $\sim$ is an equivalence relation. Each class has Δ -diameter at most ε_n . Blocks in distinct classes satisfy

$$(6.9) \quad \lambda(v_n^\alpha, v_n^\beta) > \lambda_0,$$

since otherwise their distance would be at most $\varepsilon_n < \rho$.

Step 3. In this step, we choose a representative of least dimension in each class, construct inside every member a coherent copy of the representative tuple, and obtain a projection p_n and unitaries $w_{j,n}$ satisfying (6.12) and the multiplicity form (6.13). In each class choose a representative of least dimension t , with tuple V , and consider a member of dimension $r \geq t$, with tuple U . Put $\theta_n = \varepsilon_n + 1/n$ and choose isometries ι, j in the distance definition with

$$(6.10) \quad \sum_{j=0}^h \|\iota U_j \iota^* - j V_j j^*\|_{\text{HS}}^2 \leq 4\theta_n^2(r+t).$$

Use the identity alignment for the representative itself. Set $P = \iota^*$, $Q = j j^*$, and $Z = \iota^* j$. The identity coordinate gives

$$(6.11) \quad r - t \leq 4\theta_n^2(r+t), \quad \text{Tr}(I_t - Z^* Z) = t - \text{Tr}(PQ) \leq 2\theta_n^2(r+t).$$

Extend the polar part of Z to an isometry $W : \mathbb{C}^t \rightarrow \mathbb{C}^r$. Since Z is a contraction, $(1-s)^2 \leq 1-s^2$ for its singular values yields

$$\|W - Z\|_{\text{HS}}^2 \leq 2\theta_n^2(r+t).$$

Compressing the generator differences in (6.10), and then replacing Z by W , gives

$$\begin{aligned} \sum_j \|U_j W - W V_j\|_{\text{HS}}^2 &\leq 2 \sum_j \|U_j Z - Z V_j\|_{\text{HS}}^2 + 8h \|W - Z\|_{\text{HS}}^2 \\ &\leq (8 + 16h)\theta_n^2(r+t) \leq (16 + 32h)\theta_n^2 r. \end{aligned}$$

Keep the subspace $W\mathbb{C}^t$ and define on the full member space

$$U_j^{\text{new}} = W V_j W^* + (I_r - W W^*).$$

This is unitary. Splitting the domain at $W W^*$ gives

$$\begin{aligned} \sum_j \|U_j - U_j^{\text{new}}\|_{\text{HS}}^2 &\leq \sum_j \|U_j W - W V_j\|_{\text{HS}}^2 + 4h(r-t) \\ &\leq (16 + 64h)\theta_n^2 r. \end{aligned}$$

Also (6.11) gives $r - t \leq 8\theta_n^2 r$. Summing over all retained members and dividing by d_n bounds the additional discarded trace by $8\theta_n^2$ and the summed squared generator change by $(16 + 64h)\theta_n^2$. The sum of the member dimensions is at most d_n .

Let p_n project onto the sum of the kept subspaces. Use the representative copies there, and the identity tuple on its orthogonal complement, to obtain unitaries $w_{j,n}$. Changing the tuple to the identity on the previously discarded space costs at most four times its trace per generator. Together with the original change from u_n to v_n , this proves

$$(6.12) \quad \chi_n = \text{tr}_{d_n}(1 - p_n) \rightarrow_\omega 0, \quad \eta_n = \left(\sum_j \|u_{j,n} - w_{j,n}\|_{2,d_n}^2 \right)^{1/2} \rightarrow_\omega 0, \quad [w_{j,n}, p_n] = 0.$$

For a class with representative dimension t and chosen isometries $W_1, \dots, W_m$ into its different member spaces, the operators $W_i W_j^*$ are matrix units: their products are $\delta_{jk} W_i W_\ell^*$ and their adjoints are $W_j W_i^*$. Index the classes by ν , let $m_{\nu,n}$ be the number of members in class ν , and let $t_{\nu,n}$ be its representative dimension. Write $z_{j,n}^\nu$ for the j th unitary of its unchanged representative tuple. In the resulting orthogonal coordinates,

$$(6.13) \quad \begin{aligned} p_n \mathbb{C}^{d_n} &= \bigoplus_\nu (\mathbb{C}^{m_{\nu,n}} \otimes \mathbb{C}^{t_{\nu,n}}), \\ w_{j,n}|_{p_n \mathbb{C}^{d_n}} &= \bigoplus_\nu (I_{m_{\nu,n}} \otimes z_{j,n}^\nu). \end{aligned}$$

Step 4. In this step, we bound from below the commutator energy on all rectangular sectors of $p_n M_{d_n}(\mathbb{C}) p_n$ to prove the gap (6.14) off the commutant D_n , deduce (6.15) for $A_n = D_n \oplus \mathbb{C}(1 - p_n)$, show $[A_n]_\omega = \mathcal{C}$, and conclude the proof. Decompose $p_n M_{d_n}(\mathbb{C}) p_n$ into the rectangular spaces

between individual copies in (6.13). Between distinct classes, (6.9) bounds the commutator energy below by λ_0 times the squared Hilbert–Schmidt norm. Between two copies in the same class, identify them by their chosen isometries. The resulting action is conjugation by the representative tuple. Its fixed space is the scalars, and its gap on the trace-zero subspace is c . These rectangular spaces are orthogonal, and both squared norms and commutator energies sum over them. Therefore the exact commutant on the kept space is

$$D_n = \bigoplus_{\nu} (M_{m_{\nu,n}}(\mathbb{C}) \otimes \mathbb{C}I_{t_{\nu,n}}),$$

and its gap off that commutant is $\gamma = \min\{c, \lambda_0\} > 0$. In the original normalization, for every $y \in p_n M_{d_n}(\mathbb{C}) p_n$ this says

$$(6.14) \quad \|y - E_{D_n}(y)\|_{2,d_n} \leq \frac{(\sum_j \| [w_{j,n}, y] \|_{2,d_n}^2)^{1/2}}{2\sqrt{h\gamma}}.$$

Indeed, changing the denominator from rank p_n to d_n multiplies both squared sides by the same factor.

Put $A_n = D_n \oplus \mathbb{C}(1 - p_n)$, omitting a zero summand, and take a contraction $x \in M_{d_n}(\mathbb{C})$. For $y = p_n x p_n$ we have

$$\|x - y\|_{2,d_n} \leq 2\sqrt{\chi_n}, \quad [w_{j,n}, y] = p_n [w_{j,n}, x] p_n.$$

Also, with $e_n(x) = (\sum_j \| [u_{j,n}, x] \|_{2,d_n}^2)^{1/2}$,

$$\left(\sum_j \| [w_{j,n}, x] \|_{2,d_n}^2 \right)^{1/2} \leq e_n(x) + 2\eta_n.$$

The contraction $E_{D_n}(y)$ belongs to A_n . Since E_{A_n} is the orthogonal L^2 projection, (6.14) gives

$$(6.15) \quad \|x - E_{A_n}(x)\|_{2,d_n} \leq 2\sqrt{\chi_n} + \frac{e_n(x) + 2\eta_n}{2\sqrt{h\gamma}}.$$

Every contraction $b \in A_n$ commutes with every $w_{j,n}$, so $e_n(b) \leq 2\eta_n$.

Together with (6.12), these two estimates prove both inclusions. A bounded sequence from A_n asymptotically commutes with the generator lifts, so $[A_n]_{\omega} \subseteq \mathcal{C}$. Conversely, for a bounded lift x_n of an element of $\mathcal{C}$, rescale to contractions and apply (6.15); its right side tends to zero. The bounded sequence $E_{A_n}(x_n)$ represents the same element, proving the reverse inclusion. Complete the constructions arbitrarily on the ω -small sets excluded above. Every algebra A_n is unital in the original $M_{d_n}(\mathbb{C})$. $\square$

Proposition 6.6. *Let H be a discrete group with property (T), let $s_1, \dots, s_h$ be a nonempty finite symmetric generating list, and let $\pi : H \rightarrow U(\mathcal{M})$ be a homomorphism for the specified ω and dimensions d_n . Let $u_{j,n}$ be unitary lifts of $\pi(s_j)$, and let $\varepsilon_k \in [0, 1)$ tend to zero. Suppose that for each $k \geq 1$ there are nonzero projections $p_{k,n} \in M_{d_n}(\mathbb{C})$ and unitaries $v_{j,k,n}$ in the corners $p_{k,n} M_{d_n}(\mathbb{C}) p_{k,n}$ satisfying*

$$\lim_{\omega} \operatorname{tr}_{d_n}(1 - p_{k,n}) \leq \varepsilon_k, \quad \sum_{j=1}^h \|u_{j,n} p_{k,n} - v_{j,k,n}\|_{2,d_n}^2 \rightarrow_{\omega} 0.$$

Assume that, for each fixed k , the $v_{j,k,n}$ preserve a finite orthogonal decomposition of $p_{k,n} \mathbb{C}^{d_n}$ into nonzero blocks, and the restricted list on every such block has scalar gap $c_k > 0$. Then $\pi(H)' \cap \mathcal{M}$ is internal in the dimensions d_n .

The constants c_k may tend to zero. The retained-column errors must tend to zero for each fixed k , before k is allowed to grow.

Proof. For each fixed tolerance we realize the commutant in the retained corner. We then bound the error of the full conditional expectation by the discarded trace, transfer this bound to coordinate unit balls, and diagonalize those uniform bounds.

Write $\mathcal{C} = \pi(H)' \cap \mathcal{M}$. For a linear map on $\mathcal{M}$, its $\infty \rightarrow 2$ norm is the supremum of the output 2-norm over the operator-norm unit ball. Fix once and for all the expectation lift

$$\Phi_n = T_n^{\ell_n}, \quad [\Phi_n(x_n)] = E_{\mathcal{C}}([x_n])$$

given by Proposition 3.1 for the original generator lifts. Property (T) supplies its quotient-gap hypothesis. The displayed identity holds for every uniformly bounded sequence, and the choice of Φ_n is independent of k .

Step 1. In this step, we show for fixed k that $p_k \in \mathcal{C}$, identify the retained corner $p_k \mathcal{M} p_k$ as a tracial matrix ultraproduct carrying the same action $[v_{j,k,n}] = p_k \pi(s_j) p_k$, and obtain corner algebras $D_{k,n}$ with $[D_{k,n}]_{\omega} = p_k \mathcal{C} p_k$. Fix k and set $p_k = [p_{k,n}]$. Since the corner unitaries vanish on the orthogonal complement,

$$\|(1 - p_{k,n})u_{j,n}p_{k,n}\|_{2,d_n} \leq \|u_{j,n}p_{k,n} - v_{j,k,n}\|_{2,d_n} \rightarrow_{\omega} 0.$$

For a unitary u and a projection p , the two off-diagonal corners have equal squared Hilbert–Schmidt norm: they equal $\text{Tr}(p) - \text{Tr}((pup)^*(pup))$ and $\text{Tr}(p) - \text{Tr}((pup)(pup)^*)$, respectively. Therefore $\|[u_{j,n}, p_{k,n}]\|_{2,d_n} \rightarrow_{\omega} 0$, so $p_k \in \mathcal{C}$, and

$$[v_{j,k,n}] = p_k \pi(s_j) p_k.$$

These are the generator images of a homomorphism into $U(p_k \mathcal{M} p_k)$.

Put $\mu_k = \tau(p_k) \geq 1 - \varepsilon_k > 0$ and $r_{k,n} = \text{rank } p_{k,n}$. The corner with trace τ/μ_k is the tracial ultraproduct of the $M_{r_{k,n}}(\mathbb{C})$ with their normalized traces. Indeed, on corner-supported matrices,

$$(6.16) \quad \|x_n\|_{2,r_{k,n}}^2 = \frac{\|x_n\|_{2,d_n}^2}{\text{tr}_{d_n}(p_{k,n})},$$

and the denominator tends to $\mu_k > 0$. Thus the two null ideals agree on bounded corner sequences, the traces agree in the limit, and compression of any bounded lift supplies every element of $p_k \mathcal{M} p_k$.

Apply Proposition 6.5 in this corner ultraproduct, with the $v_{j,k,n}$ themselves as both tuples and with no exceptional block. We may replace c_k by $\min\{c_k, 1/2\}$. The proposition gives corner algebras $D_{k,n}$, unital with unit $p_{k,n}$, such that

$$[D_{k,n}]_{\omega} = p_k \mathcal{C} p_k.$$

The equality holds also with the ambient normalization by (6.16). To identify its right side, observe that an element supported on p_k commutes with every $p_k \pi(s_j) p_k$ exactly when it commutes with every $\pi(s_j)$, since $p_k \in \mathcal{C}$.

Step 2. In this step, we define $B_{k,n}$ by adjoining the scalars on the discarded corner, show $B_k \subseteq \mathcal{C}$ as in (6.17), and estimate every sector meeting the discarded corner in terms of the discarded trace, which proves (6.18). Set

$$B_{k,n} = D_{k,n} \oplus \mathbb{C}(1 - p_{k,n}), \quad B_k = [B_{k,n}]_{\omega}.$$

A bounded sequence of scalar coefficients has a scalar ultralimit. Consequently

$$(6.17) \quad B_k = p_k \mathcal{C} p_k \oplus \mathbb{C}(1 - p_k) \subseteq \mathcal{C}.$$

The scalar summand is omitted when it is zero. Centrality of p_k in $\mathcal{C}$ is not needed for this inclusion.

For a contraction $b \in \mathcal{C}$, the element $p_k b p_k$ belongs to B_k , and

$$b - p_k b p_k = (1 - p_k)b + p_k b (1 - p_k).$$

Both terms have 2-norm at most $\sqrt{\tau(1-p_k)}$. Thus orthogonal projection onto $L^2(B_k)$ gives

$$\|b - E_{B_k}(b)\|_2 \leq \|b - p_k b p_k\|_2 \leq 2\sqrt{\tau(1-p_k)} \leq 2\sqrt{\varepsilon_k}.$$

The first term includes the lower-left and lower-right sectors; the second is the upper-right sector. This estimates every sector meeting the discarded corner. For an arbitrary contraction $x \in \mathcal{M}$, apply this bound to the contraction $b = E_{\mathcal{C}}(x)$. The inclusion in (6.17) gives $E_{B_k} E_{\mathcal{C}} = E_{B_k}$. We obtain

$$(6.18) \quad \|E_{\mathcal{C}} - E_{B_k}\|_{\infty \rightarrow 2} \leq 2\sqrt{\varepsilon_k}.$$

Step 3. In this step, we show for fixed k that the maps $E_{B_{k,n}}$ induce E_{B_k} and that the coordinate suprema $\|\Phi_n - E_{B_{k,n}}\|_{\infty \rightarrow 2, n}$ for the same lift Φ_n have ultralimit $\|E_{\mathcal{C}} - E_{B_k}\|_{\infty \rightarrow 2}$, and we deduce (6.19). For fixed k , the maps $E_{B_{k,n}}$ induce E_{B_k} . To see this directly, for any bounded x_n put $y_n = E_{B_{k,n}}(x_n)$. Then $[y_n] \in B_k$, and for every bounded sequence $b_n \in B_{k,n}$,

$$\text{tr}_{d_n}(b_n^*(x_n - y_n)) = 0.$$

Passing to the limit identifies $[y_n]$ as the orthogonal projection of $[x_n]$ onto $L^2(B_k)$.

Put $L_{k,n} = \Phi_n - E_{B_{k,n}}$ and $L_k = E_{\mathcal{C}} - E_{B_k}$. These coordinate maps have uniformly bounded operator and 2-norms and induce L_k . Choose a contraction x_n attaining $\|L_{k,n}\|_{\infty \rightarrow 2, n}$, which is possible in each finite-dimensional unit ball. Then

$$\lim_{\omega} \|L_{k,n}\|_{\infty \rightarrow 2, n} = \|L_k([x_n])\|_2 \leq \|L_k\|_{\infty \rightarrow 2}.$$

Conversely, every contraction $x \in \mathcal{M}$ has contraction lifts. If z_n is a bounded lift, clip its singular values at one; the squared distance to the clipped matrices has limit

$$\lim_{\omega} \text{tr}_{d_n}((|z_n| - 1)_+^2) = \tau((|x| - 1)_+^2) = 0.$$

Applying the coordinate norm bound to such lifts of every x proves the reverse inequality. Together with (6.18), this gives

$$(6.19) \quad \lim_{\omega} \|\Phi_n - E_{B_{k,n}}\|_{\infty \rightarrow 2, n} = \|E_{\mathcal{C}} - E_{B_k}\|_{\infty \rightarrow 2} \leq 2\sqrt{\varepsilon_k}.$$

The same upper bound controls the ultralimit of $\|(I - E_{B_{k,n}})\Phi_n\|_{\infty \rightarrow 2, n}$, since

$$(I - E_{B_{k,n}})\Phi_n = (I - E_{B_{k,n}})(\Phi_n - E_{B_{k,n}})$$

and the first factor is an orthogonal L^2 projection.

Step 4. In this step, we choose indices $k(n) \rightarrow_{\omega} \infty$ by diagonalizing the uniform estimates, define $A_n = B_{k(n), n}$ on an ω -large set, prove $\|\Phi_n - E_{A_n}\|_{\infty \rightarrow 2, n} \rightarrow_{\omega} 0$, deduce $[A_n]_{\omega} = \mathcal{C}$, and conclude the proof. Put $\alpha_k = 2\sqrt{\varepsilon_k} + 1/k$ and define

$$S_k = \{n : \|\Phi_n - E_{B_{k,n}}\|_{\infty \rightarrow 2, n} \leq \alpha_k\}, \quad \Omega_k = \{n : n \geq k\} \cap \bigcap_{i=1}^k S_i.$$

Equation (6.19) implies $S_k \in \omega$, so the Ω_k are decreasing ω -large sets. Let $k(n)$ be the largest $k \leq n$ with $n \in \Omega_k$, using $k(n) = 0$ if there is none. For every fixed K , membership in Ω_K implies $k(n) \geq K$, so $k(n) \rightarrow_{\omega} \infty$. Define

$$A_n = \begin{cases} B_{k(n), n}, & k(n) \geq 1, \\ \mathbb{C}I_{d_n}, & k(n) = 0. \end{cases}$$

On an ω -large set,

$$\|\Phi_n - E_{A_n}\|_{\infty \rightarrow 2, n} \leq \alpha_{k(n)} \rightarrow_{\omega} 0.$$

Proposition 3.1 now gives $[A_n]_{\omega} = \mathcal{C}$. The diagonal uses the uniform bounds on all coordinate contractions. The estimate for each fixed k depends on the discarded trace alone, and no uniform lower bound on c_k or nesting of the retained corners is required. $\square$

Theorem 6.7. *Let H be a finitely generated discrete group with property (T), let ω be a nonprincipal ultrafilter on $\mathbb{N}$, let $d_n \geq 1$, and let $\pi : H \rightarrow U(\mathcal{M})$ be a homomorphism. Then there are unital $*$ -subalgebras $A_n \subseteq M_{d_n}(\mathbb{C})$ such that*

$$\pi(H)' \cap \mathcal{M} = [A_n]_\omega.$$

The algebras realize the full relative commutant for the given ultrafilter and the original sequence of matrix dimensions.

Proof. Choose a nonempty finite symmetric generating list $s_1, \dots, s_h$ for H , using the identity alone if H is trivial, and choose unitary lifts $u_{j,n}$ of $\pi(s_j)$. Such lifts exist by unitary polar completion: if x_n is a bounded lift of a unitary and v_n is its unitary polar completion, then

$$\|x_n - v_n\|_{2,d_n}^2 = \text{tr}_{d_n}((|x_n| - I_{d_n})^2) \leq \|x_n^* x_n - I_{d_n}\|_{2,d_n}^2 \rightarrow_\omega 0.$$

By Proposition 3.1, these lifts have quotient gap $a = \kappa^2/(4h) \in (0, 1)$ for a Kazhdan displacement constant $\kappa \in (0, 1]$.

For $k \geq 1$ put $\varepsilon_k = 2^{-k-2}$. Apply Theorem 5.1 with this tolerance. Write its projections as $R_{i,k,n}$, its unitaries as $z_{j,k,n}^\pm$, and its scalar gap as $c_k > 0$. Set

$$p_{k,n} = I_{d_n} - R_{0,k,n}, \quad v_{j,\pm,k,n} = p_{k,n} z_{j,k,n}^\pm p_{k,n}.$$

These are corner unitaries because the $z_{j,k,n}^\pm$ reduce the partition. Their nonzero good blocks decompose $p_{k,n} \mathbb{C}^{d_n}$, and

$$\lim_\omega \text{tr}_{d_n}(1 - p_{k,n}) \leq \varepsilon_k, \quad \sum_{j,\pm} \|u_{j,n} p_{k,n} - v_{j,\pm,k,n}\|_{2,d_n}^2 \leq \sum_{j,\pm} \|u_{j,n} - z_{j,k,n}^\pm\|_{2,d_n}^2 \rightarrow_\omega 0.$$

For fixed k , the retained projection is nonzero on an ω -large set, since its trace has limit at least $1 - \varepsilon_k > 0$. On the remaining coordinates replace the data by a rank-one corner with the identity tuple. This does not change any ultralimit; its scalar-gap condition is vacuous.

Use Proposition 6.6 with the fixed doubled generating list $s_1, s_1, \dots, s_h, s_h$. Its block-energy normalization is $1/(4(2h)) = 1/(8h)$, precisely the normalization for the doubled list in Theorem 5.1. Both copies of s_j have the same quotient image $\pi(s_j)$, so the commutant is still $\pi(H)' \cap \mathcal{M}$. The lazy average of the doubled original lifts equals T_n ; hence its expectation lift can be fixed once for all k . All the hypotheses of the proposition hold, and it supplies the required unital algebras in $M_{d_n}(\mathbb{C})$. In particular, the final expectation error is at most $2\sqrt{\varepsilon_k}$ at fixed k , independently of the size of c_k . $\square$

7. NORMALIZATION UNDER SUBGROUP COMPRESSIONS

We retain the ultraproduct and trace conventions of Section 2. We prove that an internal subgroup commutant is normalized by a Kazhdan group generated by subgroup compressions. The first part of the argument realizes the invariant center by finite partitions. The second part compares the dimensions and multiplicities of matrix summands in a finite-dimensional dilation. Throughout this section, asymptotic estimates are along ω . Here a compressing element satisfies the one-sided subgroup inclusion in Theorem 7.4; it acts on the subgroup commutant by conjugation by the adjoint of its image. In that theorem's proof, $\mathcal{D}$ denotes the subgroup commutant and $\mathcal{F}$ denotes the ambient-group commutant.

Lemma 7.1. *Let (N, σ) be a finite von Neumann algebra with a faithful normal tracial state, let $q \geq 1$ be an integer, and let $\alpha_1, \dots, \alpha_q : N \rightarrow N$ be normal unital trace-preserving $*$ -endomorphisms. Put*

$$F = \{x \in N : \alpha_j(x) = x \text{ for } 1 \leq j \leq q\}.$$

Then $E_F(Z(N)) \subseteq Z(N)$.

Thus taking the expectation onto the common fixed algebra preserves central elements even when the endomorphisms are not onto.

Proof. We first treat one endomorphism α . It is an isometry on $L^2(N, \sigma)$. Its fixed vectors agree with those of α^* : one implication follows from $\alpha^*\alpha = I$, and if $\alpha^*\xi = \xi$, then

$$\|\alpha\xi - \xi\|_2^2 = 2\|\xi\|_2^2 - 2\operatorname{Re}\langle \xi, \alpha^*\xi \rangle = 0.$$

The Cesàro averages of α are the identity on this fixed space and tend to zero on $(I - \alpha)L^2(N, \sigma)$ by telescoping. The closure of that range is the orthogonal complement of the fixed space. Since the averages are contractions, they converge strongly to the fixed-space projection. On bounded elements they are uniformly bounded in operator norm, so their limits belong to N . The fixed bounded elements form a von Neumann algebra F_α . The limit is trace preserving and F_α -bimodular, and hence is E_{F_α} .

For $z \in Z(N)$ and $v = E_{F_\alpha}(z)$, set

$$z_m = \frac{1}{m} \sum_{k=0}^{m-1} \alpha^k(z).$$

Each summand commutes with $\alpha^m(x)$ for $x \in N$, because z commutes with $\alpha^{m-k}(x)$. Also $\alpha^m(v) = v$. Trace preservation therefore gives

$$\|[v, x]\|_{2, \sigma} = \|[v, \alpha^m(x)]\|_{2, \sigma} \leq 2\|x\|_{\text{op}}\|v - z_m\|_{2, \sigma} \longrightarrow 0.$$

This proves $v \in Z(N)$.

For the finite family, put $F_j = F_{\alpha_j}$ and $Q = q^{-1} \sum_j E_{F_j}$ on $L^2(N, \sigma)$. This is a positive self-adjoint contraction, and

$$\langle (I - Q)\xi, \xi \rangle = \frac{1}{q} \sum_j \|\xi - E_{F_j}\xi\|_{2, \sigma}^2.$$

Its fixed space is the intersection of the individual fixed spaces. Spectral calculus shows that Q^m converges strongly to the projection onto that intersection. On bounded elements the limit is bounded, belongs to F , is trace preserving and F -bimodular, and is therefore E_F . Each Q^m preserves $Z(N)$ by the first part, so the limit does too. $\square$

Lemma 7.2. *Let ω be a nonprincipal ultrafilter on $\mathbb{N}$. For every n , let (Ω_n, μ_n) be a finite probability space whose points have positive mass, and let P_n be a reversible Markov operator whose spectrum on $L^2(\Omega_n, \mu_n)$ is contained in $[0, 1]$. Let $\mathcal{Z} = \prod_{n \rightarrow \omega} (\ell^\infty(\Omega_n), \mu_n)$ be their abelian tracial ultraproduct, with traces given by integration, and let P be the induced operator on $L^2(\mathcal{Z})$. Suppose that $\mathcal{Z}_0 \subseteq \mathcal{Z}$ is a unital von Neumann subalgebra and that, for some $a \in (0, 1)$,*

$$\ker(I - P) = L^2(\mathcal{Z}_0), \quad \operatorname{spec}(P|_{L^2(\mathcal{Z}) \ominus L^2(\mathcal{Z}_0)}) \subseteq [0, 1 - a].$$

Then there are partitions $\mathcal{P}_n$ of Ω_n such that their algebras B_n of functions constant on each part satisfy $[B_n]_\omega = \mathcal{Z}_0$.

The number of parts, the atom weights, and the ratios between those weights are unrestricted.

Proof. We first obtain a uniform estimate for characteristic functions outside a small exceptional set. We then partition by sets with small boundary and use medians to approximate every bounded fixed element. Write

$$\begin{aligned} c_n(x, y) &= \mu_n(x) P_n(x, y), & b_n(S) &= \sum_{x \in S, y \notin S} c_n(x, y), \\ \epsilon_n(f) &= \frac{1}{2} \sum_{x, y} c_n(x, y) |f(x) - f(y)|^2. \end{aligned}$$

The coefficients c_n are symmetric and nonnegative, their row sums are $\mu_n(x)$, and $\epsilon_n(f) = \langle f, (I - P_n)f \rangle$. All norms in this proof use the probability measure at hand.

Step 1. In this step, we transfer the spectral assumption to the uniform estimates (7.1), bound the total mass of disjoint exceptional sets, and choose sets J_n , of mass less than r on an ω -large

set, whose complements contain no exceptional set. For every fixed integer $k \geq 1$, the spectral assumption gives

$$\|(P^{k+1} - P^k)f\|_2 \leq (1-a)^k \|(P - I)f\|_2.$$

Consequently there are $\delta_{k,n} \rightarrow_\omega 0$ such that, for every real function f on Ω_n ,

$$(7.1) \quad \|(P_n^{k+1} - P_n^k)f\|_2 \leq (1-a)^k \|(P_n - I)f\|_2 + \delta_{k,n} \|f\|_\infty.$$

Indeed, otherwise real functions in the unit ball witnessing a fixed positive defect would give an element of $\mathcal{Z}$ contradicting the preceding inequality. Homogeneity gives the asserted estimate for every real f .

We use the following packing observation. Suppose that linear maps L_1, L_2 on a finite probability space (Ω, μ) satisfy

$$\|L_1 f\|_2 \leq c \|L_2 f\|_2 + \delta \|f\|_\infty \quad (f \text{ real}),$$

where $c, \delta \geq 0$. Fix $\eta, \beta > 0$, and call a nonempty set S exceptional if

$$\|L_1 \mathbf{1}_S\|_2 \geq (c + \eta) \|L_2 \mathbf{1}_S\|_2, \quad \|L_2 \mathbf{1}_S\|_2 \geq \beta \sqrt{\mu(S)}.$$

For disjoint exceptional sets S_i , let $f = \sum_i \epsilon_i \mathbf{1}_{S_i}$ with independent uniform signs $\epsilon_i \in \{-1, 1\}$. Then $\|f\|_\infty \leq 1$. If $v = \sum_i \|L_2 \mathbf{1}_{S_i}\|_2^2$, averaging the squared inequality over the signs gives

$$(c + \eta)^2 v \leq \mathbb{E} \|L_1 f\|_2^2 \leq c^2 v + 2c\delta\sqrt{v} + \delta^2.$$

The mixed terms vanish, and Cauchy–Schwarz bounds $\mathbb{E} \|L_2 f\|_2$ by $\sqrt{v}$. Taking square roots yields $\eta\sqrt{v} \leq \delta$, and hence

$$(7.2) \quad \sum_i \mu(S_i) \leq \frac{\delta^2}{\eta^2 \beta^2}.$$

A maximal disjoint family exists because the space is finite. After deleting its union, no exceptional set is contained in the remaining space.

Fix $0 < r < 1/4$. Choose $K \geq 1$ and $\eta > 0$ so that

$$(1-a)^K \leq \frac{r^2}{96}, \quad \eta \leq \frac{r^2}{96}, \quad K\eta \leq 1.$$

Apply (7.2) for each $1 \leq k \leq K$ with $L_1 = P_n^{k+1} - P_n^k$, $L_2 = P_n - I$, $c = (1-a)^k$, $\beta = r$, and $\delta_n = \max_{k \leq K} \delta_{k,n}$. The union J_n of the deleted sets has mass at most $K\delta_n^2/(\eta^2 r^2)$, and therefore $\mu_n(J_n) < r$ on an ω -large set.

Step 2. In this step, we define γ and prove that every nonempty $S \subseteq \Omega_n \setminus J_n$ with boundary less than γ times its mass can be replaced by a set U of small boundary satisfying (7.3). Set

$$\gamma = \frac{1}{100(1/a + 1)^2}.$$

We claim that every nonempty $S \subseteq \Omega_n \setminus J_n$ with $b_n(S) < \gamma \mu_n(S)$ admits a set $U \subseteq \Omega_n$ such that

$$(7.3) \quad \mu_n(U \triangle S) \leq \frac{1}{4} \mu_n(S), \quad b_n(U) \leq r \mu_n(U).$$

Fix n and write $\mathbf{P} = P_n$, $\mu = \mu_n$, $b = b_n$, and $\epsilon = \epsilon_n$. If $\|(\mathbf{P} - I)\mathbf{1}_S\|_2 < r\sqrt{\mu(S)}$, take $U = S$, since

$$b(S) = \langle \mathbf{1}_S, (I - \mathbf{P})\mathbf{1}_S \rangle < r \mu(S).$$

Otherwise S is not exceptional for any $1 \leq k \leq K$, so

$$\|(\mathbf{P}^{k+1} - \mathbf{P}^k)\mathbf{1}_S\|_2 < ((1-a)^k + \eta) \|(\mathbf{P} - I)\mathbf{1}_S\|_2.$$

Put $f = \mathbf{P}^K \mathbf{1}_S$. Telescoping, including the term for $k = 0$, gives

$$\|f - \mathbf{1}_S\|_2 \leq (1/a + K\eta) \|(\mathbf{P} - I)\mathbf{1}_S\|_2.$$

Since $0 \leq P \leq I$ as a Hilbert-space operator, $\|(I - P)\mathbf{1}_S\|_2^2 \leq b(S)$. Thus

$$9\|f - \mathbf{1}_S\|_2^2 < \frac{9}{100}\mu(S) < \frac{1}{4}\mu(S).$$

Every $U_t = \{f > t\}$, $1/3 < t < 2/3$, differs from S on a set of mass at most $9\|f - \mathbf{1}_S\|_2^2$. Also $0 \leq f \leq 1$, $\int f d\mu = \mu(S)$, and $\|f\|_2 \leq \sqrt{\mu(S)}$. Writing $R_0 = (1 - a)^K + \eta \leq r^2/48$, we obtain

$$\mathfrak{e}(f) \leq \|f\|_2 \|(I - P)f\|_2 \leq R_0\mu(S).$$

Here the estimate with $k = K$ was used, together with $\|(P - I)\mathbf{1}_S\|_2 \leq \sqrt{\mu(S)}$. Let $L = \{f > 1/3\}$; then $\mu(L) \leq 3\mu(S)$. The coarea identity and Cauchy–Schwarz give

$$\int_{1/3}^{2/3} b(U_t) dt \leq \sqrt{\mu(L)\mathfrak{e}(f)} \leq \sqrt{3R_0\mu(S)}.$$

To see the first inequality, write the integral as one half of the sum of the oriented edge weights times the length of $[\min(f(x), f(y)), \max(f(x), f(y))] \cap (1/3, 2/3)$. Every contributing edge meets L , so the total half-weight is at most $\mu(L)$. Some t consequently satisfies $b(U_t) \leq 3\sqrt{3R_0\mu(S)}$. Since $\mu(U_t) \geq 3\mu(S)/4$, this is at most $4\sqrt{3R_0\mu(U_t)} \leq r\mu(U_t)$, proving (7.3).

Step 3. In this step, we construct the partition $\mathcal{P}_n$ from J_n and the sets U obtained from sets S of smallest mass, and prove the expansion bound (7.4) for subsets of the other parts and the total boundary bound (7.5). Make J_n the exceptional part. In the remaining set, choose a nonempty set S of smallest mass among those satisfying $b_n(S) < \gamma\mu_n(S)$. Choose U from (7.3), remove the part $C = U \cap (\text{remaining set})$, and repeat. This part has mass between $3\mu_n(S)/4$ and $5\mu_n(S)/4$. For every $Q \subseteq C$ with $0 < \mu_n(Q) \leq \mu_n(C)/2$, its mass is at most $5\mu_n(S)/8 < \mu_n(S)$. Minimality therefore gives

$$(7.4) \quad b_n(Q) \geq \gamma\mu_n(Q).$$

If no eligible S remains, take the remainder as the final part; it also satisfies (7.4). The process terminates, since each chosen part is nonempty. We omit empty parts.

Every edge separated by the partition is first cut by J_n or by one of the sets U . For a part C obtained from S , $\mu_n(U) \leq 5\mu_n(S)/4 \leq 5\mu_n(C)/3$. It follows that

$$(7.5) \quad \sum_{C \in \mathcal{P}_n} b_n(C) \leq 2b_n(J_n) + 2 \sum_U b_n(U) \leq 6r.$$

The final remainder requires no further charge: its boundary edges were cut at earlier stages. Also $\mu_n(J_n) < r$.

Step 4. In this step, we prove the median approximation (7.6) of real functions and the energy bound (7.7) for step functions on $\mathcal{P}_n$, choose $r_n \rightarrow_\omega 0$, deduce that $[B_n]_\omega$ is the algebra $\mathcal{Z}_0$ of fixed elements, and conclude the proof. Let f be real with $\|f\|_\infty \leq M$. On each nonexceptional part C , choose a median m_C of f , meaning that each of $\{f > m_C\} \cap C$ and $\{f < m_C\} \cap C$ has mass at most $\mu_n(C)/2$. Let $g = m_C$ on C and let $g = 0$ on J_n . Apply (7.4) to the superlevel sets, inside C , of $(f - m_C)_+$ and $(m_C - f)_+$. Integrating and summing gives

$$(7.6) \quad \|f - g\|_1 \leq Mr + \gamma^{-1}(\sqrt{\mathfrak{e}_n(f)/2} + 12Mr).$$

Indeed, the interior edges contribute one half of their weighted absolute differences $|f(x) - f(y)|$. Their total is at most $\sqrt{\mathfrak{e}_n(f)/2}$ by Cauchy–Schwarz and $\sum_{x,y} c_n(x,y) = 1$. Edges leaving C contribute at most $2Mb_n(C)$, whose sum is bounded by $12Mr$ using (7.5). The exceptional part contributes at most Mr . Moreover $\|f - g\|_2^2 \leq 2M\|f - g\|_1$; when $M = 0$ there is nothing to prove. Every step function v on the partition with $\|v\|_\infty \leq M$ satisfies

$$(7.7) \quad \mathfrak{e}_n(v) \leq 2M^2 \sum_{C \in \mathcal{P}_n} b_n(C) \leq 12M^2r.$$

Finally choose $r_n \rightarrow_\omega 0$ for which these finite constructions are available. Explicitly, for each $j \geq 5$ take an ω -large set of indices on which the construction works with $r = 1/j$, and replace

these sets by their successive finite intersections X_j . At index n , use the largest $j \leq n$ for which $n \in X_j$; if there is no such j , use any partition. The chosen j tends to infinity along ω . Equation (7.7) shows that $[B_n]_\omega \subseteq \mathcal{Z}_0$. Conversely, a bounded real element of $\mathcal{Z}_0$ has uniformly bounded real lifts f_n with $\epsilon_n(f_n) \rightarrow_\omega 0$. Equation (7.6) approximates them in 2-norm by step functions with the same bound. Splitting into real and imaginary parts proves the reverse inclusion. $\square$

For finite-dimensional Hilbert spaces, we write $\|\cdot\|_{\text{HS}}$ for the unnormalized Hilbert–Schmidt norm, also for rectangular operators. A prime on a matrix algebra denotes its commutant in the full algebra on its given Hilbert space.

Lemma 7.3. *Let ω be a nonprincipal ultrafilter on $\mathbb{N}$, let $d_n \geq 1$, put $\mathcal{H}_n = \mathbb{C}^{d_n}$, and let $E_n, F_n \subseteq B(\mathcal{H}_n)$ be unital $*$ -subalgebras. Suppose that*

$$\delta_n = \sup_{\substack{x \in E_n \\ \|x\|_{\text{op}} \leq 1}} \|x - E_{F_n}(x)\|_{2, d_n} \rightarrow_\omega 0,$$

and let $L_n \subseteq E_n \cap F_n$ be a unital $$ -subalgebra. Then, on an ω -large set of indices, there are finite-dimensional Hilbert spaces K_n , isometries $V_n : \mathcal{H}_n \rightarrow K_n$, projections $R_n \in B(K_n)$, and commuting unital $*$ -representations*

$$\sigma_n : E_n \rightarrow B(R_n K_n), \quad \rho_n : F'_n \rightarrow B(R_n K_n).$$

With $P_n = V_n V_n^$ and $D_n = \text{rank } R_n$, they satisfy*

$$(7.8) \quad \|R_n - P_n\|_{\text{HS}} = o(\sqrt{d_n}), \quad D_n/d_n \rightarrow_\omega 1.$$

There are partial isometries $W_n, Z_n : \mathcal{H}_n \rightarrow R_n K_n$ satisfying

$$(7.9) \quad \begin{aligned} & \|W_n - V_n\|_{\text{HS}} + \|Z_n - V_n\|_{\text{HS}} = o(\sqrt{d_n}), \\ & d_n - \text{rank } W_n = o(d_n), \quad d_n - \text{rank } Z_n = o(d_n), \\ & W_n x = \sigma_n(x) W_n \quad (x \in E_n), \quad Z_n y = \rho_n(y) Z_n \quad (y \in F'_n), \\ & Z_n l = \sigma_n(l) Z_n \quad (l \in L_n). \end{aligned}$$

The dilation turns the approximate inclusion into the exact inclusion $\sigma_n(E_n) \subseteq \rho_n(F'_n)'$. Both representations retain all but $o(d_n)$ dimensions of their original modules.

Proof. We construct commuting representations by averaging over a unitary group, cut down by a spectral projection, and take polar decompositions to obtain the two intertwiners. Fix a coordinate and omit its subscript. On $\mathcal{K} = L^2(U(E), \nu; \mathcal{H})$, with normalized Haar measure, define

$$\begin{aligned} (V\xi)(u) &= u^* \xi, \\ (\lambda(c)f)(u) &= f(c^{-1}u) \quad (c \in U(E)), \\ (r(y)f)(u) &= yf(u) \quad (y \in F'). \end{aligned}$$

The map V is an isometry, $\lambda(c)V = Vc$, and $\lambda(c)$ commutes with $r(F')$. The subspace

$$K = \text{span}\{r(y)V\xi : y \in F', \xi \in \mathcal{H}\}$$

is finite dimensional, contains $V\mathcal{H}$, and is reducing for both actions. The action of λ on K extends to a $*$ -representation of E . Indeed, express $x \in E$ as a linear combination of unitaries. The corresponding linear combination of the operators $\lambda(c)$ sends every spanning vector $r(y)V\xi$ to $r(y)Vx\xi$. A zero linear combination in E therefore acts as zero on K , and multiplication and adjoints follow from the unitary action. We denote this extension by λ .

Step 1. In this step, we construct R as a spectral projection of the average Q of $P = VV^*$ over the unitary group of F' , prove (7.11), and define σ and ρ by restriction to the nearly unchanged

finite corner RK . Conditional expectation onto F is averaging over $U(F')$. For $c \in U(E)$ and $y \in U(F')$,

$$\|[c, y]\|_{2,d} \leq 2\|c - E_F(c)\|_{2,d} \leq 2\delta.$$

Consequently

$$(7.10) \quad \|r(y)V - Vy\|_{\text{HS}}^2 = \int_{U(E)} \|yu^* - u^*y\|_{\text{HS}}^2 d\nu(u) \leq 4\delta^2 d.$$

The projection $P = VV^*$ commutes with $\lambda(E)$. Put

$$Q = \int_{U(F')} r(y)Pr(y)^* d\nu(y).$$

This positive contraction has trace d and commutes with $\lambda(E)$ and $r(F')$. Since

$$\|P - r(y)Pr(y)^*\|_{\text{HS}}^2 = 2\|(I - P)r(y)P\|_{\text{HS}}^2,$$

equation (7.10) gives $\|Q - P\|_{\text{HS}} \leq 2\sqrt{2}\delta\sqrt{d}$. Let $R = \mathbf{1}_{[1/2,1]}(Q)$. In an eigenbasis of Q , the squared distance from Q to a projection is linear in each diagonal entry of that projection, after adding the constant $\text{Tr}(Q^2)$. Thresholding at $1/2$ minimizes this distance. Hence

$$(7.11) \quad \|R - P\|_{\text{HS}} \leq 4\sqrt{2}\delta\sqrt{d}, \quad |\text{rank } R - d| \leq \|R - P\|_{\text{HS}}^2 \leq 32\delta^2 d.$$

Set $\sigma = \lambda|_{RK}$ and $\rho = r|_{RK}$. These representations commute and have identity R . We restrict to the ω -large set on which $32\delta_n^2 < 1/2$, so $R_n K_n$ has positive dimension.

Step 2. In this step, we define the intertwiners W and Z as the polar partial isometries of RV and of the averaged contraction A_0 , bound their deficiencies by (7.12), and prove the assertions of Lemma 7.3. The contraction RV intertwines E and is within $4\sqrt{2}\delta\sqrt{d}$ of V . Its polar partial isometry W also intertwines E . We use the following elementary estimates: if $A : \mathcal{H} \rightarrow K$ is a contraction, $V : \mathcal{H} \rightarrow K$ is an isometry, and J is the polar partial isometry of A , then

$$(7.12) \quad \|J - A\|_{\text{HS}} \leq \|A - V\|_{\text{HS}}, \quad \dim \ker A \leq \|A - V\|_{\text{HS}}^2.$$

For the first inequality, $\text{Re Tr}(V^*A) \leq \text{Tr}|A|$ bounds $\|A - V\|_{\text{HS}}^2$ below by the sum of $(1 - t)^2$ over all singular values t of A . The expression $\|J - A\|_{\text{HS}}^2$ is the same sum over the nonzero singular values. For the second inequality, restrict $A - V$ to $\ker A$. These estimates give the assertions for W .

Define a second contraction

$$A_0 = \int_{U(F')} \rho(y)RVy^* d\nu(y) : \mathcal{H} \rightarrow RK.$$

It intertwines F' , and

$$\|A_0 - V\|_{\text{HS}} \leq \|RV - V\|_{\text{HS}} + 2\delta\sqrt{d}.$$

Indeed, R commutes with $r(F')$, and (7.10) bounds the difference between each averaged term and RV . Its polar partial isometry Z satisfies the asserted norm and rank estimates by (7.12). If $l \in L$, then l commutes with F' , $Vl = \lambda(l)V$, and $\lambda(l)$ commutes with $r(F')$ and R . It follows that $A_0 l = \sigma(l)A_0$. Applying the same identity to l^* shows that polar decomposition gives $Zl = \sigma(l)Z$. Together with (7.11), this proves (7.8) and (7.9). In particular, $D_n - \text{rank } W_n$ and $D_n - \text{rank } Z_n$ are also $o(d_n)$.

Step 3. In this step, we prove uniform comparisons of the represented algebras with the originals transported by V_n , namely (7.13), its general form (7.14), and the closeness in K_n of the unit balls of $F_n^\sharp$ and F_n . Extend operators on $R_n K_n$ by zero on K_n . The preceding estimates imply, uniformly for $x \in E_n$ with $\|x\|_{\text{op}} \leq 1$,

$$(7.13) \quad \|\sigma_n(x) - V_n x V_n^*\|_{\text{HS}} = o(\sqrt{d_n}).$$

In fact, $W_n x W_n^* = \sigma_n(x) W_n W_n^*$; the omitted range has dimension $D_n - \text{rank } W_n = o(d_n)$, and replacing W_n by V_n costs at most $2\|W_n - V_n\|_{\text{HS}}$. More generally, suppose x on $\mathcal{H}_n$ and $\hat{x}$ on $R_n K_n$ are contractions, and J_n is either W_n or Z_n , with

$$J_n x = \hat{x} J_n, \quad J_n x^* = \hat{x}^* J_n.$$

The same argument gives the uniform estimate

$$(7.14) \quad \|\hat{x} - V_n x V_n^*\|_{\text{HS}} \leq \sqrt{D_n - \text{rank } J_n} + 2\|J_n - V_n\|_{\text{HS}} = o(\sqrt{d_n}).$$

Put $F_n^\# = \rho_n(F_n') \cap B(R_n K_n)$. For every contraction $a \in F_n$, the contraction $Z_n a Z_n^*$ belongs to $F_n^\#$ and is within $2\|Z_n - V_n\|_{\text{HS}}$ of $V_n a V_n^*$. Conversely, if $b \in F_n^\#$ is a contraction, then $a = Z_n^* b Z_n \in F_n$ is a contraction, and

$$\|b - Z_n a Z_n^*\|_{\text{HS}} = \|b - (Z_n Z_n^*) b (Z_n Z_n^*)\|_{\text{HS}} \leq 2\sqrt{D_n - \text{rank } Z_n}.$$

Replacing $Z_n a Z_n^*$ by $V_n a V_n^*$ adds $2\|Z_n - V_n\|_{\text{HS}}$. Thus the two unit balls are uniformly $o(\sqrt{d_n})$ apart in the common space K_n . $\square$

Theorem 7.4. *Let H be a countable group with property (T), let $\Lambda \leq H$, and suppose that $H = \langle \Lambda, t_1, \dots, t_s \rangle$ for finitely many t_i satisfying $t_i \Lambda t_i^{-1} \leq \Lambda$. Let $\pi : H \rightarrow U(\mathcal{M})$ be a homomorphism into a tracial matrix ultraproduct with specified ω and dimensions d_n . Suppose $\mathcal{D} = \pi(\Lambda)' \cap \mathcal{M} = [A_n]_\omega$ is internal. Then $\pi(H)$ normalizes $\mathcal{D}$, and there are unital abelian subalgebras $B_n \subseteq Z(A_n)$ satisfying*

$$Z(\mathcal{D}) \cap \pi(H)' = [B_n]_\omega.$$

Only the subgroup commutant is assumed internal. In particular, the algebras B_n will be constructed before normalization is known.

Proof. We first realize the invariant center by applying the weighted partition lemma. For each compressor we then dilate the resulting near inclusion. Two bounded transforms of the matrix sizes and their multiplicities force a matching of almost all summands, which gives the reverse inclusion. Put

$$\mathcal{F} = \pi(H)' \cap \mathcal{M}, \quad \mathcal{Z} = Z(\mathcal{D}), \quad \mathcal{Z}_0 = \mathcal{Z} \cap \mathcal{F}.$$

Step 1. In this step, we prove the center lift (7.15), show that the coordinate expectations onto the centers of A_n induce $E_{\mathcal{Z}}$, prove the theorem when $s = 0$, and obtain the preservation (7.16) of $\mathcal{Z}$ under $E_{\mathcal{F}}$. First,

$$(7.15) \quad \mathcal{Z} = [Z(A_n)]_\omega.$$

One inclusion holds coordinatewise. For the other, let $z \in \mathcal{Z}$ be a contraction and choose contraction lifts $z_n \in A_n$. Centrality implies

$$\sup_{v \in U(A_n)} \|[z_n, v]\|_{2, d_n} \rightarrow_\omega 0 :$$

otherwise a sequence of unitary witnesses would contradict $z \in Z(\mathcal{D})$. Averaging z_n over $U(A_n)$ gives $E_{Z(A_n)}(z_n)$ within this error of z_n and proves (7.15). Furthermore, the coordinate expectations onto $Z(A_n)$ induce $E_{\mathcal{Z}}$ on all of $\mathcal{M}$: their images lie in $\mathcal{Z}$, and their differences from their inputs are orthogonal to every bounded sequence from $Z(A_n)$. If $s = 0$, then $H = \Lambda$, normalization is pointwise, and $B_n = Z(A_n)$ proves the theorem. We henceforth assume $s \geq 1$.

For each ℓ , the compressor relation gives the endomorphism

$$\alpha_\ell : \mathcal{D} \rightarrow \mathcal{D}, \quad \alpha_\ell(x) = \pi(t_\ell)^* x \pi(t_\ell).$$

Indeed, for $\lambda \in \Lambda$, the element x commutes with $\pi(t_\ell \lambda t_\ell^{-1})$. The common fixed algebra of these endomorphisms is $\mathcal{F}$, by the generation hypothesis. Lemma 7.1 gives

$$(7.16) \quad E_{\mathcal{F}}(\mathcal{Z}) \subseteq \mathcal{Z}, \quad E_{\mathcal{F}}|_{\mathcal{Z}} = E_{\mathcal{Z}_0}|_{\mathcal{Z}}.$$

The second identity follows because the restriction is trace preserving and has range $\mathcal{Z}_0$.

Step 2. In this step, we construct the Markov operator Q on $\mathcal{Z}$ from T_H , show that it fixes exactly $\mathcal{Z}_0$ with spectral gap a , and obtain algebras B_n realizing the invariant center (7.18) from the weighted partition lemma. Choose a nonempty finite symmetric generating list $\mathcal{S}$ for H and a Kazhdan displacement constant $\kappa \in (0, 1]$. Such a list exists for a countable discrete group with property (T). For the conjugation representation on $L^2(\mathcal{M})$, put

$$T_H = \frac{1}{2}I + \frac{1}{2|\mathcal{S}|} \sum_{g \in \mathcal{S}} \text{Ad}(\pi(g)).$$

This is a positive self-adjoint contraction. The property-(T) clause of Proposition 3.1, applied to this symmetric list, gives

$$(7.17) \quad \langle (I - T_H)\xi, \xi \rangle \geq a \|\xi - E_{\mathcal{F}}\xi\|_2^2, \quad a = \frac{\kappa^2}{4|\mathcal{S}|} \in (0, 1).$$

Compress to $L^2(\mathcal{Z})$ by setting $Q = E_{\mathcal{Z}}T_H|_{L^2(\mathcal{Z})}$. It is a positive self-adjoint Markov operator. Equations (7.16) and (7.17) imply

$$\langle (I - Q)z, z \rangle \geq a \|z - E_{\mathcal{Z}_0}z\|_2^2.$$

Every element of $L^2(\mathcal{Z}_0)$ is fixed by Q ; the displayed inequality identifies its fixed space and bounds its remaining spectrum by $1 - a$.

Choose unitary representatives $u_{g,n}$ of $\pi(g)$ with $u_{g^{-1},n} = u_{g,n}^*$ for paired elements. For an involution, spectral rounding supplies a self-adjoint unitary representative. Define $T_{H,n}$ by the same lazy formula and put

$$Q_n = E_{Z(A_n)}T_{H,n}|_{Z(A_n)}.$$

Identify $Z(A_n)$ with functions on its minimal central projections, giving an atom p the positive weight $\text{tr}_{d_n}(p)$. Then Q_n is a Markov kernel reversible for these weights, and its Hilbert-space spectrum is in $[0, 1]$. The preceding description of the coordinate expectations shows that its induced operator is Q . Lemma 7.2 therefore provides unital abelian algebras

$$(7.18) \quad B_n \subseteq Z(A_n), \quad [B_n]_{\omega} = \mathcal{Z}_0.$$

Step 3. In this step, we construct compatible unitary lifts $U_{\ell,n} \in B'_n$ of the compressors as in (7.19), prove the uniform near inclusion (7.20) of $E_{\ell,n}$ in A_n , and obtain a dilation for every compressor from the dilation lemma. Since each $\pi(t_{\ell})$ commutes with $[B_n]_{\omega}$, any unitary lifts $v_{\ell,n}$ satisfy

$$\sup_{b \in U(B_n)} \|[v_{\ell,n}, b]\|_{2,d_n} \rightarrow_{\omega} 0.$$

Failure would again give unitary witnesses contradicting commutation in the ultraproduct. Haar averaging gives contractions $a_{\ell,n} = E_{B'_n}(v_{\ell,n})$ with $\|a_{\ell,n} - v_{\ell,n}\|_{2,d_n} \rightarrow_{\omega} 0$. Extend their polar parts to unitaries on each matrix summand of B'_n . The resulting unitaries satisfy

$$(7.19) \quad U_{\ell,n} \in B'_n \cap U(d_n), \quad [U_{\ell,n}]_{\omega} = \pi(t_{\ell}).$$

For the estimate, the averaged contraction has $\text{tr}_{d_n}(I - a_{\ell,n}^* a_{\ell,n}) \rightarrow_{\omega} 0$, and the squared distance from it to the polar unitary is bounded by this quantity, since $(1 - t)^2 \leq 1 - t^2$ for $0 \leq t \leq 1$.

Set $E_{\ell,n} = U_{\ell,n}^* A_n U_{\ell,n}$. The inclusion $\alpha_{\ell}(\mathcal{D}) \subseteq \mathcal{D}$ implies

$$(7.20) \quad \sup_{\substack{x \in E_{\ell,n} \\ \|x\|_{\text{op}} \leq 1}} \|x - E_{A_n}(x)\|_{2,d_n} \rightarrow_{\omega} 0.$$

If this failed, contraction witnesses would represent an element of $\mathcal{D}$ at a positive distance from $[A_n]_{\omega}$. The conditional expectations supply contractions in the target algebra. Also $B_n \subseteq E_{\ell,n} \cap A_n$ by (7.19). Apply Lemma 7.3 with $E_n = E_{\ell,n}$, $F_n = A_n$, and $L_n = B_n$, separately for every ℓ . Since there are finitely many compressors, we may use one ω -large set of indices for all their dilations. Arbitrary choices on its complement will not affect any conclusion.

Step 4. In this step, we compare the matrix sizes and multiplicities of E and A through the dilation, proving (7.21) and the branching relations (7.22), and obtain the one-sided inequalities (7.23) after deleting blocks of total dimension $o(d_n)$. Fix one compressor and suppress ℓ, n temporarily. Write the matrix algebra decompositions on $\mathcal{H} = \mathbb{C}^d$ as

$$A = \bigoplus_j (M_{b_j}(\mathbb{C}) \otimes I_{s_j}), \quad E = U^* A U = \bigoplus_i (M_{a_i}(\mathbb{C}) \otimes I_{r_i}),$$

and denote the respective minimal central projections by f_j and e_i . Define

$$\begin{aligned} h_A &= \sum_j b_j f_j, & k_A &= \sum_j s_j f_j, \\ h_E &= \sum_i a_i e_i = U^* h_A U, & k_E &= \sum_i r_i e_i = U^* k_A U. \end{aligned}$$

Use the notation of the dilation lemma, and put $A^\sharp = \rho(A')' \cap B(RK)$. Let $r_i^\sharp$ be the multiplicity of the i th simple E -module in σ , and let $b_j^\sharp$ be the multiplicity of the j th simple A' -module in ρ . Some of these multiplicities may be zero. The two partial isometries give

$$(7.21) \quad \sum_i a_i |r_i^\sharp - r_i| = o(d), \quad \sum_j s_j |b_j^\sharp - b_j| = o(d).$$

For example, if q_i is the multiplicity retained by W , then $q_i \leq r_i, r_i^\sharp$, and

$$\sum_i a_i |r_i^\sharp - r_i| \leq \sum_i a_i (r_i + r_i^\sharp - 2q_i) = d + D - 2 \text{rank } W = o(d).$$

Using the intertwining of A' by Z proves the second estimate.

The exact inclusion $\sigma(E) \subseteq A^\sharp$ has a matrix of nonnegative integer multiplicities m_{ij} satisfying

$$(7.22) \quad b_j^\sharp = \sum_i m_{ij} a_i, \quad r_i^\sharp = \sum_j m_{ij} s_j.$$

Indeed, on the j th A' -isotypic subspace, $\sigma(E)$ acts on the multiplicity space of dimension $b_j^\sharp$; decompose that space into its simple E -modules. Thus $\sigma(e_i)$ and $\rho(f_j)$ commute, and their intersection has rank $a_i m_{ij} s_j$. On a nonzero intersection,

$$a_i \leq b_j^\sharp, \quad s_j \leq r_i^\sharp.$$

These projections also commute with $\sigma(B)$.

Choose $0 < \varepsilon_n < 1$ tending to zero slowly enough that each error in (7.21), divided by $\varepsilon_n d_n$, tends to zero. Delete the j -blocks with $|b_j^\sharp - b_j| > \varepsilon_n b_j$ and the i -blocks with $|r_i^\sharp - r_i| > \varepsilon_n r_i$. The deleted subspace in RK has dimension $o(d_n)$. For the first family this follows from

$$b_j^\sharp \leq b_j + |b_j^\sharp - b_j| \leq (1 + \varepsilon_n^{-1}) |b_j^\sharp - b_j|,$$

after multiplying by s_j and summing. Multiply the analogous estimate for the second family by a_i . On every remaining nonzero intersection we have

$$(7.23) \quad a_i \leq (1 + \varepsilon_n) b_j, \quad r_i \geq (1 + \varepsilon_n)^{-1} s_j.$$

Step 5. In this step, we define the two bounded median transforms $F_{A_n}^h$ and $F_{A_n}^k$ of the size and multiplicity operators, prove their asymptotic invariance (7.26) under every compressor by comparing them in the dilation, and deduce their concentration (7.28) at $1/2$. For every coordinate, make the following choices using only A_n and B_n , independently of the compressor. On each minimal projection z of $B = B_n$, choose a positive eigenvalue median m_z^h of h_A on $z\mathcal{H}$ and a positive eigenvalue median m_z^k of k_A there, counting eigenvalues with their multiplicities. Put

$$M_B^h = \sum_z m_z^h z, \quad M_B^k = \sum_z m_z^k z, \quad F_A^h = h_A (h_A + M_B^h)^{-1}, \quad F_A^k = k_A (k_A + M_B^k)^{-1}.$$

These are central positive contractions in A , and each has median $1/2$ on every $z\mathcal{H}$. Since U commutes with B , define

$$F_E^h = h_E(h_E + M_B^h)^{-1} = U^* F_A^h U, \quad F_E^k = k_E(k_E + M_B^k)^{-1} = U^* F_A^k U.$$

Only these bounded transforms, rather than the possibly unbounded size and multiplicity operators, will pass to the ultraproduct.

On RK , form the commuting operators

$$\begin{aligned} \hat{h}_E &= \sum_i a_i \sigma(e_i), & \hat{k}_E &= \sum_i r_i \sigma(e_i), \\ \hat{h}_A &= \sum_j b_j \rho(f_j), & \hat{k}_A &= \sum_j s_j \rho(f_j), \\ \widehat{M}_B^\theta &= \sigma(M_B^\theta) \quad (\theta = h, k). \end{aligned}$$

Use them to define

$$\begin{aligned} \hat{F}_E^h &= \hat{h}_E(\hat{h}_E + \widehat{M}_B^h)^{-1}, & \hat{F}_E^k &= \hat{k}_E(\hat{k}_E + \widehat{M}_B^k)^{-1}, \\ \hat{F}_A^h &= \hat{h}_A(\hat{h}_A + \widehat{M}_B^h)^{-1}, & \hat{F}_A^k &= \hat{k}_A(\hat{k}_A + \widehat{M}_B^k)^{-1}. \end{aligned}$$

All inverses are taken on RK . The coefficients in these formulas are the original a_i, r_i, b_j, s_j , not the dilation multiplicities. All four operators are commuting positive contractions. The source transforms are intertwined with the originals by W . For the target transforms, $h_A, k_A \in A'$ are intertwined by Z with their images under ρ , while $M_B^\theta \in B$ is intertwined with its image under σ . These exact identities pass to the displayed rational functional calculi and their adjoints. Equation (7.14) gives

$$(7.24) \quad \|\hat{F}_E^\theta - V F_E^\theta V^*\|_{\text{HS}} + \|\hat{F}_A^\theta - V F_A^\theta V^*\|_{\text{HS}} = o(\sqrt{d}) \quad (\theta = h, k).$$

This uses the two intertwiners and the bounds by one on the transforms; it requires neither bounds on the central labels nor equality of the kernels of the two joint representations.

For $x, m > 0$ and $0 < \varepsilon < 1$,

$$\begin{aligned} 0 &\leq \frac{(1+\varepsilon)x}{(1+\varepsilon)x+m} - \frac{x}{x+m} \leq \varepsilon, \\ 0 &\leq \frac{x}{x+m} - \frac{x/(1+\varepsilon)}{x/(1+\varepsilon)+m} \leq \varepsilon. \end{aligned}$$

Combining these inequalities with (7.23) by simultaneous diagonalization shows, on the complement of the deleted subspace,

$$(7.25) \quad \hat{F}_E^h - \hat{F}_A^h \leq \varepsilon_n R, \quad \hat{F}_E^k - \hat{F}_A^k \geq -\varepsilon_n R.$$

The deleted subspace has dimension $o(d)$ and commutes with all four transforms. Their differences have operator norm at most one. Also

$$\text{Tr}(\hat{F}_E^\theta - \hat{F}_A^\theta) = o(d),$$

by (7.24) and $\text{Tr}(F_E^\theta) = \text{Tr}(F_A^\theta)$. To justify this use of trace, the comparison errors are supported on $V\mathcal{H} + RK$, of dimension at most $d + D = O(d)$. Cauchy–Schwarz converts their $o(\sqrt{d})$ Hilbert–Schmidt norms into $o(d)$ trace errors, irrespective of $\dim K$.

Equation (7.25) bounds the trace of the positive part of the h -difference, and of the negative part of the k -difference, by $\varepsilon_n D + o(d) = o(d)$. The trace identities bound the other parts by $o(d)$ as well. Since the differences are contractions,

$$\|\hat{F}_E^\theta - \hat{F}_A^\theta\|_{\text{HS}}^2 \leq \text{Tr}|\hat{F}_E^\theta - \hat{F}_A^\theta| = o(d).$$

Transferring back and restoring the indices gives

$$(7.26) \quad \|U_{\ell,n}^* F_{A_n}^\theta U_{\ell,n} - F_{A_n}^\theta\|_{2,d_n} \longrightarrow_\omega 0 \quad (\theta = h, k, 1 \leq \ell \leq s).$$

The sequences $F_{A_n}^h, F_{A_n}^k$ depend only on A_n, B_n , so the same sequences satisfy this conclusion for all compressors. Their classes lie in $\mathcal{Z}$, and now commute with each $\pi(t_\ell)$. They already commute with $\pi(\Lambda)$, hence belong to $\mathcal{Z}_0 = [B_n]_\omega$. Orthogonal projection gives

$$(7.27) \quad \|F_{A_n}^\theta - E_{B_n}(F_{A_n}^\theta)\|_{2,d_n} \rightarrow_\omega 0 \quad (\theta = h, k).$$

On each minimal B_n -block, this expectation is the eigenvalue mean of the transform. A real distribution with median $1/2$, mean c , and variance v satisfies $v \geq (c - 1/2)^2/2$: at least half its mass lies on the side of $1/2$ opposite c . Its mean squared distance from $1/2$ is therefore at most $3v$. Apply this in every block, with eigenvalues counted according to their actual multiplicities, and sum with the block trace weights. Equation (7.27) yields

$$(7.28) \quad \|F_{A_n}^h - \tfrac{1}{2}I_{d_n}\|_{2,d_n} \rightarrow_\omega 0, \quad \|F_{A_n}^k - \tfrac{1}{2}I_{d_n}\|_{2,d_n} \rightarrow_\omega 0.$$

Step 6. In this step, we prove the two-sided inequalities (7.29) on almost all intersections, deduce from the branching relations that the good pairs form a partial matching of the summands, and obtain the projection S_n satisfying (7.30). Fix a compressor again and use its dilation. Equations (7.24), (7.28), and (7.8) show that all four hatted transforms are within $o(\sqrt{d_n})$ of $R_n/2$. Choose $0 < \zeta_n < 1/2$ tending to zero slowly enough that the spectral projection in $B(R_n K_n)$ of each transform outside $[1/2 - \zeta_n, 1/2 + \zeta_n]$ has rank $o(d_n)$. This follows by bounding each such rank by its squared Hilbert–Schmidt error divided by ζ_n^2 . The four transforms commute with one another and with every $\sigma(e_i)\rho(f_j)$, so their common good spectral subspace has codimension $o(d_n)$ in $R_n K_n$.

Define

$$\chi_n = \left(\frac{1 + 2\zeta_n}{1 - 2\zeta_n} \right)^2 \rightarrow_\omega 1.$$

If an intersection $\sigma(e_i)\rho(f_j)$ has a nonzero vector in this common good subspace, simultaneous diagonalization, including the median operators, gives

$$(7.29) \quad \chi_n^{-1} \leq a_i/b_j \leq \chi_n, \quad \chi_n^{-1} \leq r_i/s_j \leq \chi_n.$$

Indeed, if $x/(x + m)$ lies in the indicated interval and $m > 0$, then

$$\frac{1 - 2\zeta_n}{1 + 2\zeta_n} \leq \frac{x}{m} \leq \frac{1 + 2\zeta_n}{1 - 2\zeta_n}.$$

Use the same median eigenvalue for the source and target size transforms, and then for the two multiplicity transforms. The ratios in (7.29) are constant on the entire intersection. Thus the sum of the intersections failing these inequalities has rank $o(d_n)$. Discard also those incident to the blocks removed in proving (7.23). Call every remaining nonzero intersection good.

On a good intersection,

$$\frac{a_i}{b_j^\sharp} \geq \frac{1}{\chi_n(1 + \varepsilon_n)} > \frac{1}{2}, \quad \frac{s_j}{r_i^\sharp} \geq \frac{1}{\chi_n(1 + \varepsilon_n)} > \frac{1}{2}$$

on an ω -large set. Equation (7.22) implies that a good pair has $m_{ij} = 1$, that one target j has at most one good source, and that one source i has at most one good target. For the first two conclusions use $b_j^\sharp = \sum_i m_{ij}a_i$; for the third use $r_i^\sharp = \sum_j m_{ij}s_j$. The good pairs therefore form a partial matching. Let $S_n \leq R_n$ be the sum of their intersection projections. Then

$$(7.30) \quad D_n - \text{rank } S_n = o(d_n), \quad S_n \sigma_n(E_{\ell,n}) S_n = S_n A_n^\sharp S_n.$$

For the equality, within a matched target block j , the unique source i occupies one copy of $\mathbb{C}^{a_i}$ inside the multiplicity space $\mathbb{C}^{b_j^\sharp}$. Both compressed algebras are $M_{a_i}(\mathbb{C}) \otimes I_{s_j}$ there. Distinct good pairs have distinct sources and distinct targets, so these matrices can be chosen independently on both sides. Also S_n commutes with $\sigma_n(E_{\ell,n})$. Every contraction in the common corner lifts to a contraction in $\sigma_n(E_{\ell,n})$: use the chosen $M_{a_i}(\mathbb{C})$ matrix on the full multiplicity space of a retained source and zero on all other source summands.

Step 7. In this step, we prove the reverse near inclusion (7.31) of A_n in $E_{\ell,n}$ from the matched corner, deduce that every compressor, and hence the image of H under π , normalizes $\mathcal{D}$, and conclude the proof. If $x \in A_n^\sharp$ is a contraction, choose a contraction $c^\sharp \in \sigma_n(E_{\ell,n})$ with $S_n x S_n = S_n c^\sharp S_n$. For any contraction y on $R_n K_n$,

$$\|y - S_n y S_n\|_{\text{HS}} \leq 2\sqrt{D_n - \text{rank } S_n}.$$

Applying this to x and $c^\sharp$ shows $\|x - c^\sharp\|_{\text{HS}} = o(\sqrt{d_n})$, uniformly over x . Given a contraction $x_0 \in A_n$, take $x = Z_n x_0 Z_n^* \in A_n^\sharp$. The lift just described can be written $c^\sharp = \sigma_n(c_0)$ with $c_0 \in E_{\ell,n}$ a contraction, by filling unused simple summands with zero. The uniform estimates in the dilation lemma give

$$\begin{aligned} & \|V_n x_0 V_n^* - V_n c_0 V_n^*\|_{\text{HS}} \\ & \leq 2\|Z_n - V_n\|_{\text{HS}} + \|x - c^\sharp\|_{\text{HS}} + \|\sigma_n(c_0) - V_n c_0 V_n^*\|_{\text{HS}} \\ & = o(\sqrt{d_n}). \end{aligned}$$

Since V_n is an isometry, this proves

$$(7.31) \quad \sup_{\substack{x_0 \in A_n \\ \|x_0\|_{\text{op}} \leq 1}} \inf_{\substack{c_0 \in E_{\ell,n} \\ \|c_0\|_{\text{op}} \leq 1}} \|x_0 - c_0\|_{2,d_n} \xrightarrow{\omega} 0.$$

Together with (7.20), this identifies the internal algebras in the original dimensions:

$$\mathcal{D} = [A_n]_\omega = [E_{\ell,n}]_\omega = \pi(t_\ell)^* \mathcal{D} \pi(t_\ell).$$

Every compressor normalizes $\mathcal{D}$, and $\pi(\Lambda)$ fixes it pointwise. The generation hypothesis gives normalization by $\pi(H)$. Equation (7.18) already supplies the required invariant-center realization. $\square$

8. A NONHYPERLINEAR WREATH PRODUCT

We retain the normalized trace convention of Section 2. We apply internality and normalization to a polynomial elementary subgroup of a Laurent semidirect product. The resulting obstruction is a product of two lamps at distinct cosets. For $v = (v_1, v_2, v_3) \in \mathbb{Z}^3$ we write $x^v = x_1^{v_1} x_2^{v_2} x_3^{v_3}$, and E_{ij} denotes the usual matrix unit.

Definition 8.1. Put $R_+ = \mathbb{F}_2[x_1, x_2, x_3]$ and $R = \mathbb{F}_2[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}]$. For a unital ring B , let $\text{EL}_3(B)$ be the subgroup of $\text{GL}_3(B)$ generated by $I_3 + bE_{ij}$ with $b \in B$ and $i \neq j$. For $A \in \text{SL}_3(\mathbb{Z})$ let $\alpha_A(x^v) = x^{Av}$, extended to R and its matrix entries. Define

$$\Gamma = \text{EL}_3(R_+), \quad G = \text{EL}_3(R) \rtimes_\alpha \text{SL}_3(\mathbb{Z}), \quad W = \left(\bigoplus_{G/\Gamma} C_2 \right) \rtimes G.$$

The subgroup Γ lies in the first factor of G , the cosets are left cosets, and the lamp action is left translation. For a coset x let a_x be its nonidentity lamp. Put

$$t = (I_3, I_3 + E_{12}), \quad \gamma_0 = (I_3 + x_2 E_{12}, I_3), \quad b = a_{\gamma_0 t \Gamma} a_{t \Gamma}.$$

For $A, B \in \text{SL}_3(\mathbb{Z})$, the maps α_A fix the coefficient field and satisfy $\alpha_A \alpha_B = \alpha_{AB}$. Their inverses are $\alpha_{A^{-1}}$. Thus, for $u, v \in \text{EL}_3(R)$, the semidirect product has multiplication

$$(u, A)(v, B) = (u\alpha_A(v), AB).$$

We identify its first factor with its image in G , and identify G with its image in W . The lamp relations are

$$a_x^2 = e, \quad a_x a_y = a_y a_x, \quad g a_x g^{-1} = a_{gx} \quad (x, y \in G/\Gamma, g \in G).$$

The group-theoretic input is the following result [KT26, Theorem E and Section 5.2].

Theorem 8.2 (Kun–Thom). *Let q be a prime power and let $r, d \geq 3$. With the monomial substitution action of $\mathrm{SL}_d(\mathbb{Z})$, both*

$$\mathrm{EL}_r(\mathbb{F}_q[x_1, \dots, x_d]) \quad \text{and} \quad \mathrm{EL}_r(\mathbb{F}_q[x_1^{\pm 1}, \dots, x_d^{\pm 1}]) \rtimes \mathrm{SL}_d(\mathbb{Z})$$

have property (T). The first group is a nonnormal subgroup of the second, and the elements g satisfying $g\Lambda g^{-1} \leq \Lambda$ generate the second group, with Λ denoting the first.

At $q = 2$ and $r = d = 3$, the action in this theorem is exactly the action in Definition 8.1. In particular, Γ and G have property (T). We will verify finite generation and the required finite list of compressing elements directly. The matrix approximation obstruction then follows from Theorems 6.7 and 7.4.

Theorem 8.3. *The group W of Definition 8.1 is countable and $b \neq e$. For every nonprincipal ultrafilter ω on $\mathbb{N}$, every sequence $d_n \geq 1$, and every homomorphism $\rho : W \rightarrow U(\prod_{n \rightarrow \omega} (M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n}))$, one has $\rho(b) = I$.*

There exist a finite symmetric $F \subset W$ containing e and $\eta > 0$ such that for every $d \geq 1$ and every $\phi : F \rightarrow U(d)$ with $\phi(e) = I_d$, at least one of the following holds:

- (i) *Some $g, h, gh \in F$ satisfy $\|\phi(gh) - \phi(g)\phi(h)\|_{2,d} \geq \eta$.*
- (ii) *Some distinct $g, h \in F$ satisfy $\|\phi(g) - \phi(h)\|_{2,d} - \sqrt{2} \geq \eta$.*

In particular, W is not hyperlinear.

The same nonidentity element is killed by every homomorphism into a tracial matrix ultraproduct. The finite set and tolerance in the second assertion are independent of the matrix dimension.

Proof. We first distinguish the two lamp coordinates and verify the generation hypotheses. We then use normalization to identify their images, and pass from this identity to finitely many local approximation constraints.

Step 1. In this step, we prove that W is countable and that the two cosets $\gamma_0 t\Gamma$ and $t\Gamma$ are distinct, and deduce that $b \neq e$. Both rings are countable: their elements are finite sums of monomials over a finite field. Their matrix groups, the integer matrix group, and hence G and G/Γ are countable. There are countably many finite subsets of a countable set, and each finite support permits only finitely many lamp configurations. The direct sum of the lamps, and therefore W , is countable.

The shear defining t sends x_1 to x_1 , x_2 to $x_1 x_2$, and x_3 to x_3 . Consequently

$$\alpha_{I_3 + E_{12}}(R_+) = \mathbb{F}_2[x_1, x_1 x_2, x_3].$$

Every monomial in this subring has its x_1 exponent at least its x_2 exponent. Linear independence of Laurent monomials gives $x_2 \notin \mathbb{F}_2[x_1, x_1 x_2, x_3]$. Every matrix entry of an element of $t\Gamma t^{-1}$ belongs to this subring, so the $(1, 2)$ entry of γ_0 shows that $\gamma_0 \notin t\Gamma t^{-1}$. Hence

$$\gamma_0 t\Gamma \neq t\Gamma.$$

The two lamps defining b are supported at these distinct coordinates. Their product is nonidentity in the direct sum, which embeds in W . Thus $b \neq e$.

Step 2. In this step, we prove that the finite symmetric set S of elementary matrices generates Γ , that the six elements t_{ij} satisfy $t_{ij}\Gamma t_{ij}^{-1} \leq \Gamma$, and that these compressing elements together with Γ generate G . The group Γ is generated by the following 24 matrices:

$$S = \{I_3 + cE_{ij} : i, j \in \{1, 2, 3\}, i \neq j, c \in \{1, x_1, x_2, x_3\}\}.$$

Each is an involution. For coefficients $f_1, f_2 \in R_+$,

$$(I_3 + f_1 E_{ij})(I_3 + f_2 E_{ij}) = I_3 + (f_1 + f_2)E_{ij}.$$

For three distinct indices i, k, j , direct matrix multiplication gives

$$(I_3 + f_1 E_{ik})(I_3 + f_2 E_{kj})(I_3 - f_1 E_{ik})(I_3 - f_2 E_{kj}) = I_3 + f_1 f_2 E_{ij}.$$

Induction on total degree now gives every monomial in every off-diagonal position. Indeed, remove one variable from a monomial, use the induction hypothesis for the remaining monomial in position (i, k) , and use that variable in position (k, j) . The first identity then gives every polynomial coefficient. This proves the generating claim, including symmetry of S .

For $A \in \text{SL}_3(\mathbb{Z})$ put $t_A = (I_3, A)$, and put $t_{ij} = t_{I_3 + E_{ij}}$ for $i \neq j$. The entries of $I_3 + E_{ij}$ are nonnegative integers, so its monomial substitution maps R_+ into R_+ . Thus

$$t_{ij} \Gamma t_{ij}^{-1} \leq \Gamma \quad (i \neq j).$$

These six elements, together with Γ , generate G . To check the second factor, the powers of $I_3 + E_{ij}$ perform elementary row additions by arbitrary integers. On two coordinates, the product

$$(I_3 + E_{ij})(I_3 - E_{ji})(I_3 + E_{ij})$$

is a signed row exchange; its square changes both signs. The Euclidean algorithm, followed by elimination, therefore reduces every determinant-one integer matrix to the identity using these row additions. It follows that the six positive elementary matrices generate $\text{SL}_3(\mathbb{Z})$ as a group.

To obtain the first factor, let $\nu \in \mathbb{Z}^3 \setminus \{0\}$. Write $\nu = mu$ with $m \geq 1$ and u primitive. The same integer row reduction sends u to the first standard basis vector e_1 . The inverse of that row reduction is a matrix $A \in \text{SL}_3(\mathbb{Z})$ with $A(me_1) = \nu$. For every $i \neq j$ we have, identifying the first factor as above,

$$t_A(I_3 + x_1^m E_{ij})t_A^{-1} = I_3 + x^\nu E_{ij}.$$

The matrix inside the conjugation belongs to Γ . Constant elementary matrices also belong to Γ . Adding coefficients in a fixed position now gives every Laurent polynomial coefficient, and these elementary matrices generate $\text{EL}_3(R)$ by definition. We have proved

$$G = \langle \Gamma, t_{ij} : i \neq j \rangle.$$

Step 3. In this step, we show that $\rho(G)$ normalizes the commutant $\mathcal{D}$ of $\rho(\Gamma)$ in $\mathcal{M}$, deduce that the two lamps defining b have the same image under ρ , and conclude that the lamp product collapses, namely $\rho(b) = I$. Fix ω , the dimensions $d_n \geq 1$, and ρ as in the statement, and write $\mathcal{M}$ for its target algebra. Theorem 8.2 gives property (T) for both groups. Since S is finite, Theorem 6.7 applied to $\rho|_\Gamma$ gives unital $*$ -subalgebras $A_n \subseteq M_{d_n}(\mathbb{C})$ such that

$$\mathcal{D} = \rho(\Gamma)' \cap \mathcal{M} = [A_n]_\omega.$$

We can apply Theorem 7.4 with $H = G$, $\Lambda = \Gamma$, and the six t_{ij} . Countability, property (T), finite generation by these compressing elements and Γ , and internality have all been established. It follows that $\rho(G)$ normalizes $\mathcal{D}$.

The subgroup Γ fixes the coset Γ under left translation. Thus $\rho(a_\Gamma) \in \mathcal{D}$. Normalization and lamp covariance give

$$\rho(a_{t\Gamma}) = \rho(t)\rho(a_\Gamma)\rho(t)^* \in \mathcal{D}.$$

This unitary commutes with $\rho(\gamma_0)$ because $\gamma_0 \in \Gamma$. Therefore

$$\rho(a_{\gamma_0 t \Gamma}) = \rho(\gamma_0)\rho(a_{t\Gamma})\rho(\gamma_0)^* = \rho(a_{t\Gamma}).$$

Both lamp images square to the identity, so $\rho(b) = I$. This argument applies to every specified ultrafilter and every sequence of positive dimensions.

Step 4. In this step, we prove by contradiction, using that b has trivial image in every tracial matrix ultraproduct, that finitely many relators $\mathcal{R}_0 \subseteq \mathcal{R}$ and some $\delta > 0$ suffice for the implication (8.1) in every dimension d . We convert the identity just proved into a bound for finite words, uniformly over all matrix dimensions. Use the countable presentation

$$\mathcal{S} = \{s_g : g \in W\}, \quad \mathcal{R} = \{s_e\} \cup \{s_g s_h s_{gh}^{-1} : g, h \in W\}.$$

This presents W : the map $s_g \mapsto g$ respects the relators, and the relations make the map sending g to the class of s_g a homomorphism into the presented group inverse to this map. Every relator has a displayed spelling of length at most three. For a word v in these generators and their inverses, write $\bar{v}$ for its group image and $|v|$ for the length of its spelling. For an assignment $U : \mathcal{S} \rightarrow U(d)$, let $v(U)$ be its evaluation, with negative letters evaluated by adjoints. Set $w = s_b$, so $|w| = 1$ and $\bar{w} = b \neq e$.

There are a finite $\mathcal{R}_0 \subseteq \mathcal{R}$ and $\delta > 0$ such that, for every $d \geq 1$ and every such assignment,

$$(8.1) \quad (\|r(U) - I_d\|_{2,d} < \delta \text{ for all } r \in \mathcal{R}_0) \implies \|w(U) - I_d\|_{2,d} \leq 1.$$

Suppose this were false. Choose increasing finite subsets $\mathcal{R}_n$ exhausting $\mathcal{R}$. For each $n \geq 1$, the failure of the assertion gives a positive integer D_n and an assignment $V^{(n)} : \mathcal{S} \rightarrow U(D_n)$ satisfying

$$\|r(V^{(n)}) - I_{D_n}\|_{2,D_n} < \frac{1}{n} \quad (r \in \mathcal{R}_n), \quad \|w(V^{(n)}) - I_{D_n}\|_{2,D_n} > 1.$$

Every fixed relator defect tends to zero. All assigned matrices have operator norm one, so their classes define a homomorphism from the free group into $U(\prod_{n \rightarrow \omega} (M_{D_n}(\mathbb{C}), \text{tr}_{D_n}))$. It kills all the relators and therefore factors to a homomorphism σ from W . By the trace formula in the ultraproduct,

$$\|\sigma(b) - I\|_2 = \lim_{\omega} \|w(V^{(n)}) - I_{D_n}\|_{2,D_n} \geq 1.$$

The universal collapse already proved gives $\sigma(b) = I$, a contradiction. This establishes (8.1) with no restriction on d .

Step 5. In this step, we define the finite symmetric subset F of W and the tolerance η , compare word evaluation with ϕ along prefixes to obtain (8.2), deduce the second assertion of Theorem 8.3, and conclude the proof. Let F be the symmetric closure of the set consisting of e , the group images of all generator symbols occurring in $\mathcal{R}_0$ or w and their inverses, and the images of all prefixes of their displayed spellings. This is a finite symmetric subset of W and contains both e and b . Put

$$\eta = \min \left\{ \frac{\delta}{12}, \frac{\sqrt{2}-1}{6} \right\} > 0.$$

Suppose that some $d \geq 1$ and some $\phi : F \rightarrow U(d)$ with $\phi(e) = I_d$ have every multiplicative defect and every separation deviation in the statement strictly less than η . Assign $U_s = \phi(\bar{s})$ to each generator symbol occurring in $\mathcal{R}_0$ or w , and assign I_d to every other symbol. For every occurring symbol s , the triple $\bar{s}, \bar{s}^{-1}, e$ lies in F , so

$$\|\phi(\bar{s}^{-1}) - U_s^*\|_{2,d} < \eta.$$

We compare word evaluation with ϕ along prefixes. Let $v = s_1^{\epsilon_1} \cdots s_\ell^{\epsilon_\ell}$ be one of the chosen words, with $\epsilon_j \in \{1, -1\}$. Put $g_j = \bar{s}_j^{\epsilon_j}$, $p_j = g_1 \cdots g_j$, and $P_j = U_{s_1}^{\epsilon_1} \cdots U_{s_j}^{\epsilon_j}$, with $p_0 = e$ and $P_0 = I_d$. Each p_{j-1}, g_j, p_j belongs to F . Unitary invariance of the normalized 2-norm and the triangle inequality give

$$\begin{aligned} \|P_j - \phi(p_j)\|_{2,d} &\leq \|P_{j-1} - \phi(p_{j-1})\|_{2,d} + \|U_{s_j}^{\epsilon_j} - \phi(g_j)\|_{2,d} \\ &\quad + \|\phi(p_{j-1})\phi(g_j) - \phi(p_j)\|_{2,d}. \end{aligned}$$

The last two terms have sum less than 2η . Induction yields

$$(8.2) \quad \|v(U) - \phi(\bar{v})\|_{2,d} \leq 2|v|\eta.$$

For a relator $r \in \mathcal{R}_0$ its image is e , and hence

$$\|r(U) - I_d\|_{2,d} \leq 2|r|\eta \leq 6\eta \leq \frac{\delta}{2} < \delta.$$

Equation (8.1) now implies $\|w(U) - I_d\|_{2,d} \leq 1$. On the other hand, separation for the distinct elements b, e gives $\|\phi(b) - I_d\|_{2,d} > \sqrt{2} - \eta$. Using (8.2) for w gives

$$\|w(U) - I_d\|_{2,d} > \sqrt{2} - 3\eta \geq \frac{\sqrt{2} + 1}{2} > 1,$$

a contradiction. Thus every identity-preserving map on this F , in every positive matrix dimension, has a multiplicative defect or a separation deviation at least η . By Definition 2.6, W is not hyperlinear. $\square$

APPENDIX A. DETAILS OF THE USE OF GENERATIVE AI

The results of this paper were obtained using Danus [Liu⁺26], an automated mathematical reasoning system built on Rethlas [Ju⁺26]. For transparency we describe how the system was run (§A.1) and reproduce the original problem statement (§A.2) and instruction record (§A.3), with only direct human quotations replaced by privacy notices.

A.1. How the system was run. The project used the base model `gpt-6-astra`. The main coordinator ran at reasoning effort `ultra`; the seven proof workers comprised four at `max` and three at `xhigh`. The verifier ran at `max`. Separate helper sessions investigated literature, proof architecture and independent audits. The project was created on 19 September 2026 at 15:18:31 UTC. Its complete target proof was accepted by the project’s verifier at 22:29:37 UTC on the same day, after 7 hours, 11 minutes and 6 seconds.

Independent automated audits then reviewed the complete dependency arguments and their primary sources. Six separate writer–verifier loops prepared the mathematical sections, after which a whole-manuscript verifier checked their proofs and interfaces. The introduction and disclosure appendix were completed after that check. These are automated textual mathematical reviews; the records contain no formal proof-assistant certification or external human verification. No comparison experiment or mathematical computation was used. No human author identity, affiliation or funding information is recorded for this local working draft.

A.2. The problem as posed. The following is the problem statement supplied to the project, with direct human quotations replaced by privacy notices. Line wrapping is for display only.

```
# Problem: construct a nonhyperlinear discrete group

## Verbatim operator authorization

> [28 bytes of discussion have been omitted because they contain private information.]

> [139 bytes of discussion have been omitted because they contain private information.]

The precise formulation below preserves the full problem selected in the preceding
↪ discussion.

Date: 2026-09-19. This is the binding mathematical target for this research pod. Read
↪ `RESEARCH_GUIDANCE.md` and complete its mandatory source-reading checkpoint before
↪ committing to a research strategy.

## Original goal

Construct, or rigorously specify by an unconditional existence construction, a discrete
↪ group  $G$  and prove that  $G$  is not hyperlinear. One such group resolves the target.
↪ Finite generation, a finite presentation, simplicity, torsion-freeness, or another
↪ desirable extra property may be pursued when useful, but none is an additional
↪ requirement. The construction and the nonhyperlinearity theorem must both be
↪ unconditional.
```

The choice of group and proof method is open. The pod is not assigned to prove a
 ↪ particular sufficient stability conjecture. It may change candidate groups or routes
 ↪ while preserving this original goal.

Exact approximation convention

For $d \geq 1$ and complex d -by- d matrices, use the ****normalized Hilbert-Schmidt norm****

$$\|A\|_{(2,d)} = (\text{Tr}(A^* A) / d)^{1/2},$$

$$\text{dist}_{(2,d)}(U,V) = \|U - V\|_{(2,d)}.$$

Here Tr is the ordinary, unnormalized matrix trace; $U(d)$ is the complex unitary group.
 ↪ All occurrences of the target approximation norm mean this norm, with its
 ↪ dimension-dependent normalization included.

Use the finite-local definition, which applies without a cardinality assumption on G .
 ↪ The group G is hyperlinear if, for every finite symmetric F subset G containing the
 ↪ identity e and every $\epsilon > 0$, there are an integer $d \geq 1$ and a map $\phi: F \rightarrow$
 ↪ $U(d)$ such that:

1. $\phi(e) = I_d$;
2. $\|\phi(gh) - \phi(g)\phi(h)\|_{(2,d)} < \epsilon$ whenever g, h , and gh belong to F ;
3. $|\|\phi(g) - \phi(h)\|_{(2,d)} - \sqrt{2}| < \epsilon$ for all distinct g, h in F .

For a countable group, this is equivalent to the usual existence of a sequence of maps
 ↪ $\pi_n: G \rightarrow U(d_n)$ with multiplicative defects tending to zero for every fixed pair
 ↪ and pairwise distances tending to $\sqrt{2}$ for every fixed pair of distinct elements.
 ↪ There is no upper bound on d or d_n . Do not use failure of a sequence formulation
 ↪ for an uncountable group as a cardinality shortcut.

Complete success criterion

A complete solution gives a rigorously specified group G and proves the negation of the
 ↪ finite-local property. Equivalently, it proves that there exist a finite symmetric F
 ↪ subset G containing e and an $\epsilon > 0$ such that ****for every integer $d \geq 1$ and every**
 ↪ **map $\phi: F \rightarrow U(d)$ with $\phi(e) = I_d$ **, at least one of the following holds:**

1. Some g, h, gh in F satisfy $\|\phi(gh) - \phi(g)\phi(h)\|_{(2,d)} \geq \epsilon$; or
2. Some distinct g, h in F satisfy $|\|\phi(g) - \phi(h)\|_{(2,d)} - \sqrt{2}| \geq \epsilon$.

This finite obstruction must exclude all matrix dimensions, not just dimensions below an
 ↪ experimental cutoff. A rigorous existence proof for F and ϵ is acceptable; an
 ↪ explicit numerical ϵ or effective finite certificate is welcome but is not an
 ↪ extra completion requirement.

An equivalent proof through tracial matrix ultraproducts or the non-Connes-embeddability
 ↪ of the group von Neumann algebra with its canonical trace is acceptable, provided
 ↪ the equivalence with this group target is stated and justified. If the proof uses a
 ↪ distinguished group element or word, prove that it is nonidentity in the constructed
 ↪ group; a presentation alone does not establish that fact.

The final theorem must have a closed proof chain. Check the hypotheses and normalization
 ↪ of every imported theorem. Identify the genuinely new argument and all its
 ↪ dependencies. The status of a conjectural, draft, or not-yet-checked dependency must
 ↪ remain visible until resolved. Standard established theorems may be cited precisely;
 ↪ a claim merely appearing in an AI repository or an internal verifier output is not
 ↪ itself a mathematical justification.

Results that do not complete this goal

- Nonsoficity, failure of LEF or residual finiteness, or exclusion of permutation models
 ↪ alone.

- Non-MF results, operator-norm approximation obstructions, or obstructions in
 - ↪ unnormalized Schatten norms without a proved implication to the target normalized
 - ↪ norm.
- A non-co-hyperlinear invariant random subgroup, a nonembeddable equivalence-relation
 - ↪ algebra, or a non-Connes-embeddable tracial algebra without the required bridge to a
 - ↪ group and its canonical trace.
- A conditional nonhyperlinearity theorem whose stability, lifting, compatibility,
 - ↪ completeness, or uniformity hypothesis remains unproved.
- Success in fixed or bounded matrix dimension, dimension-dependent estimates with no
 - ↪ adequate uniform conclusion, computational failures to find approximations, or
 - ↪ finitely many checked cases.
- Disproof of one proposed sufficient route, clarification of existing literature, or a
 - ↪ collection of local lemmas without a complete nonhyperlinear group construction.

Such work may be useful internal research, but it must be labeled accordingly. Do not
 ↪ replace the original target by a nearby problem or redefine completion to match an
 ↪ intermediate achievement.

Research behavior and completion record

Read primary sources first, then choose the strongest available route. The mandatory
 ↪ checkpoint in `RESEARCH\_GUIDANCE.md` requires comprehension, not merely downloading
 ↪ files or recording their titles. Research may use any valid method and may abandon
 ↪ an unproductive route without changing the goal.

Keep theorem-level progress, conditional reductions, failed approaches, and remaining
 ↪ gaps distinct. Before announcing completion, write a concise target check containing
 ↪ the exact group construction, the all-dimensions normalized-HS obstruction or
 ↪ equivalent theorem, the full dependency chain, and the resolution of every
 ↪ previously open hypothesis on that chain. A full mathematical proof is required; a
 ↪ promising strategy is not completion.

A.3. Guidance given during the run. The instruction record contains the following three operator messages, all dated 19 September 2026. The first, recorded at launch, required complete resolution:

[28 bytes of discussion have been omitted because they contain private
 ↪ information.]

It fixed complete resolution as the success criterion; conditional and partial results remained research support.

The second launch instruction authorized three standard research pods:

[139 bytes of discussion have been omitted because they contain private
 ↪ information.]

This project concerned only the nonhyperlinear-group problem, with the seven-worker configuration described above and no dimension bound.

The message starting the main session was:

You are the main agent for Danus project nonhyperlinear (this
 ↪ directory). Read AGENTS.md and begin.

It began the authorized research under the fixed problem statement. No mathematical route, proof hint or preferred theorem was supplied in these instructions. The initial literature memo was assistant-prepared context, rather than an additional operator instruction.

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